

**Tendances design et couleurs**

**Décor bois**



## Prêts à collaborer ?

Créons ensemble la solution adaptée à votre prochain projet.

## Contactez-nous

Joey: +86 187 5138 2597

Adam: +86 152 0514 9312

Calina: +86 153 7177 5487

Adam: [marketing31@changlongflor.com](mailto:marketing31@changlongflor.com)

Calina: [md06@changlongflor.com](mailto:md06@changlongflor.com)

Joey: [sales01@changlongflor.com](mailto:sales01@changlongflor.com)



# Changlong Floor

Un catalogue complet de revêtements de sol haute performance  
aux décors contemporains.

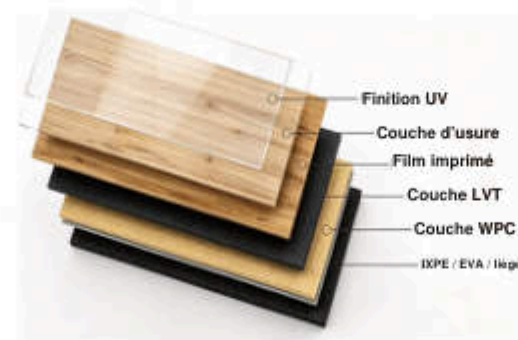
# WPC



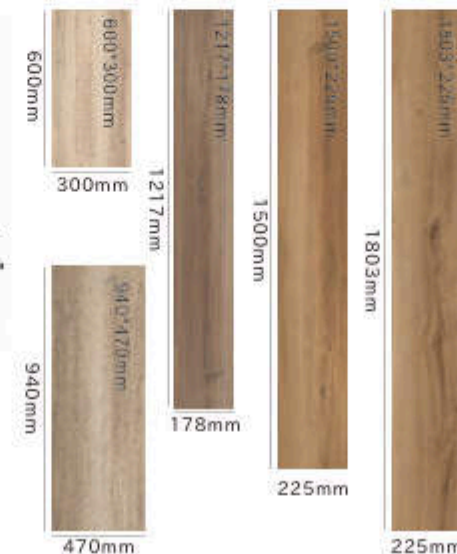
## Notre sélection WPC

|                      | Plus                             | Prime              | Premium            |
|----------------------|----------------------------------|--------------------|--------------------|
| Épaisseur totale     | 6.5mm                            | 8.0mm              | 10.0-14.0mm        |
| Épaisseur d'usure    | 0.50mm                           | 0.50mm             | 0.50-0.70mm        |
| Sous-couche          | 1mmIXPE                          | 1.5mmIXPE          | 2.0mmIXPE          |
| Format lames bois    | 1217*178                         | 1500*225           | 1803*225           |
| Format dalles pierre | 600*300mm                        | 940*470mm          | 940*470mm          |
| Relief               | Veinage bois / EIR               | Veinage bois / EIR | Veinage bois / EIR |
| Clipsage             | I4F/U Click/V Click              |                    |                    |
| Finition de surface  | Finition UV / traitement excimer |                    |                    |
| Chanfreins           | Chanfrein peint                  | Chanfrein peint    | Bord façonné       |

## Structure du WPC



## Présentation des formats



## Environnement / Certifications



## Chanfreins



Microchanfrein

Le microchanfrein est une légère inclinaison discrète sur le bord de la lame. Il souligne les contours de chaque lame pour mieux les distinguer après la pose, tout en masquant les petites imperfections entre les lames.



Chanfrein peint

Le chanfrein peint consiste à appliquer une couche de peinture sur le bord du revêtement pour créer un aspect biseauté. Il apporte de la profondeur et du relief à la surface et peut atténuer les différences visuelles entre les lames.



Bord façonné

Élégance et richesse visuelle

Le bord façonné est obtenu par moulage d'un effet biseauté sur les bords du revêtement. Il accentue la séparation entre les lames, souligne leur individualité et reproduit l'aspect naturellement paillé du bois véritable.

Chanfrein à 45° (âme apparente)



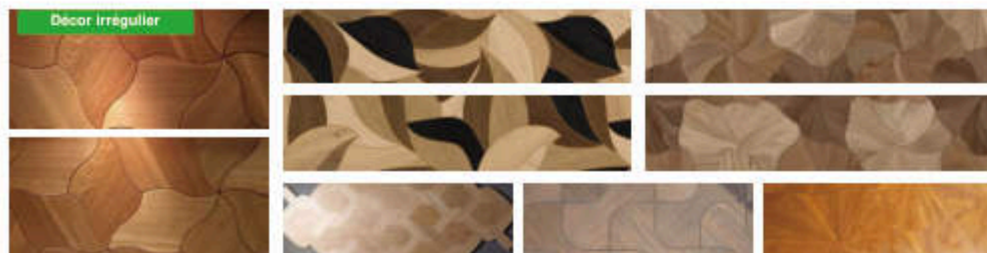
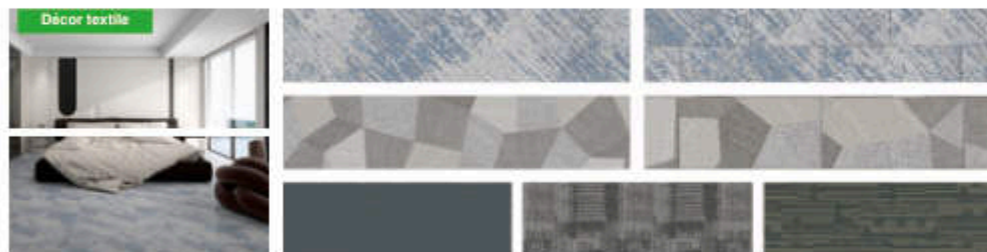
Le chanfrein à 45° est une coupe franche à 45 degrés sur le bord des lames, assez profonde pour révéler leur âme. Il crée des lignes de séparation marquées et un relief prononcé, tout en masquant les irrégularités plus visibles entre les lames.

Bord arrondi (rayon R)



Le bord à rayon R présente une courbe lisse et arrondie sur le pourtour de la lame. Il adoucit les contours entre les lames pour un aspect plus doux, aide à éviter les dommages sur les bords et masque les petites imperfections entre les lames.

## Tendances design et couleurs



## Prêts à collaborer ?

Créons ensemble la solution adaptée à votre prochain projet.

## Contactez-nous

Joey: +86 187 5138 2597

Adam: +86 152 0514 9312

Calina: +86 153 7177 5487

Adam: [marketing31@changlongflor.com](mailto:marketing31@changlongflor.com)

Calina: [md06@changlongflor.com](mailto:md06@changlongflor.com)

Joey: [sales01@changlongflor.com](mailto:sales01@changlongflor.com)



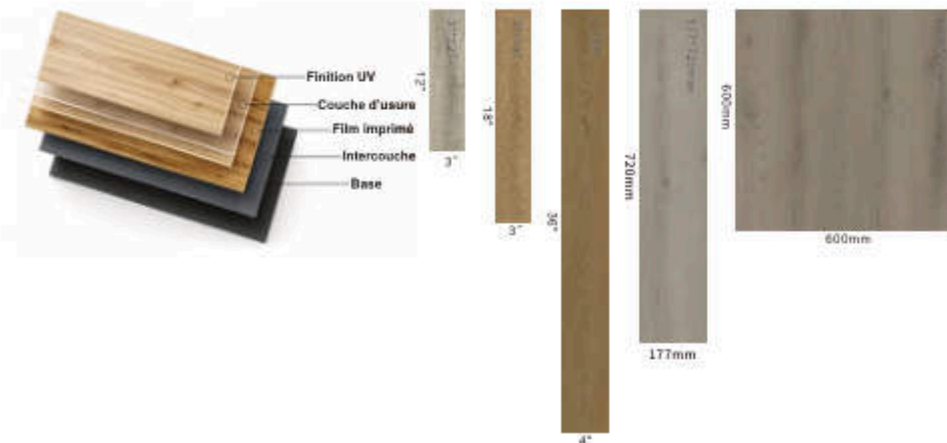
# Changlong Floor

Un catalogue complet de revêtements de sol haute performance  
aux décors contemporains.

## Notre sélection LVT

|                                                             |                                 |             |             |              |              |          |     |
|-------------------------------------------------------------|---------------------------------|-------------|-------------|--------------|--------------|----------|-----|
| Épaisseur totale                                            | 2.0                             | 2.5         | 3.0         | 3.5          | 4.0          | 4.5      | 5.0 |
| Couche décorative                                           | Impression standard / numérique |             |             |              |              |          |     |
| Clipsage                                                    | I4F/U Click/V Click             |             |             |              |              |          |     |
| Chanfreins : pressé / microchanfrein / peint / bord façonné |                                 |             |             |              |              |          |     |
| Groupe d'usure                                              | T                               | T           | T           | T            | T            | T        | T   |
| Réaction au feu                                             | Bfl-S1                          |             |             |              |              |          |     |
| Relief                                                      | Veinage bois / pierre / EIR     |             |             |              |              |          |     |
| Format lames bois                                           | 76.2*457.2                      | 101.6*914.4 | 152.4*609.6 | 184.2*1219.2 | 228.6*1422.4 | 235*1524 |     |
| Format dalles pierre                                        | 304.8*304.8                     | 406.4*406.4 | 457.2*914.4 | 609.6*609.6  | 914.4*914.4  | 600*600  |     |
| Format bâtons rompus                                        | 76.2*228.6                      | 76.2*457.2  | 114.3*457.2 | 177.8*711.2  |              |          |     |
| Brillance                                                   | Mat / brillant / mat excimer    |             |             |              |              |          |     |

## Une variété de formats



## Groupe d'usure

- Toutes les éprouvettes ont été découpées dans l'échantillon.
- Critères de classement des groupes d'usure selon EN 649:2011 :

| Caractéristique                                    | Critères par groupe d'usure |                |                |                 |
|----------------------------------------------------|-----------------------------|----------------|----------------|-----------------|
|                                                    | T                           | P              | M              | F               |
| Perte de volume, Fv (mm <sup>3</sup> ) / 100 tours | Fv ≤ 2.0                    | 2.0 < Fv ≤ 4.0 | 4.0 < Fv ≤ 7.5 | 7.5 < Fv ≤ 15.0 |

Résultat de l'essai :

| Essai                | Résultat                       |                |
|----------------------|--------------------------------|----------------|
|                      | Perte de volume, Fv            | Groupe d'usure |
| Résistance à l'usure | 2.0 mm <sup>3</sup> /100 tours | T              |

## Chanfreins



Microchanfrein

Le microchanfrein est une légère inclinaison discrète sur le bord de la lame. Il souligne les contours de chaque lame pour mieux les distinguer après la pose, tout en masquant les petites imperfections entre les lames.



Chanfrein peint

Le chanfrein peint consiste à appliquer une couche de peinture sur le bord du revêtement pour créer un aspect biseauté. Il apporte de la profondeur et du relief à la surface et peut atténuer les différences visuelles entre les lames.



Bord façonné

Élégance et richesse visuelle

Le bord façonné est obtenu par moulage d'un effet biseauté sur les bords du revêtement. Il accentue la séparation entre les lames, souligne leur individualité et reproduit l'aspect naturellement patiné du bois véritable.



Chanfrein à 45° (âme apparente)

Le chanfrein à 45° est une coupe franche à 45 degrés sur le bord des lames, assez profonde pour révéler leur âme. Il crée des lignes de séparation marquées et un relief prononcé, tout en masquant les irrégularités plus visibles entre les lames.



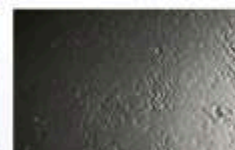
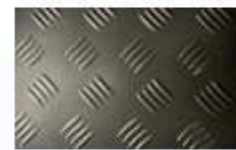
Bord arrondi (rayon R)

Le bord à rayon R présente une courbe lisse et arrondie sur le pourtour de la lame. Il adoucit les contours entre les lames pour un aspect plus doux, aide à éviter les dommages sur les bords et masque les petites imperfections entre les lames.

## Environnement / Certifications



## Relief



## Clipsage

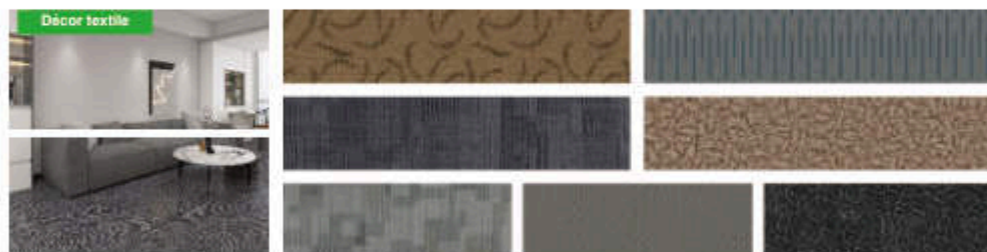
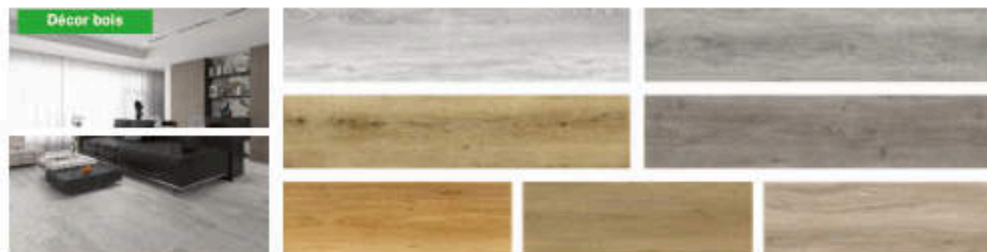
CLIC  
I4F

La technologie de clipsage 2G assure une pose solide, stable et sûre. Les lames sont d'abord inclinées, puis verrouillées grâce au mécanisme d'emboîtement.

CLIC  
Unilin

Lors de l'abaissement de la lame, la languette souple en plastique 5G s'enfonce dans sa rainure. Une fois la lame bien posée, elle reprend sa place avec un « clic », bloquant le revêtement horizontalement et verticalement.

## Tendances design et couleurs



## Prêts à collaborer ?

Créons ensemble la solution adaptée à votre prochain projet.

## Contactez-nous

Joey: +86 187 5138 2597

Adam: +86 152 0514 9312

Calina: +86 153 7177 5487

Adam: [marketing31@changlongflor.com](mailto:marketing31@changlongflor.com)

Calina: [md06@changlongflor.com](mailto:md06@changlongflor.com)

Joey: [sales01@changlongflor.com](mailto:sales01@changlongflor.com)



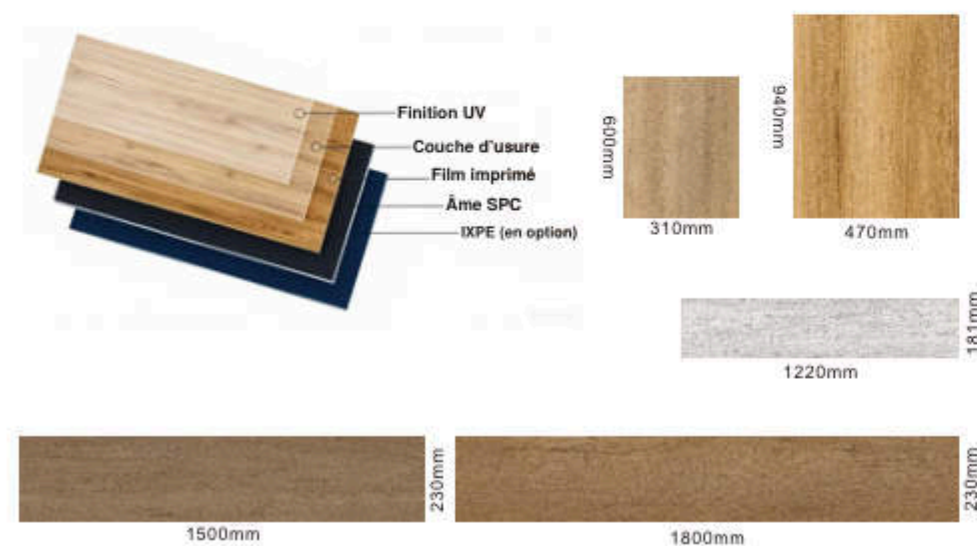
# Changlong Floor

Un catalogue complet de revêtements de sol haute performance  
aux décors contemporains.

## Notre sélection SPC

|                      | Classique | Plus                                      | Prime     | Premium   |
|----------------------|-----------|-------------------------------------------|-----------|-----------|
| Épaisseur totale     | 3.2-5.2   | 6.5                                       | 7.0-8.0   | 10.0-12.0 |
| Couche d'usure       | 0.3       | 0.55                                      | 0.55      | 0.55-0.70 |
| Sous-couche          | 1mmIXPE   | 1.5mmIXPE                                 | 1.5mmIXPE | 2.0mmIXPE |
| Format lames bois    | 1220*181  | 1500*230                                  | 1500*230  | 1800*230  |
| Format dalles pierre | 600*310   | 940*470                                   | 940*470   | 940*470   |
| Format bâtons rompus | 600*100   | 630*126                                   | 750*150   | 750*150   |
| Format chevrons      | 625*177   | 625*177                                   | 625*177   | 625*177   |
| Relief               |           | Veinage bois / EIR / pierre               |           |           |
| Clipsage             |           | I4F/U Click/V Click                       |           |           |
| Finition de surface  |           | Finition UV / traitement excimer          |           |           |
| Chanfreins           |           | Pressé / microchanfrein / peint / façonné |           |           |

## Une variété de formats



## Environnement / Certifications



## Chanfreins



Microchanfrein

Le microchanfrein est une légère inclinaison discrète sur le bord de la lame. Il souligne les contours de chaque lame pour mieux les distinguer après la pose, tout en masquant les petites imperfections entre les lames.



Chanfrein peint

Le chanfrein peint consiste à appliquer une couche de peinture sur le bord du revêtement pour créer un aspect biseauté. Il apporte de la profondeur et du relief à la surface et peut atténuer les différences visuelles entre les lames.



Bord façonné

Elegance et richesse visuelle

Le bord façonné est obtenu par moulage d'un effet biseauté sur les bords du revêtement. Il accentue la séparation entre les lames, souligne leur individualité et reproduit l'aspect naturellement patiné du bois véritable.

Chanfrein à 45° (âme apparente)



Le chanfrein à 45° est une coupe franche à 45 degrés sur le bord des lames, assez profonde pour révéler leur âme. Il crée des lignes de séparation marquées et un relief prononcé, tout en masquant les irrégularités plus visibles entre les lames.

Bord arrondi (rayon R)



Le bord à rayon R présente une courbe lisse et arrondie sur le pourtour de la lame. Il adoucit les contours entre les lames pour un aspect plus doux, aide à éviter les dommages sur les bords et masque les petites imperfections entre les lames.

## Traitement excimer

La technologie excimer offre un rendu naturel et un confort visuel exceptionnel.

## Limiter les reflets gênants

Le sol Changlong traité par excimer offre une finition ultra-mate, réduit les reflets gênants et procure un confort visuel respectueux des yeux.

Ultra-mat  
Brillance uniforme

La brillance naturelle et équilibrée reste uniforme sous tous les angles de vue, éliminant les écarts de brillance et les réclamations des clients.

Aspect bois  
naturel et authentique

Visuellement identique au bois véritable, il supprime l'aspect plastique artificiel des finitions très brillantes et crée un espace confortable, chaleureux et élégant.

Toucher soyeux  
et doux pour la peau

La surface ultra-lisse et douce procure un confort agréable, idéal pour marcher pieds nus.

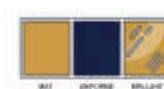
Résistance aux  
agents chimiques

La couche protectrice dense empêche la pénétration des solvants et agents chimiques : solution iodée, mercurochrome, acétone, acide nitrique (16 mol/L), chlorhydrique (12 mol/L), sulfurique (12 mol/L), peroxyde d'hydrogène à 30 %, agents de blanchiment, chloroforme, huile lubrifiante, toluène, etc. Un simple essuyage ne laisse aucune trace.

Les atouts du traitement excimer :



Retats réduits



Matité uniforme

Aspect bois  
naturel et authentique

Toucher soyeux

Résistance aux  
agents chimiques

Une structure stable, une résistance exceptionnelle aux taches et aux microrayures offrent des performances inédites.

## Résistance aux microrayures et à l'usure

Très résistant aux rayures et à l'usure, ce sol convient aux foyers avec animaux : il résiste aux pattes boueuses et aux griffes. Il est aussi adapté aux espaces commerciaux très fréquentés.

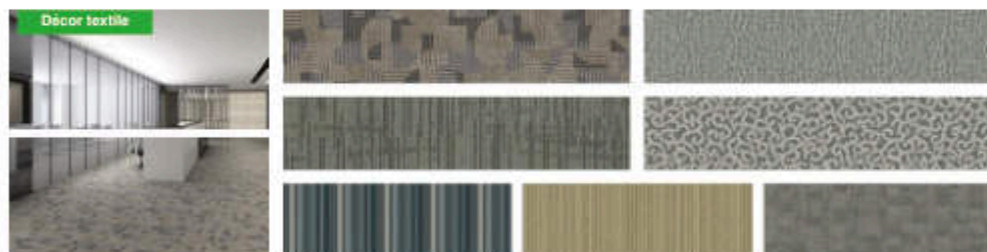
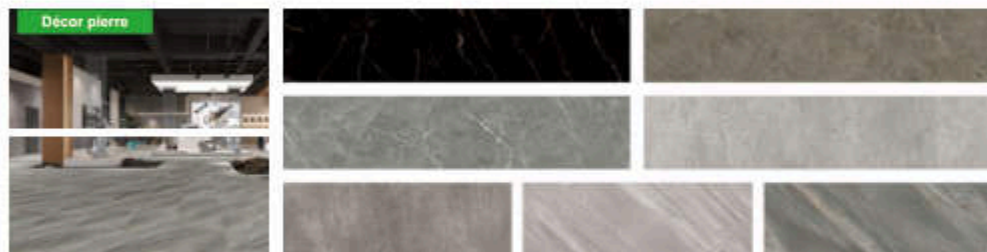
## Anti-traces de doigts et antitaches

Grâce au traitement excimer avancé, cette surface résiste efficacement aux taches tenaces.

## Antimicrobien

Inhibe efficacement les bactéries et les champignons, pour une protection active et durable et un cadre de vie plus sain et plus sûr.

## Tendances design et couleurs



## Prêts à collaborer ?

Créons ensemble la solution adaptée à votre prochain projet.

## Contactez-nous

Joey: +86 187 5138 2597

Adam: +86 152 0514 9312

Calina: +86 153 7177 5487

Adam: [marketing31@changlongflor.com](mailto:marketing31@changlongflor.com)

Calina: [md06@changlongflor.com](mailto:md06@changlongflor.com)

Joey: [sales01@changlongflor.com](mailto:sales01@changlongflor.com)



# Changlong Floor

Un catalogue complet de revêtements de sol haute performance  
aux décors contemporains.

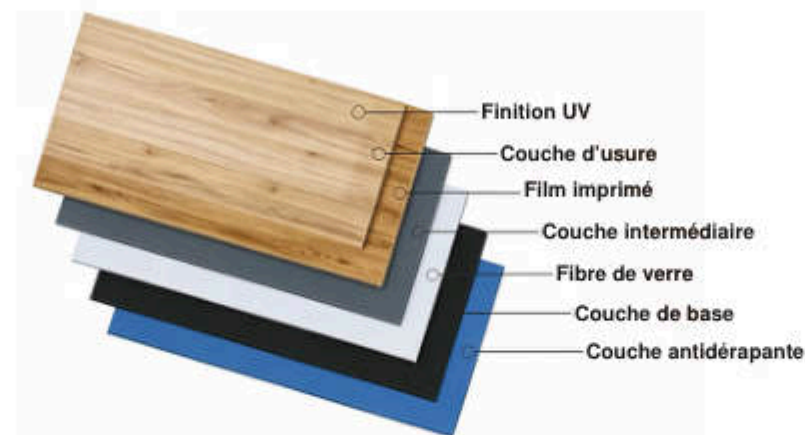
## Notre sélection LooseLay

|                                                             |                                 |              |              |              |              |     |
|-------------------------------------------------------------|---------------------------------|--------------|--------------|--------------|--------------|-----|
| Couche d'usure                                              | 0.2                             | 0.3          | 0.4          | 0.5          | 0.6          | 0.7 |
| Épaisseur totale                                            | 4.0                             | 4.5          | 5.0          | 5.5          | 6.0          |     |
| Couche décorative                                           | Impression standard / numérique |              |              |              |              |     |
| Chanfreins : pressé / microchanfrein / peint / bord façonné |                                 |              |              |              |              |     |
| Groupe d'usure                                              | T                               | T            | T            | T            | T            |     |
| Réaction au feu                                             | Bfl-S1                          |              |              |              |              |     |
| Relief                                                      | Veinage bois / pierre / EIR     |              |              |              |              |     |
| Format lames bois                                           | 152.4*914.4                     | 177.8*1219.2 | 228.6*1219.2 | 228.6*1422.4 | 457.2*1219.2 |     |
| Format dalles pierre                                        | 457.2*457.2                     | 457.2*914.4  | 457.2*1219.2 | 914.4*914.4  | 600*600      |     |
| Format bâtons rompus                                        | 76.2*228.6                      | 76.2*457.2   |              | 177.8*711.2  |              |     |
| Brillance                                                   | Mat / brillant / mat excimer    |              |              |              |              |     |

## Présentation des formats



## Structure du produit



## Sol en pose libre

### Pose sans colle, efficacité accrue

Pose libre, sans colle ni clips. Rapide et facile sur un support plan, elle réduit fortement le temps de pose, quelle que soit la taille du chantier.

### Réutilisable et économique

Se démonte et se réutilise sans abîmer le sol existant ni laisser de résidus de colle. Réduit les coûts de matériaux et de rénovation pour un meilleur rapport coût-performance.

### Antidérapant et stable

Le support haute densité et l'envers antidérapant alvéolé assurent une forte adhérence. Ils résistent au déplacement, au gauchissement et au desserrage lors d'un usage fréquent et du déplacement des meubles.

### Souple et silencieux

Ce sol souple offre un grand confort sous les pieds. Son excellente absorption acoustique réduit les bruits de pas et d'impact pour un intérieur calme.

### Écologique et sûr

PVC de qualité supérieure, sans formaldéhyde ni substances nocives. Étanche, résistant à l'usure, au feu et aux moisissures, pour un usage sûr et durable.

### De multiples applications

Réunit de multiples performances. Idéal pour les hôpitaux, écoles, bureaux et commerces, ainsi que pour l'aménagement des logements.

## Environnement / Certifications



## Chanfreins



Microchanfrein

Le microchanfrein est une légère inclinaison discrète sur le bord de la lame. Il souligne les contours de chaque lame pour mieux les distinguer après la pose, tout en masquant les petites imperfections entre les lames.



Chanfrein peint

Le chanfrein peint consiste à appliquer une couche de peinture sur le bord du revêtement pour créer un aspect biseauté. Il apporte de la profondeur et du relief à la surface et peut atténuer les différences visuelles entre les lames.

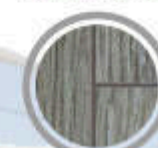


Bord façonné

Elegance et richesse visuelle

Le bord façonné est obtenu par moulage d'un effet biseauté sur les bords du revêtement. Il accentue la séparation entre les lames, souligne leur individualité et reproduit l'aspect naturellement patiné du bois véritable.

Chanfrein à 45° (âme apparente)



Le chanfrein à 45° est une coupe franche à 45 degrés sur le bord des lames, assez profonde pour révéler leur âme. Il crée des lignes de séparation marquées et un relief prononcé, tout en masquant les irrégularités plus visibles entre les lames.

Bord arrondi (rayon R)



Le bord à rayon R présente une courbe lisse et arrondie sur le pourtour de la lame. Il adoucit les contours entre les lames pour un aspect plus doux, aide à éviter les dommages sur les bords et masque les petites imperfections entre les lames.



# CERTIFICATE OF CONFORMITY

**№ V-24/015**

(Issue №1)

The Certification company "NJN" Ltd.  
certifies that the product

**Vinyl Tile Flooring**

Model: LVT,Click,Loose Lay,SPC

Is in compliance with **AVCP system 4 of Regulation 305/2011/EU** of the European Parliament and of the Council of 9 March 2011 (the Construction Products Regulation or CPR)

**Produced by:**

**JIANGSU CHANGLONG DECORATIVE MATERIAL TECHNICAL CO., LTD.**

NO. 168, BaiChuan Rd., Haian City, Jiangsu,China

**Placed on the market by**

**JIANGSU CHANGLONG DECORATIVE MATERIAL TECHNICAL CO., LTD.**

NO. 168, BaiChuan Rd., Haian City, Jiangsu,China

is submitted by the manufacturer of the initial product type test, production control and on-going trial of samples in accordance with the requirements of

**EN 14041:2018**

First issued on: 09.04.2024

Current issue on: 09.04.2024

Expiry date: 08.04.2027



A handwritten signature in blue ink, appearing to read 'Nikolay Atanasov', is written over a horizontal line.

eng. Nikolay Atanasov  
Manager

Haskovo  
29.02.2024



# ATTESTATION OF CONFORMITY

The Company mentioned below has been audited by the Notified Body UDEM, according to the Regulation (EU) No 305/2011 - Construction Products Annex V Clause 1.5

Company Name : **JIANGSU CHANGLONG DECORATIVE MATERIAL TECHNICAL CO., LTD.**

Company Address : **No.168, Baichuan Road, Haian City, Jiangsu Province, China**

Related Directives and Annex : **Regulation (EU) No 305/2011 - Construction Products Annex V Clause 1.5**

Related Standards : **EN 14041:2018**

Product Name : **PVC floor**

Report No and Date : **TCSH25042369649**

Product Brand/Model/Type : **LVT, CLICK, LOOSELAY, SPC**

Certificate Number : **M.2025.206.C118349**

Initial Assessment Date : **29.04.2025**

Registration Date : **30.04.2025**

Reissue Date/No : **-**

Expiry Date : **29.04.2030**

UDEM Uluslararası Belgelendirme  
Denetim Eğitim Merkezi  
San. ve Tic. A.Ş.  
Genel Müdür

The validity of the certificate can be checked through [www.udem.com.tr](http://www.udem.com.tr). Upon completion of EC declaration of conformity, it is used solely at the manufacturer's responsibility. This certificate remains the property of UDEM International Certification Auditing Training Centre Industry and Trade Inc. Co. to whom it must be returned upon request. The above named firm must keep a copy of this certificate for 15 years from the registration of certificate. This certificate only covers the product(s) stated above and UDEM must be noticed in case of any changes on the product(s)

Address: **Mutlukent Mahallesi 2073 Sokak (Eski 93 Sokak) No:10 Çankaya - Ankara - Türkiye**

Phone: **+90 0312 443 03 90** Fax: **+90 0312 443 03 76**

E-mail: **info@udem.com.tr** [www.udem.com.tr](http://www.udem.com.tr)



|                                                                                                                                                                                |                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Technical Construction File</b><br><b>EN 14041:2018</b><br><b>Resilient, textile, laminate and modular multilayer floor coverings</b><br><b>- Essential characteristics</b> |                                                                                   |
| Report Reference No. ....:                                                                                                                                                     | TCSH25042369649                                                                   |
| Reviewed by ( name + signature).....:                                                                                                                                          | Stephen Zhang / Test Engineer                                                     |
| Approved by (name + signature) :                                                                                                                                               | Kosco Vent / Project Manager                                                      |
| Date of issue.....:                                                                                                                                                            | April 29, 2025                                                                    |
| Reviewing Laboratory Name.....:                                                                                                                                                | Shanghai Global Testing Services Co., Ltd.                                        |
| Address: .....                                                                                                                                                                 | Floor 3rd, Building D-1, No. 128, Shenfu Road, Minhang District, Shanghai, China. |
| Review specification:                                                                                                                                                          |                                                                                   |
| Standard                                                                                                                                                                       | EN 14041:2018                                                                     |
| Non-standard Review method                                                                                                                                                     | None                                                                              |
| Applicant's name.....:                                                                                                                                                         | JIANGSU CHANGLONG DECORATIVE MATERIAL TECHNICAL CO., LTD.                         |
| Address: .....                                                                                                                                                                 | No.168, Baichuan Road, Haian City, Jiangsu Province                               |
| Review Report Form No                                                                                                                                                          | TTRF EN 14041                                                                     |
| TTRF Originator                                                                                                                                                                | GTS                                                                               |
| Master TTRF                                                                                                                                                                    | Dated 2018-01                                                                     |
| Review item description                                                                                                                                                        | PVC floor                                                                         |
| Trade Mark.....:                                                                                                                                                               | /                                                                                 |
| Model and/or type reference                                                                                                                                                    | LVT, CLICK, LOOSELAY, SPC                                                         |
| Manufacturer.....:                                                                                                                                                             | The same as applicant                                                             |
| Rating(s).....:                                                                                                                                                                | /                                                                                 |



Possible review case verdicts:

- review case does not apply to the Review object.....: N(.A.)
- review object does meet the requirement.....: P(ass)
- review object does not meet the requirement.....: F(ail)

General remarks:

"(see remark #)" refers to a remark appended to the report.

"(see appended table)" refers to a table appended to the report.

Throughout this report a comma is used as the decimal separator.

The review results presented in this report relate only to the object reviewed.

This report shall not be reproduced except in full without the written approval of the third party.

**Testing:**

Date of receipt of review item:

April 23, 2025

Date(s) of performance of review:

April 23, 2025 to April 29, 2025

**General product information:**

PVC floor

**Summary of reviewing:**

This review report includes:

Annex I: 3 page(s) of photo documentation.

Copy of marking plate

PVC floor

Model

LVT, CLICK, LOOSELAY, SPC

JIANGSU CHANGLONG DECORATIVE  
MATERIAL TECHNICAL CO., LTD.



| EN 14041 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |         |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------|
| Clause   | Requirement - Review                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Result - Remark | Verdict |
| 4        | REQUIREMENTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                 | —       |
| 4.1      | <p>Reaction to fire</p> <p>When declared, the floor covering shall be tested and classified according to the requirement of EN 13501-1 and resulting class and subclass shall be declared.</p> <p>If it is decided to make no claim for reaction for fire performance, which marked Class F, no testing is required .....</p> <p>If the product listed in Table 1, 2 or 3, in the end uses identified in the tables, are classified without further testing in the classes shown and do not require testing in respect of these end uses and classes.....:</p> |                 | P       |
| 4.2      | <p>Content of pentachlorophenol (PCP)</p> <p>Resilient, textile and laminate floor coverings shall not contain PCP or derivative thereof as a component in the production process of the product or of its raw materials.....:</p>                                                                                                                                                                                                                                                                                                                             |                 | P       |
| 4.3      | <p>Formaldehyde emission</p> <p>When formaldehyde-containing materials have been added to the product as a part of the production process, the product shall be tested and classified in to one of two classes: E1 or E2 :</p>                                                                                                                                                                                                                                                                                                                                 |                 | P       |
| 4.4      | <p>Water-tightness</p> <p>Where required, resilient floor coverings shall meet the requirement of EN 13553..... :</p>                                                                                                                                                                                                                                                                                                                                                                                                                                          |                 | N/A     |
| 4.5      | <p>Slip resistance</p> <p>When declared, the floor covering intended to be used in dry and non-contaminated conditions shall have a dynamic coefficient of friction of &gt; 0,30 when tested ex- factory under dry conditions per EN 13893 and shall be declared as technical class DS.....:</p>                                                                                                                                                                                                                                                               |                 | P       |
| 4.6      | <p>Electrical behaviour (static electricity)</p> <p>When declared, antistatic floor coverings body voltage shall not exceed 2,0 kV per EN 1815 .....</p> <p>When declared, static dissipative floor coverings vertical resistance shall not exceed <math>10^9 \Omega</math> per EN 1081.....:</p> <p>When declared, conductive floor coverings vertical resistance shall not exceed <math>10^6 \Omega</math> per EN 1081.. :</p>                                                                                                                               |                 | P       |

|     |                                                                                                                                                                                                                                                                                                                                                        |                            |   |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---|
| 4.7 | Thermal conductivity<br>If required, the thermal conductivity values shall be verify per EN 12524 or EN 12667..... :                                                                                                                                                                                                                                   |                            | P |
| 5   | EVALUATION OF CONFORMITY                                                                                                                                                                                                                                                                                                                               |                            | — |
| 5.1 | General..... :                                                                                                                                                                                                                                                                                                                                         |                            | P |
| 5.2 | Type testing..... :                                                                                                                                                                                                                                                                                                                                    | Refer to Clause 4.1 to 4.7 | P |
| 5.3 | Factory production control..... :                                                                                                                                                                                                                                                                                                                      |                            | P |
| 6   | MARKING                                                                                                                                                                                                                                                                                                                                                |                            | — |
|     | Product which conform to the requirements of this document shall be clearly and indelibly marked by the manufacturer either on their package or on an adhesive label with following information:<br>a) the number and the year of this European Standard<br>b) the manufacturer's or supplier's identification<br>c) the product name and batch number |                            | P |

\*\*\*\*\*End of Review Report\*\*\*\*\*

Type of equipment,  
model:

PVC floor  
LVT, CLICK, LOOSELAY, SPC

Details of: LVT

View:

☒ general

☐ front

☐ rear

☐ right

☐ left

☐ top

☐ bottom



Details of: SPC

View:

☒ general

☐ front

☐ rear

☐ right

☐ left

☐ top

☐ bottom



Details of: **LOOSELAY**

View:

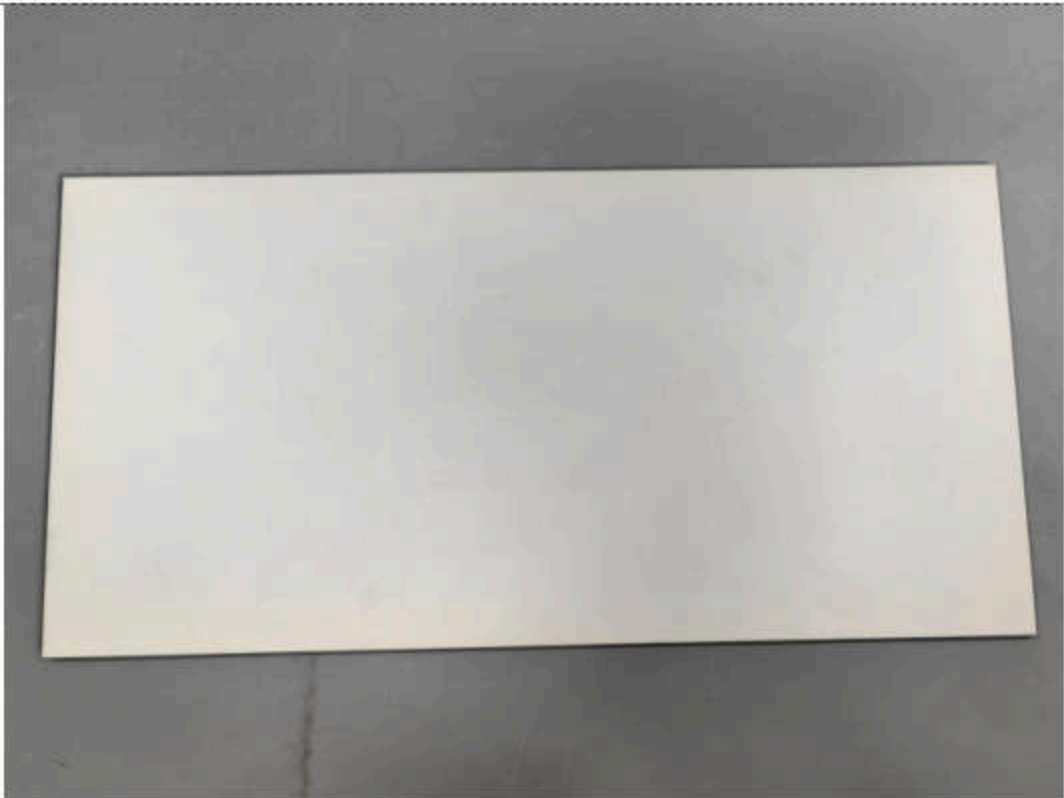
☒ general☐ front☐ rear☐ right☐ left☐ top☐ bottomDetails of: **CLICK**

View:

☒ general☐ front☐ rear☐ right☐ left☐ top☐ bottom

Details of: [CLICK](#)

View:

☒ general☐ front☐ rear☐ right☐ left☐ top☐ bottom

- End of Annex I -



# ATTESTATION OF CONFORMITY

The Company mentioned below has been audited by the Notified Body UDEM, according to the Regulation (EU) No 305/2011 - Construction Products Annex V Clause 1.5

Company Name : **JIANGSU CHANGLONG DECORATIVE MATERIAL TECHNICAL CO., LTD.**

Company Address : **No.168, Baichuan Road, Haian City, Jiangsu Province, China**

Related Directives and Annex : **Regulation (EU) No 305/2011 - Construction Products Annex V Clause 1.5**

Related Standards : **EN 14041:2018**

Product Name : **PET floor**

Report No and Date : **TCSH25042369650**

Product Brand/Model/Type : **PET**

Certificate Number : **M.2025.206.C118684**

Initial Assessment Date : **09.05.2025**

Registration Date : **12.05.2025**

Reissue Date/No : **-**

Expiry Date : **11.05.2030**

UDEM Uluslararası Belgelendirme  
Denetim Eğitim Merkezi  
San. ve Tic. A.Ş.  
Genel Müdür

The validity of the certificate can be checked through [www.udem.com.tr](http://www.udem.com.tr). Upon completion of EC declaration of conformity, it is used solely at the manufacturer's responsibility. This certificate remains the property of UDEM International Certification Auditing Training Centre Industry and Trade Inc. Co. to whom it must be returned upon request. The above named firm must keep a copy of this certificate for 15 years from the registration of certificate. This certificate only covers the product(s) stated above and UDEM must be noticed in case of any changes on the product(s).

Address: **Mutlukent Mahallesi 2073 Sokak (Eski 93 Sokak) No:10 Çankaya Ankara - Türkiye**

Phone: **+90 0312 443 03 90** Fax: **+90 0312 443 03 76**

E-mail: **info@udem.com.tr** [www.udem.com.tr](http://www.udem.com.tr)



|                                                                                                                                                                                |                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Technical Construction File</b><br><b>EN 14041:2018</b><br><b>Resilient, textile, laminate and modular multilayer floor coverings</b><br><b>- Essential characteristics</b> |                                                                                   |
| Report Reference No. ....:                                                                                                                                                     | TCSH25042369650                                                                   |
| Reviewed by ( name + signature).....:                                                                                                                                          | Stephen Zhang / Test Engineer                                                     |
| Approved by (name + signature) :                                                                                                                                               | Kosco Vent / Project Manager                                                      |
| Date of issue.....:                                                                                                                                                            | April 29, 2025                                                                    |
| Reviewing Laboratory Name.....:                                                                                                                                                | Shanghai Global Testing Services Co., Ltd.                                        |
| Address: .....                                                                                                                                                                 | Floor 3rd, Building D-1, No. 128, Shenfu Road, Minhang District, Shanghai, China. |
| Review specification:                                                                                                                                                          |                                                                                   |
| Standard                                                                                                                                                                       | EN 14041:2018                                                                     |
| Non-standard Review method                                                                                                                                                     | None                                                                              |
| Applicant's name.....:                                                                                                                                                         | JIANGSU CHANGLONG DECORATIVE MATERIAL TECHNICAL CO., LTD.                         |
| Address: .....                                                                                                                                                                 | No.168, Baichuan Road, Haian City, Jiangsu Province                               |
| Review Report Form No                                                                                                                                                          | TTRF EN 14041                                                                     |
| TTRF Originator                                                                                                                                                                | GTS                                                                               |
| Master TTRF                                                                                                                                                                    | Dated 2018-01                                                                     |
| Review item description                                                                                                                                                        | PET floor                                                                         |
| Trade Mark.....:                                                                                                                                                               | /                                                                                 |
| Model and/or type reference                                                                                                                                                    | PET                                                                               |
| Manufacturer.....:                                                                                                                                                             | The same as applicant                                                             |
| Rating(s).....:                                                                                                                                                                | /                                                                                 |



Possible review case verdicts:

- review case does not apply to the Review object.....: N(.A.)
- review object does meet the requirement.....: P(ass)
- review object does not meet the requirement.....: F(ail)

General remarks:

"(see remark #)" refers to a remark appended to the report.

"(see appended table)" refers to a table appended to the report.

Throughout this report a comma is used as the decimal separator.

The review results presented in this report relate only to the object reviewed.

This report shall not be reproduced except in full without the written approval of the third party.

## Testing:

Date of receipt of review item:

April 23, 2025

Date(s) of performance of review:

April 23, 2025 to April 29, 2025

## General product information:

PET floor

## Summary of reviewing:

This review report includes:

Annex I: 1 page(s) of photo documentation.

Copy of marking plate

PET floor

Model

PET

JIANGSU CHANGLONG DECORATIVE  
MATERIAL TECHNICAL CO., LTD.



| EN 14041 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |         |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------|
| Clause   | Requirement - Review                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Result - Remark | Verdict |
| 4        | REQUIREMENTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                 | —       |
| 4.1      | <p>Reaction to fire</p> <p>When declared, the floor covering shall be tested and classified according to the requirement of EN 13501-1 and resulting class and subclass shall be declared.</p> <p>If it is decided to make no claim for reaction for fire performance, which marked Class F, no testing is required .....</p> <p>If the product listed in Table 1, 2 or 3, in the end uses identified in the tables, are classified without further testing in the classes shown and do not require testing in respect of these end uses and classes.....:</p> |                 | P       |
| 4.2      | <p>Content of pentachlorophenol (PCP)</p> <p>Resilient, textile and laminate floor coverings shall not contain PCP or derivative thereof as a component in the production process of the product or of its raw materials.....:</p>                                                                                                                                                                                                                                                                                                                             |                 | P       |
| 4.3      | <p>Formaldehyde emission</p> <p>When formaldehyde-containing materials have been added to the product as a part of the production process, the product shall be tested and classified in to one of two classes: E1 or E2 :</p>                                                                                                                                                                                                                                                                                                                                 |                 | P       |
| 4.4      | <p>Water-tightness</p> <p>Where required, resilient floor coverings shall meet the requirement of EN 13553..... :</p>                                                                                                                                                                                                                                                                                                                                                                                                                                          |                 | N/A     |
| 4.5      | <p>Slip resistance</p> <p>When declared, the floor covering intended to be used in dry and non-contaminated conditions shall have a dynamic coefficient of friction of &gt; 0,30 when tested ex- factory under dry conditions per EN 13893 and shall be declared as technical class DS.....:</p>                                                                                                                                                                                                                                                               |                 | P       |
| 4.6      | <p>Electrical behaviour (static electricity)</p> <p>When declared, antistatic floor coverings body voltage shall not exceed 2,0 kV per EN 1815 .....</p> <p>When declared, static dissipative floor coverings vertical resistance shall not exceed <math>10^9 \Omega</math> per EN 1081.....:</p> <p>When declared, conductive floor coverings vertical resistance shall not exceed <math>10^6 \Omega</math> per EN 1081.. :</p>                                                                                                                               |                 | P       |

|     |                                                                                                                                                                                                                                                                                                                                                        |                            |   |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---|
| 4.7 | Thermal conductivity<br>If required, the thermal conductivity values shall be verify per EN 12524 or EN 12667..... :                                                                                                                                                                                                                                   |                            | P |
| 5   | EVALUATION OF CONFORMITY                                                                                                                                                                                                                                                                                                                               |                            | — |
| 5.1 | General..... :                                                                                                                                                                                                                                                                                                                                         |                            | P |
| 5.2 | Type testing..... :                                                                                                                                                                                                                                                                                                                                    | Refer to Clause 4.1 to 4.7 | P |
| 5.3 | Factory production control..... :                                                                                                                                                                                                                                                                                                                      |                            | P |
| 6   | MARKING                                                                                                                                                                                                                                                                                                                                                |                            | — |
|     | Product which conform to the requirements of this document shall be clearly and indelibly marked by the manufacturer either on their package or on an adhesive label with following information:<br>a) the number and the year of this European Standard<br>b) the manufacturer's or supplier's identification<br>c) the product name and batch number |                            | P |

\*\*\*\*\*End of Review Report\*\*\*\*\*

Type of equipment,  
model:

PET floor  
PET

Details of: PET

View:

☒ general

☐ front

☐ rear

☐ right

☐ left

☐ top

☐ bottom



- End of Annex I -



# Certificate

## Indoor Air Comfort Gold

LVT

*Certified Product*

Jiangsu Changlong Decorative Material Technical Co. Ltd

NO. 168, BaiChuan Rd., 226600 Hai'An City, Jiangsu, China

*Applicant*

The above product complies with the Indoor Air Comfort Gold specifications, version 9.0 (2023). These include both inspection of factory production and VOC emissions testing according to EN 16516, at regular intervals. Indoor Air Comfort Gold combines all key European and selected global requirements on VOC product emissions. Additional requirements not related to VOC product emissions, for example content of certain substances or odour are not combined or evaluated. The following VOC emission requirements are combined and the certified product shows compliance with these VOC emission related limit values:

- |                          |                            |                               |                              |
|--------------------------|----------------------------|-------------------------------|------------------------------|
| - Belgian VOC regulation | - LEED (ACP)               | - Blue Angel DE-UZ 120        | - Eco Product Norway         |
| - France VOC class A+    | - BREEAM New Construction  | - Austrian Ecolabel UZ 42     | - SINTEF (Norway)            |
| - Germany (AgBB/ABG)     | - WELL Building            | - Austrian Baubook            | - Cradle to Cradle           |
| - Italian CAM Edilizia   | - DGNB                     | - M1                          | - very low emitting products |
| - EU Taxonomy Regulation | - SKA Rating               | - Danish Indoor Climate Label | according to EN 16798-1      |
|                          | - French HQE certification | (Emission Class 1)            | - Singapore Green Label      |
|                          |                            | - BVB (Sweden)                | - Global GreenTag            |
|                          |                            | - Miljöbyggnad (Sweden)       | - Declare 2.0                |

Issue date: 31 October 2025

Product type: Resilient Floorings

Validity date: 31 October 2030

Certificate number: IACG-432-01-01-2025

This certificate is valid as specified if regular surveillance and testing is done.

**Thomas Neuhaus**

Head of Certification Body

Eurofins Product Testing A/S  
Smedeskovvej 38, Gate 9  
8464 Galten  
Denmark

 **DANAK**  
PROD Reg.no. 7054  
Member EA MLA



# Certificate

## Indoor Air Comfort Gold

Loose lay

*Certified Product*

Jiangsu Changlong Decorative Material Technical Co. Ltd

NO. 168, BaiChuan Rd., 226600 Hai'An City, Jiangsu, China

*Applicant*

The above product complies with the Indoor Air Comfort Gold specifications, version 9.0 (2023). These include both inspection of factory production and VOC emissions testing according to EN 16516, at regular intervals. Indoor Air Comfort Gold combines all key European and selected global requirements on VOC product emissions. Additional requirements not related to VOC product emissions, for example content of certain substances or odour are not combined or evaluated. The following VOC emission requirements are combined and the certified product shows compliance with these VOC emission related limit values:

- |                          |                            |                               |                              |
|--------------------------|----------------------------|-------------------------------|------------------------------|
| - Belgian VOC regulation | - LEED (ACP)               | - Blue Angel DE-UZ 120        | - Eco Product Norway         |
| - France VOC class A+    | - BREEAM New Construction  | - Austrian Ecolabel UZ 42     | - SINTEF (Norway)            |
| - Germany (AgBB/ABG)     | - WELL Building            | - Austrian Baubook            | - Cradle to Cradle           |
| - Italian CAM Edilizia   | - DGNB                     | - M1                          | - very low emitting products |
| - EU Taxonomy Regulation | - SKA Rating               | - Danish Indoor Climate Label | according to EN 16798-1      |
|                          | - French HQE certification | (Emission Class 1)            | - Singapore Green Label      |
|                          |                            | - BVB (Sweden)                | - Global GreenTag            |
|                          |                            | - Miljöbyggnad (Sweden)       | - Declare 2.0                |

Issue date: 31 October 2025

Product type: Resilient Floorings

Validity date: 31 October 2030

Certificate number: IACG-432-01-03-2025

This certificate is valid as specified if regular surveillance and testing is done.

**Thomas Neuhaus**

Head of Certification Body

Eurofins Product Testing A/S  
Smedeskovvej 38, Gate 9  
8464 Galten  
Denmark

 **DANAK**  
PROD Reg.no. 7054  
Member EA MLA



# Certificate

## Indoor Air Comfort Gold

SPC

*Certified Product*

Jiangsu Changlong Decorative Material Technical Co. Ltd

NO. 168, BaiChuan Rd., 226600 Hai'An City, Jiangsu, China

*Applicant*

The above product complies with the Indoor Air Comfort Gold specifications, version 9.0 (2023). These include both inspection of factory production and VOC emissions testing according to EN 16516, at regular intervals. Indoor Air Comfort Gold combines all key European and selected global requirements on VOC product emissions. Additional requirements not related to VOC product emissions, for example content of certain substances or odour are not combined or evaluated. The following VOC emission requirements are combined and the certified product shows compliance with these VOC emission related limit values:

- |                          |                            |                               |                              |
|--------------------------|----------------------------|-------------------------------|------------------------------|
| - Belgian VOC regulation | - LEED (ACP)               | - Blue Angel DE-UZ 120        | - Eco Product Norway         |
| - France VOC class A+    | - BREEAM New Construction  | - Austrian Ecolabel UZ 42     | - SINTEF (Norway)            |
| - Germany (AgBB/ABG)     | - WELL Building            | - Austrian Baubook            | - Cradle to Cradle           |
| - Italian CAM Edilizia   | - DGNB                     | - M1                          | - very low emitting products |
| - EU Taxonomy Regulation | - SKA Rating               | - Danish Indoor Climate Label | according to EN 16798-1      |
|                          | - French HQE certification | (Emission Class 1)            | - Singapore Green Label      |
|                          |                            | - BVB (Sweden)                | - Global GreenTag            |
|                          |                            | - Miljöbyggnad (Sweden)       | - Declare 2.0                |

Issue date: 31 October 2025

Product type: Resilient Floorings

Validity date: 31 October 2030

Certificate number: IACG-432-01-04-2025

This certificate is valid as specified if regular surveillance and testing is done.

**Thomas Neuhaus**

Head of Certification Body

Eurofins Product Testing A/S  
Smedeskovvej 38, Gate 9  
8464 Galten  
Denmark

 **DANAK**  
PROD Reg.no. 7054  
Member EA MLA



# Certificate

## Indoor Air Comfort Gold

PET floor

*Certified Product*

Jiangsu Changlong Decorative Material Technical Co. Ltd

NO. 168, BaiChuan Rd., 226600 Hai'An City, Jiangsu, China

*Applicant*

The above product complies with the Indoor Air Comfort Gold specifications, version 9.0 (2023). These include both inspection of factory production and VOC emissions testing according to EN 16516, at regular intervals. Indoor Air Comfort Gold combines all key European and selected global requirements on VOC product emissions. Additional requirements not related to VOC product emissions, for example content of certain substances or odour are not combined or evaluated. The following VOC emission requirements are combined and the certified product shows compliance with these VOC emission related limit values:

- |                          |                            |                               |                              |
|--------------------------|----------------------------|-------------------------------|------------------------------|
| - Belgian VOC regulation | - LEED (ACP)               | - Blue Angel DE-UZ 120        | - Eco Product Norway         |
| - France VOC class A+    | - BREEAM New Construction  | - Austrian Ecolabel UZ 42     | - SINTEF (Norway)            |
| - Germany (AgBB/ABG)     | - WELL Building            | - Austrian Baubook            | - Cradle to Cradle           |
| - Italian CAM Edilizia   | - DGNB                     | - M1                          | - very low emitting products |
| - EU Taxonomy Regulation | - SKA Rating               | - Danish Indoor Climate Label | according to EN 16798-1      |
|                          | - French HQE certification | (Emission Class 1)            | - Singapore Green Label      |
|                          |                            | - BVB (Sweden)                | - Global GreenTag            |
|                          |                            | - Miljöbyggnad (Sweden)       | - Declare 2.0                |

Issue date: 31 October 2025

Product type: Resilient Floorings

Validity date: 31 October 2030

Certificate number: IACG-432-01-05-2025

This certificate is valid as specified if regular surveillance and testing is done.

**Thomas Neuhaus**

Head of Certification Body

Eurofins Product Testing A/S  
Smedeskovvej 38, Gate 9  
8464 Galten  
Denmark

 **DANAK**  
PROD Reg.no. 7054  
Member EA MLA



# Certificate

## Indoor Air Comfort Gold

LVT

*Certified Product*

Jiangsu Changlong Decorative Material Technical Co. Ltd

NO. 168, BaiChuan Rd., 226600 Hai'An City, Jiangsu, China

*Applicant*

The above product complies with the Indoor Air Comfort Gold specifications, version 9.0 (2023). These include both inspection of factory production and VOC emissions testing according to EN 16516, at regular intervals. Indoor Air Comfort Gold combines all key European and selected global requirements on VOC product emissions. Additional requirements not related to VOC product emissions, for example content of certain substances or odour are not combined or evaluated. The following VOC emission requirements are combined and the certified product shows compliance with these VOC emission related limit values:

- |                          |                            |                               |                              |
|--------------------------|----------------------------|-------------------------------|------------------------------|
| - Belgian VOC regulation | - LEED (ACP)               | - Blue Angel DE-UZ 120        | - Eco Product Norway         |
| - France VOC class A+    | - BREEAM New Construction  | - Austrian Ecolabel UZ 42     | - SINTEF (Norway)            |
| - Germany (AgBB/ABG)     | - WELL Building            | - Austrian Baubook            | - Cradle to Cradle           |
| - Italian CAM Edilizia   | - DGNB                     | - M1                          | - very low emitting products |
| - EU Taxonomy Regulation | - SKA Rating               | - Danish Indoor Climate Label | - according to EN 16798-1    |
|                          | - French HQE certification | - (Emission Class 1)          | - Singapore Green Label      |
|                          |                            | - BVB (Sweden)                | - Global GreenTag            |
|                          |                            | - Miljöbyggnad (Sweden)       | - Declare 2.0                |

Issue date: 19 November 2024

Product type: Resilient Floorings

Validity date: 19 November 2029

Certificate number: IACG-432-01-01-2024

This certificate is valid as specified if regular surveillance and testing is done.

**Thomas Neuhaus**

Head of Certification Body

Eurofins Product Testing A/S  
Smedeskovvej 38, Gate 9  
8464 Galten  
Denmark

 **DANAK**  
PROD Reg.no. 7054  
Member EA MLA



# Certificate

## Indoor Air Comfort Gold

Loose lay

*Certified Product*

Jiangsu Changlong Decorative Material Technical Co. Ltd

NO. 168, BaiChuan Rd., 226600 Hai'An City, Jiangsu, China

*Applicant*

The above product complies with the Indoor Air Comfort Gold specifications, version 9.0 (2023). These include both inspection of factory production and VOC emissions testing according to EN 16516, at regular intervals. Indoor Air Comfort Gold combines all key European and selected global requirements on VOC product emissions. Additional requirements not related to VOC product emissions, for example content of certain substances or odour are not combined or evaluated. The following VOC emission requirements are combined and the certified product shows compliance with these VOC emission related limit values:

- |                          |                            |                               |                              |
|--------------------------|----------------------------|-------------------------------|------------------------------|
| - Belgian VOC regulation | - LEED (ACP)               | - Blue Angel DE-UZ 120        | - Eco Product Norway         |
| - France VOC class A+    | - BREEAM New Construction  | - Austrian Ecolabel UZ 42     | - SINTEF (Norway)            |
| - Germany (AgBB/ABG)     | - WELL Building            | - Austrian Baubook            | - Cradle to Cradle           |
| - Italian CAM Edilizia   | - DGNB                     | - M1                          | - very low emitting products |
| - EU Taxonomy Regulation | - SKA Rating               | - Danish Indoor Climate Label | - according to EN 16798-1    |
|                          | - French HQE certification | - (Emission Class 1)          | - Singapore Green Label      |
|                          |                            | - BVB (Sweden)                | - Global GreenTag            |
|                          |                            | - Miljöbyggnad (Sweden)       | - Declare 2.0                |

Issue date: 19 November 2024

Product type: Resilient Floorings

Validity date: 19 November 2029

Certificate number: IACG-432-01-03-2024

This certificate is valid as specified if regular surveillance and testing is done.

**Thomas Neuhaus**

Head of Certification Body

Eurofins Product Testing A/S  
Smedeskovvej 38, Gate 9  
8464 Galten  
Denmark

 **DANAK**  
PROD Reg.no. 7054  
Member EA MLA



# Certificate

## Indoor Air Comfort Gold

SPC

*Certified Product*

Jiangsu Changlong Decorative Material Technical Co. Ltd

NO. 168, BaiChuan Rd., 226600 Hai'An City, Jiangsu, China

*Applicant*

The above product complies with the Indoor Air Comfort Gold specifications, version 9.0 (2023). These include both inspection of factory production and VOC emissions testing according to EN 16516, at regular intervals. Indoor Air Comfort Gold combines all key European and selected global requirements on VOC product emissions. Additional requirements not related to VOC product emissions, for example content of certain substances or odour are not combined or evaluated. The following VOC emission requirements are combined and the certified product shows compliance with these VOC emission related limit values:

- |                          |                            |                               |                              |
|--------------------------|----------------------------|-------------------------------|------------------------------|
| - Belgian VOC regulation | - LEED (ACP)               | - Blue Angel DE-UZ 120        | - Eco Product Norway         |
| - France VOC class A+    | - BREEAM New Construction  | - Austrian Ecolabel UZ 42     | - SINTEF (Norway)            |
| - Germany (AgBB/ABG)     | - WELL Building            | - Austrian Baubook            | - Cradle to Cradle           |
| - Italian CAM Edilizia   | - DGNB                     | - M1                          | - very low emitting products |
| - EU Taxonomy Regulation | - SKA Rating               | - Danish Indoor Climate Label | - according to EN 16798-1    |
|                          | - French HQE certification | - (Emission Class 1)          | - Singapore Green Label      |
|                          |                            | - BVB (Sweden)                | - Global GreenTag            |
|                          |                            | - Miljöbyggnad (Sweden)       | - Declare 2.0                |

Issue date: 19 November 2024

Product type: Resilient Floorings

Validity date: 19 November 2029

Certificate number: IACG-432-01-04-2024

This certificate is valid as specified if regular surveillance and testing is done.

**Thomas Neuhaus**

Head of Certification Body

Eurofins Product Testing A/S  
Smedeskovvej 38, Gate 9  
8464 Galten  
Denmark

 **DANAK**  
PROD Reg.no. 7054  
Member EA MLA



# Certificate

## Indoor Air Comfort Gold

PET floor

*Certified Product*

Jiangsu Changlong Decorative Material Technical Co. Ltd

NO. 168, BaiChuan Rd., 226600 Hai'An City, Jiangsu, China

*Applicant*

The above product complies with the Indoor Air Comfort Gold specifications, version 9.0 (2023). These include both inspection of factory production and VOC emissions testing according to EN 16516, at regular intervals. Indoor Air Comfort Gold combines all key European and selected global requirements on VOC product emissions. Additional requirements not related to VOC product emissions, for example content of certain substances or odour are not combined or evaluated. The following VOC emission requirements are combined and the certified product shows compliance with these VOC emission related limit values:

- |                          |                            |                               |                              |
|--------------------------|----------------------------|-------------------------------|------------------------------|
| - Belgian VOC regulation | - LEED (ACP)               | - Blue Angel DE-UZ 120        | - Eco Product Norway         |
| - France VOC class A+    | - BREEAM New Construction  | - Austrian Ecolabel UZ 42     | - SINTEF (Norway)            |
| - Germany (AgBB/ABG)     | - WELL Building            | - Austrian Baubook            | - Cradle to Cradle           |
| - Italian CAM Edilizia   | - DGNB                     | - M1                          | - very low emitting products |
| - EU Taxonomy Regulation | - SKA Rating               | - Danish Indoor Climate Label | - according to EN 16798-1    |
|                          | - French HQE certification | - (Emission Class 1)          | - Singapore Green Label      |
|                          |                            | - BVB (Sweden)                | - Global GreenTag            |
|                          |                            | - Miljöbyggnad (Sweden)       | - Declare 2.0                |

Issue date: 19 November 2024

Product type: Resilient Floorings

Validity date: 19 November 2029

Certificate number: IACG-432-01-05-2024

This certificate is valid as specified if regular surveillance and testing is done.

**Thomas Neuhaus**

Head of Certification Body

Eurofins Product Testing A/S  
Smedeskovvej 38, Gate 9  
8464 Galten  
Denmark

 **DANAK**  
PROD Reg.no. 7054  
Member EA MLA

Jiangsu Changlong Decorative Material Technical Co. Ltd  
NO. 168, BaiChuan Rd.  
226600 Hai'An City, Jiangsu  
China

Eurofins Product Testing Denmark A/S  
Smedeskovvej 38  
8464 Galten  
Denmark

Tel: +45 70 22 42 76  
ccs@eurofins.com

## CERTIFICATION REPORT

**S-432-01-01-2024**

**19/11/2024**

This certification report combines the outcome of the external factory audit and of the lab tests into a final evaluation to show compliance with the requirements of the certification scheme.

|                             |                                                            |
|-----------------------------|------------------------------------------------------------|
| <b>CERTIFICATION SCHEME</b> | Indoor Air Comfort and Indoor Air Comfort Gold Version 9.0 |
| <b>CERTIFIED GROUPS</b>     |                                                            |
| IACG-432-01-01              | LVT                                                        |
| IACG-432-01-03              | Loose lay                                                  |
| IACG-432-01-04              | SPC                                                        |
| IACG-432-01-05              | PET floor                                                  |

### CONCLUSIONS:

- The factory production control and all the documentation fulfil the requirements to a large extent.
- The surveyed products fulfilled the requirements of the Indoor Air Comfort GOLD label.
- The Indoor Air Comfort GOLD certificates remain valid.

## 1 AUDIT OF THE FACTORY PRODUCTION CONTROL

The audit included surveillance of the factory production control, staff and technical equipment, and documentation of any deviations.

|                |                                                                                                                          |
|----------------|--------------------------------------------------------------------------------------------------------------------------|
| REPORT NUMBER  | I-432-01-01-2024                                                                                                         |
| TYPE OF AUDIT  | <input type="checkbox"/> Initial <input checked="" type="checkbox"/> Routine <input type="checkbox"/> Follow-up          |
| AUDITED SITE   | Jiangsu Changlong Decorative Material Technical Co. Ltd<br>NO. 168, BaiChuan Rd.<br>226600 Hai'An City, Jiangsu<br>China |
| CONTACT PERSON | Cici Xu                                                                                                                  |
| AUDIT DATE     | 12.09.2024 9:00 AM                                                                                                       |
| LEAD AUDITOR   | Caesar Xue                                                                                                               |
| PARTICIPANTS   | David Xi/ Quality manager; Sun Wei/ Vice-general manager;<br>Zhihong Duan/ Production manager                            |
| REMARKS        |                                                                                                                          |
| EVALUATION     | The factory production control and all the documentation fulfil the requirements to a large extent.                      |

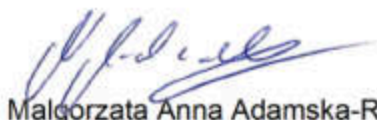
## 2 VOC TESTING

VOC testing of the selected test samples was investigated and evaluated according to the respective certification scheme.

Details of the tests were documented in the laboratory test reports:

| CERTIFIED GROUP | PRODUCT NAME       | TEST REPORT NUMBER     | REQUIREMENTS FULFILLMENT |
|-----------------|--------------------|------------------------|--------------------------|
| IACG-432-01-01  | LVT (3.0 mm)       | 392-2024-00264601_rev1 | IAC GOLD                 |
| IACG-432-01-03  | Loose lay (5.0 mm) | 392-2024-00264602_rev1 | IAC GOLD                 |
| IACG-432-01-04  | SPC (6.0 mm)       | 392-2024-00264603      | IAC GOLD                 |
| IACG-432-01-05  | PET floor (4.2 mm) | 392-2024-00264604_rev1 | IAC GOLD                 |

This report includes 3 pages.



Malgorzata Anna Adamska-Reiche

Certification Body

Jiangsu Changlong Decorative Material Technical Co.,Ltd  
NO. 168, Baichuan Road,  
Haian City, Jiangsu Province  
China

Eurofins Product Testing Denmark A/S  
Smedeskovvej 38  
8464 Galten  
Denmark

Tel: +45 70 22 42 76  
ccs@eurofins.com

---

## **CERTIFICATION REPORT**

**S-432-01-01-2023**

**21/11/2023**

---

This certification report combines the outcome of the external factory audit and of the lab tests into a final evaluation to show compliance with the requirements of the certification scheme.

|                             |                           |
|-----------------------------|---------------------------|
| <b>CERTIFICATION SCHEME</b> | Indoor Air Comfort V. 9.0 |
| <b>CERTIFIED GROUPS</b>     |                           |
| IACG-432-01-01              | LVT                       |
| IACG-432-01-03              | Loose Lay                 |
| IACG-432-01-04              | SPC                       |
| IACG-432-01-05              | PET                       |

### **CONCLUSIONS:**

- The factory production control and all the documentation fulfil the requirements to a large extent.
- The surveyed products fulfilled the requirements of the Indoor Air Comfort GOLD label.
- The Indoor Air Comfort GOLD certificates remain valid.

## 1 AUDIT OF THE FACTORY PRODUCTION CONTROL

The audit included surveillance of the factory production control, staff and technical equipment, and documentation of any deviations.

|                |                                                                                                                             |
|----------------|-----------------------------------------------------------------------------------------------------------------------------|
| REPORT NUMBER  | I-432-01-01-2023                                                                                                            |
| TYPE OF AUDIT  | <input type="checkbox"/> Initial <input checked="" type="checkbox"/> Routine <input type="checkbox"/> Follow-up             |
| AUDITED SITE   | Jiangsu Changlong Decorative Material Technical Co.,Ltd<br>NO. 168, Baichuan Road,<br>Haian City, Jiangsu Province<br>China |
| CONTACT PERSON | David Xi/Quality manager                                                                                                    |
| AUDIT DATE     | 18.09.2023 8:30                                                                                                             |
| LEAD AUDITOR   | Lili Cai                                                                                                                    |
| PARTICIPANTS   | David Xi/Quality manager                                                                                                    |
| REMARKS        | -                                                                                                                           |
| EVALUATION     | The factory production control and all the documentation fulfil the requirements to a large extent.                         |

## 2 VOC TESTING

VOC testing of the selected test samples was investigated and evaluated according to the Indoor Air Comfort GOLD specifications.

Details of the tests were documented in the laboratory test reports:

| CERTIFIED GROUP | PRODUCT NAME | TEST REPORT NUMBER | REQUIREMENTS FULFILLMENT |
|-----------------|--------------|--------------------|--------------------------|
| IACG-432-01-01  | LVT          | 392-2023-00309301  | IAC GOLD                 |
| IACG-432-01-03  | Loose Lay    | 392-2023-00309302  | IAC GOLD                 |
| IACG-432-01-04  | SPC          | 392-2023-00309303  | IAC GOLD                 |
| IACG-432-01-05  | PET          | 392-2023-00309304  | IAC GOLD                 |

This report includes 3 pages.



Thomas Neuhaus  
Certification Body

**SCS Global Services** does hereby certify that an independent assessment has been conducted on behalf of:

# Jiangsu Changlong Decorative Material Technical Co., Ltd.

No 168, BaiChuan Road, Hai'an City, Jiangsu, China

For the following product(s):

**Vinyl Tile:**

Click, Loose Lay, LVT, SPC Flooring  
(Maximum thickness: 6.0mm)

The product(s) meet(s) all of the necessary qualifications to be certified for the following claim(s):

**FloorScore®**

Indoor Air Quality Certified to SCS-105 Version 4.2 – 2023.

Conforms to the CDPH/EHLB Standard Method v1.2-2017 (California Section 01350), effective April 1, 2017, for the school classroom and private office parameters when modeled as Flooring.

Measured Concentration of Total Volatile Organic Compounds (TVOC): Less than/equal to 0.5 mg/m<sup>3</sup> (in compliance with CDPH/EHLB Standard Method v1.2-2017)

Registration # SCS-FS-07212

Valid from: August 01, 2026 to July 31, 2027

SCS Global Services is currently the only certification body approved by the Resilient Floor Covering Institute (RFCI) to provide FloorScore® product certification; certified products are only listed on the SCS Green Products Guide, <http://www.scsglobalservices.com/certified-green-products-guide>.



A handwritten signature in black ink.

Diana Kirsanova Phillips, Chief of Staff to the CEO / Chief Assurance Officer  
SCS Global Services

**FloorScore®**

## **2026 Renewal Certification Assessment**

---

*SCS Environmental Certification Services*

Prepared for:

**Jiangsu Changlong Decorative Material  
Technical Co., Ltd.**

**Vinyl Tile**

*Manufactured in Jiangsu, China*

**Project Work Order: # WO- 332373**

**July 17, 2026**



**SCS**global  
SERVICES

*Setting the standard for sustainability™*

2000 Powell Street, Ste. 600, Emeryville, CA 94608 USA  
+1.510.452.8000 main | +1.510.452.8001 fax  
[www.SCSglobalServices.com](http://www.SCSglobalServices.com)

## FloorScore® Renewal Certification Assessment

SCS Global Services, Environmental Certification Services completed a renewal FloorScore certification assessment of Jiangsu Changlong Decorative Material Technical Co., Ltd.'s Vinyl Tile products manufactured in Haian, Jiangsu. The lead auditor has completed a full assessment of the quality management assessment of each manufacturing site and product emissions review to determine conformance to SCS-105 Indoor Air Quality Performance Standard and CA 01350 (Standard Method v1.2, effective April 1, 2017). The results of the assessment are included within this assessment report.

This certification assessment has been completed in accordance with SCS' quality management system, ISO 19011 standard auditing practices, ISO/IEC 17065 and has been approved by the program manager. If there is a disagreement with the certification decision, please contact the program manager/quality manager directly.

## Authorized Signatures

| AUDITOR INFORMATION          |                                                                                                    |      |               |
|------------------------------|----------------------------------------------------------------------------------------------------|------|---------------|
| Lead Auditor Name            | Bridgette Degenhardt, Technical Specialist<br>bdegenhardt@scsglobalservices.com<br>+1.510.495.6052 | Date | July 16, 2026 |
| SCS FINAL REVIEWER SIGNATURE |                                                                                                    |      |               |
| SCS Decision Maker           | Michael Hubbell, Program Manager<br>mhubbell@scsglobalservices.com<br>+1.510.838.1071              |      |               |
| Signature                    |                 | Date | July 17, 2026 |

## EXECUTIVE SUMMARY

### ASSESSMENT SCOPE

#### Changes to Client Certificates

Jiangsu Changlong's Vinyl Tile assessment scope is detailed in the work order WO-332373.

#### Changes to Private Label Certificates

A change of scope was required to add an item to a current private label certificate.

| COMPANY NAME      | CERTIFICATE NUMBER | PRODUCT LINE(S) ADDED | PRODUCT LINE(S) REMOVED |
|-------------------|--------------------|-----------------------|-------------------------|
| Karndean Flooring | SCS-FS-03277       | Palio Essenza         | NA                      |

### INDOOR AIR QUALITY CERTIFICATION SUMMARY

The following table details the conformance for products that have met FloorScore® product certification requirements. Products that have met FloorScore® certification also comply with the following criteria: USGBC LEED 2008, 2009 and v4 for Low-Emitting Materials (Flooring); Collaborative for High Performance Schools (CHPS) 2009 Criteria, EQ2.2.3; ANSI/ASHRAE/USGBC/IES Standard 189.1-2014, Section 8.4.2.3: Floor Covering Materials; BREEAM International and BREEAM UK; Green Guide for Health Care 2.2 EP 3.3 and EQ 4.3; and WELL Building Standard 4.3.a: Air.

| PRODUCT CATEGORY | PRODUCT LINE(S)                                                   | QUALIFICATIONS                            | CERTIFICATE NUMBER | CERTIFICATION PERIOD               |
|------------------|-------------------------------------------------------------------|-------------------------------------------|--------------------|------------------------------------|
| Vinyl Tile       | Click, Loose Lay, LVT, SPC Flooring<br>(Maximum thickness: 6.0mm) | TVOC Range: 0.5 mg/m <sup>3</sup> or less | SCS-FS-07212       | August 1, 2026 to<br>July 31, 2027 |

## PRODUCT ASSESSMENT SUMMARY

The following table indicates if a Non-Conformance Report (NCR) / Opportunity for Improvement (OFI) /New Information Request (NIR) is issued based on deficiencies found in the documentation or other supplemental information provided to meet both Standard Method v1.2 and the Quality Management System requirements. All NCRs must be closed before certification may be issued.

| NCR/<br>OFI/<br>NIR-# | STANDARD<br>REQUIREMENT    | AUDITOR FINDINGS                                                                                                                 | DUE DATE      | STATUS                     |
|-----------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------|----------------------------|
| NIR-25-01             | SCS 105 Section<br>7.4.1.1 | The client's ISO9001 certification expires in August of 2025. An updated version of the certificate is requested once available. | July 31, 2026 | CLOSED-<br>See Section 1.4 |

## MANUFACTURING ONSITE AUDIT SUMMARY

The table below indicates the entire scope of applicable site audits, the status of site audits completed, conformance status, and rationale for site audit selection for initial and renewal years.

| MANUFACTURING SITE                                                                                                | DATE OF LAST ONSITE<br>AUDIT                          | STATUS OF<br>NCR/NIR | AUDITOR RATIONALE                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>Jiangsu Changlong<br/>Decorative Material<br/>Technical Co., Ltd.</u><br>No 168, BaiChuan Road<br>Hai'an China | May 18, 2021<br><br>Last Full Quality Review:<br>2025 | NA                   | This site continues to demonstrate the ability to control production through a quality management system and manufactures products considered to be a low priority for emissions testing. Therefore, the site is also considered a low priority and does not require a renewal onsite audit currently. The site will be re-evaluated for a renewal onsite audit during the 2029 renewal assessment. |

## 1.0 Product Category Assessment

The following tables provide a progressive review on how products were selected to represent the emission profile for the product category, actual emissions results, and overall product certification results.

### 1.1 PRODUCT TESTING SAMPLE SELECTION

The following product sample selection is based on the product formulations, ingredients/materials, production processes and known production variability.

| PRODUCT LINE | PRODUCT DETAILS | SUPPORTING DOCUMENTATION                                         | AUDITOR RATIONALE                                                                                                                                         |
|--------------|-----------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Click        | 5.0mm           | Email: Re: Jiangsu Changlong 2026 FloorScore Renewal Sample Plan | Click flooring has had the highest emissions results in the past. Therefore, it was chosen for testing. 6mm was not available, so 5.0mm was sent instead. |

### 1.2 PRODUCT EMISSIONS RESULTS

The table below summarizes the product emissions history results for product category for the individual volatile organic compounds (VOCs) tested for compliance to CA Specification 01350 (Standard Method v1.2, effective April 1, 2017) private office and school classroom parameters when modeled as flooring.

### 1.2.1 VINYL TILE

| MANUFACTURING SITE | PRODUCT LINE           | REPORT #/DATE               | CREL VOCs    | EF (ug/m <sup>2</sup> -h) at 96-hrs | Private Office* Results | School Classroom Results | TVOC RANGE                   |
|--------------------|------------------------|-----------------------------|--------------|-------------------------------------|-------------------------|--------------------------|------------------------------|
| Shanghai           | Easy Street (Glueless) | TUV 143264956-Sep2713       | Toluene      | 46.00                               | PASS                    | PASS                     | -                            |
|                    |                        |                             | Phenol       | 13.00                               | PASS                    | PASS                     | -                            |
|                    |                        |                             | TVOC         | 190.00                              | -                       | -                        | 0.5mg/m <sup>3</sup> or less |
| Shanghai           | Click                  | 392-2015-00282601-Oct2015   | Phenol       | 16.00                               | PASS                    | PASS                     | -                            |
|                    |                        |                             | TVOC         | 90.00                               | -                       | -                        | 0.5mg/m <sup>3</sup> or less |
| Shanghai           | PVC Floor (Glueless)   | 392-2017-00310401 - Oct2017 | Not Detected | 0.00                                | PASS                    | PASS                     | -                            |
|                    |                        |                             | TVOC         | 52.00                               | -                       | -                        | 0.5mg/m <sup>3</sup> or less |
| Shanghai           | Click 5.0              | 392-2019-00338901 - Dec2019 | Phenol       | 18.00                               | PASS                    | PASS                     | -                            |
|                    |                        |                             | TVOC         | 91.00                               | -                       | -                        | 0.5mg/m <sup>3</sup> or less |
| Jiangsu, China     | SPC Flooring 6.0mm     | 244326875a 001 - May2021    | Not Detected | 0.00                                | PASS                    | PASS                     | -                            |
|                    |                        |                             | TVOC         | 0.00                                | -                       | -                        | 0.5mg/m <sup>3</sup> or less |
| Jiangsu, China     | Click 5.0              | 158250719a 001 - June 2022  | Formaldehyde | 0.00                                | PASS                    | PASS                     | -                            |
|                    |                        |                             | TVOC         | 20.70                               | -                       | -                        | 0.5mg/m <sup>3</sup> or less |
| Jiangsu, China     | Click 5.0              | 326024571a 001 - July 2024  | Formaldehyde | 0.00                                | PASS                    | PASS                     | -                            |
|                    |                        |                             | TVOC         | 10.00                               | -                       | -                        | 0.5mg/m <sup>3</sup> or less |
| Jiangsu, China     | Click 5.0              | 304176174a 001 7 July 2026  | Not Detected | 0.00                                | PASS                    | PASS                     | -                            |
|                    |                        |                             | TVOC         | 0.00                                | -                       | -                        | 0.5mg/m <sup>3</sup> or less |

The results for Jiangsu Changlong's Vinyl Tile products manufactured in Jiangsu demonstrate that all 35 targeted CREL VOCs listed in the Standard Method V1.2 are below 70% of the maximum allowable limits. Therefore, these products are considered **Low Priority**. Low Priority products require renewal testing every two years after initial certification.

### 1.3 Lab Testing Requirements

The table below is a checklist of the standard requirements for product testing and report results according to CA 01350 (Standard Method v1.2) and SCS-105.

|   | REQUIREMENT                                                                                                                                                                                                                             | SUPPORTING DOCUMENTATION                                                | AUDITOR KEY FINDINGS                                                                                              | ASSESSMENT RESULT                                                                                                                             |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Laboratory is a qualified ISO 17025 certified laboratory.<br>[Reference: SCS-105 Section 6.2]                                                                                                                                           | 304176174a 001_extsigned.pdf                                            | The sample was tested by TUV Rheinland Suzhou, an ISO 17025 certified lab.                                        | <input checked="" type="checkbox"/> No Issues<br><input type="checkbox"/> NCR<br><input type="checkbox"/> NIR<br><input type="checkbox"/> OFI |
| 2 | The manufacturer has submitted results in accordance with CDPH/EHLB/Standard Method V1.2 (January 2017),<br>[Reference: SCS-105 Section 6.2]                                                                                            | 304176174a 001_extsigned.pdf                                            | The manufacturer submitted results in accordance with the stated standards.                                       | <input checked="" type="checkbox"/> No Issues<br><input type="checkbox"/> NCR<br><input type="checkbox"/> NIR<br><input type="checkbox"/> OFI |
| 3 | Samples for testing are selected by a SCS auditor or a person authorized by SCS.<br>[Reference: SCS-105 Section 6.1]                                                                                                                    | Email: Email: Re: Jiangsu Changlong 2026 FloorScore Renewal Sample Plan | Samples sent for testing were chosen by an SCS auditor.                                                           | <input checked="" type="checkbox"/> No Issues<br><input type="checkbox"/> NCR<br><input type="checkbox"/> NIR<br><input type="checkbox"/> OFI |
| 4 | The samples are selected from a production lot that is representative of the processes and quality the manufacturer intends to present to the market.<br>[Reference: SCS-105 Section 6.2]                                               | 304176174a 001_extsigned.pdf<br>Chain of Custody Forms                  | Sample selections were representative of processes and quality the manufacturer intends to present to the market. | <input checked="" type="checkbox"/> No Issues<br><input type="checkbox"/> NCR<br><input type="checkbox"/> NIR<br><input type="checkbox"/> OFI |
| 5 | Manufacturer has submitted their nominations for test groups and worst case models accompanied by supportive data in the form of calculations, test results, formulations, and written explanation.<br>[Reference: SCS-105 Section 6.2] | Jiangsu Changlong_2026_FS_Renewal_DRF_auditor.xls                       | Sample recommendation was based on current scope and previous emissions history.                                  | <input checked="" type="checkbox"/> No Issues<br><input type="checkbox"/> NCR<br><input type="checkbox"/> NIR<br><input type="checkbox"/> OFI |
| 6 | Samples submitted to the laboratory are in conformance with CDPH/EHLB/Standard Method V1.2-January 2017, Section 2.<br>[Reference: SCS-105 Section 6.2]                                                                                 | 304176174a 001_extsigned.pdf<br>Chain of Custody Forms                  | Samples submitted were in conformance with the stated standards.                                                  | <input checked="" type="checkbox"/> No Issues<br><input type="checkbox"/> NCR<br><input type="checkbox"/> NIR<br><input type="checkbox"/> OFI |

## 1.4 Quality Management System

The table below includes a review of the ongoing maintenance of the manufacturer's quality management system in accordance to SCS-105.

|   | REQUIREMENT                                                                                          | SUPPORTING DOCUMENTATION                                         | AUDITOR KEY FINDINGS                                        | CONFORMANCE ISSUE                                                                                                                            |
|---|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>QMS Certification</b><br>Valid ISO 9001 or comparable certification was submitted (if applicable) | ISO9001+ISO14001 (2).pdf                                         | Site is ISO 9001 certified valid until August 21, 2028.     | <input checked="" type="checkbox"/> No issue<br><input type="checkbox"/> OFI<br><input type="checkbox"/> NIR<br><input type="checkbox"/> NCR |
| 2 | <b>Internal/External Audits</b><br>Evidence of no outstanding NCRs were provided.                    | CN-SZH-247286_RAU2.1__GS0304A Stage 2 Client Report-issue 2b.pdf | The ISO 9001 audit was submitted with no findings included. | <input checked="" type="checkbox"/> No issue<br><input type="checkbox"/> OFI<br><input type="checkbox"/> NIR<br><input type="checkbox"/> NCR |
| 3 | <b>Production Flow Chart</b><br>[Reference: SCS-105 Section 7.3]                                     | Production Flow Chart.docx                                       | An updated production flow chart was submitted.             | <input checked="" type="checkbox"/> No issue<br><input type="checkbox"/> OFI<br><input type="checkbox"/> NIR<br><input type="checkbox"/> NCR |
| 4 | <b>Production Variability</b><br>[Reference: SCS-105 Section 7.3]                                    | Jiangsu Changlong_2026_FS_Renewal_DRF_auditor.xls                | No changes reported.                                        | <input checked="" type="checkbox"/> No issue<br><input type="checkbox"/> OFI<br><input type="checkbox"/> NIR<br><input type="checkbox"/> NCR |
| 5 | <b>Product Line Identification and Traceability</b><br>[Reference: SCS-105 Section 7.3]              | Jiangsu Changlong_2026_FS_Renewal_DRF_auditor.xls                | Updated product line traceability documents were submitted. | <input checked="" type="checkbox"/> No issue<br><input type="checkbox"/> OFI<br><input type="checkbox"/> NIR<br><input type="checkbox"/> NCR |
| 6 | <b>Production Control</b><br>[Reference: SCS-105 Section 7.3]                                        | Customer Complaint.docx                                          | The customer complaint protocol and example were provided.  | <input checked="" type="checkbox"/> No issue<br><input type="checkbox"/> OFI<br><input type="checkbox"/> NIR<br><input type="checkbox"/> NCR |

## 1.5 FloorScore® Logo Review

The table below includes a review of the ongoing FloorScore® logo usage

|   | REQUIREMENT                                                                                                                                           | SUPPORTING DOCUMENTATION                                                                                                                                                                            | AUDITOR KEY FINDINGS                                                                                             | CONFORMANCE ISSUE                                                                                                                            |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Logo Use</b><br>a. Logo Use: Website<br>b. Logo Use: Marketing Collateral<br>c. Logo Use: On Product Packaging<br>[Reference: SCS-105 Section 8.2] | <a href="https://www.changlongflor.com/F1_UK_DFI_VG_GD_7x28_x_3.0_(KD6400GLO_08-21)_27-8-21_(3).pdf">https://www.changlongflor.com/F1_UK_DFI_VG_GD_7x28_x_3.0_(KD6400GLO_08-21)_27-8-21_(3).pdf</a> | FloorScore logo is used appropriately on websites and product packaging. Not submitted for marketing collateral. | <input checked="" type="checkbox"/> No issue<br><input type="checkbox"/> OFI<br><input type="checkbox"/> NIR<br><input type="checkbox"/> NCR |

## APPENDIX

### A.1 Renewal Audit Recommendations

#### A.1.1 Renewal Sample Selection

The following table represents suggested sample selections for emissions testing. The frequency and required number of samples is dependent on the quality management systems review per manufacturing site and emissions history review completed during each renewal assessment by the auditor.

| PRODUCT CATEGORY | PRODUCT LINE       | RENEWAL YEAR | AUDITOR RATIONALE                                                                                                           |
|------------------|--------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------|
| Vinyl Tile       | LVT Flooring 6.0mm | 2028         | LVT Flooring has not been tested in recent years. It is requested for emissions testing in 2028 or the next testing period. |

## A.2 Private Labels

### A.2.1 Summary of Private Label Certificates

The following table summarizes privately labeled products of Jiangsu Changlong.

| PRIVATE LABEL APPLICANT      | PRIVATE LABEL PRODUCT LINE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | PRIVATE LABEL CERTIFICATE | CLIENT CERTIFIED PRODUCT                                       | CLIENT CERTIFICATE |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------|--------------------|
| Green-Flor                   | Nature Fit 30/55, Young Living, Authentic Mood, Pure Plaza, Master Collection, Master Trend, True Spirit, New Square, New Square Piazza, Modern Vintage, Stylish Grid, Grand Class, Performance 55/70, Performance XL, Performance XXL, Modern Vintage Smart, Modern Vintage Chic, Nature Fit 30/50 Stark, Nature Living, Pure Spectra Collection, Solid Essence Collection, Essential Discovery, Prime Inspiration, Tierra Nuera, True Hive, Royal Touch, Silent Elegance, Pure and Powerful 55, Pure and Powerful 30, Authentic Mood 30, Authentic Mood 55 Click, Stone Design, Wood Design, Authentic Mood 55, Prime Inspiration Click, Performance 50/70 XXL, Performance 50/70 XL | SCS-FS-03968              | Click, Loose Lay, LVT, SPC Flooring (Maximum thickness: 6.0mm) | SCS-FS-07212       |
| DesignFlooring International | Art Select , Designflooring Residence , K-Trade, Monet, Opus, Opus Enhance, Palio Gluedown , Palio Gluedown Plus , Rubens, Van Gogh                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | SCS-FS-03278              | LVT (Maximum thickness: 6.0mm)                                 | SCS-FS-07212       |
| Karndean Designflooring      | Art Select, Capitol Luxury Vinyl Plank, Da Vinci, Knight Tile, K-Trade, Michelangelo, Opus, Opus Enhance, <b>Palio</b> <b>Essenza</b> , Palio Gluedown 2.0mm, Palio Gluedown 2.5mm, Van Gogh                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | SCS-FS-03277              | LVT (Maximum thickness: 6.0mm)                                 | SCS-FS-07212       |
| TOLI Corporation             | ROYAL STONE, ROYAL WOOD, TOLI select, CLASIAL TILE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | SCS-FS-08252              | LVT, SPC Flooring                                              | SCS-FS-07212       |

**SCS Global Services** does hereby certify that an independent assessment has been conducted on behalf of:

# Jiangsu Changlong Decorative Material Technical Co., Ltd.

No 168, BaiChuan Road, Hai'an City, Jiangsu, China

For the following product(s):

**Vinyl Tile:**

Click, Loose Lay, LVT, SPC Flooring  
(Maximum thickness: 6.0mm)

The product(s) meet(s) all of the necessary qualifications to be certified for the following claim(s):

**FloorScore®**

Indoor Air Quality Certified to SCS-105 Version 4.2 – 2023.

Conforms to the CDPH/EHLB Standard Method v1.2-2017 (California Section 01350), effective April 1, 2017, for the school classroom and private office parameters when modeled as Flooring.

Measured Concentration of Total Volatile Organic Compounds (TVOC): Less than/equal to 0.5 mg/m<sup>3</sup> (in compliance with CDPH/EHLB Standard Method v1.2-2017)

Registration # SCS-FS-07212

Valid from: August 01, 2025 to July 31, 2026

SCS Global Services is currently the only certification body approved by the Resilient Floor Covering Institute (RFCI) to provide FloorScore® product certification; certified products are only listed on the SCS Green Products Guide, <http://www.scsglobalservices.com/certified-green-products-guide>.



A handwritten signature in black ink.

Diana Kirsanova Phillips, Chief Assurance Officer,  
SCS Global Services



## Jiangsu Changlong Decorative Materials Technical Co., Ltd

No.168 Baichuan Road, Hai'An City, Jiangsu Province, China

This is to Certify that the following Product/s have been found in conformance with the Global GreenTagCERT™ Scheme Standard for the Tier and Level noted herein:

**LVT, Loose Lay, Click**

**GreenRate Level A**

Licensed Sites:

Jiangsu, China

Licence No.:

JDC:RF02:2022:GR

Licence Date:

4th November 2022

Latest Revision Date:

19 February 2025

Valid to:

4th November 2025

GreenTag<sup>CERT™</sup> Standard:

Standard Version 4.0



[www.globalgreentag.com](http://www.globalgreentag.com)

David Baggs  
Global GreenTag<sup>Cert™</sup> Chief Executive Officer



green product certification  
trust brands

## *Conditions of Licence*

The conditions of licence are contained in full in the Global GreenTag<sup>CERT™</sup> Standard, Terms and Conditions, and Rules for the Use of the Mark Documents as sighted and/or executed by the Licensee.

In summary it is the responsibility of the licensee in particular to:

1. always comply with the relevant provisions of the GreenTag certification program;
2. make all necessary arrangements for the conduct of the future evaluation, including provision for examining documentation and access to all areas, records (including internal audit reports) and personnel for the purposes of evaluation (e.g. testing, inspection, assessment surveillance, reassessment) and resolution of complaints;
3. make claims regarding certification only in respect of the scope for which certification has been granted;
4. not use its product certification in such a manner as to bring the GreenTag into disrepute and not make any statement regarding its product certification which the certification body may consider misleading or unauthorized;
5. upon suspension or cancellation of certification, discontinue its use of all advertising matter that contains any reference thereto and returns any certification documents as required by GreenTag;
6. use certification only to indicate that products are certified as being in conformity with specified GreenTag standards;
7. endeavour to ensure that no certificate or report nor any part thereof is used in a misleading manner;
8. make comment or inclusions solely in accordance with license requirements in making reference to its product certification in communication media such as online, emails, documents, brochures or advertising;
9. Inform GreenTag of any change in the Certified product or manufacturing process that is likely to significantly affect the product's design or specification, or changes in the ownership, structure or management of the Licensee, if relevant, or any other information that indicates the product may no longer comply with the requirements of this Standard;
10. In the event of GreenTag determining changes have been made to product or supplier details and not notified to GreenTag, the Licensee will, on receipt of a GreenTag 'Notice to Rectify', immediately provide GreenTag with the required details and any fees necessary to allow recertification. Failure to do so may result in the withdrawal of the Licence. If the product Licence is withdrawn, the Licensee must, within 7 days, cease to further promulgate all product marketing, packaging, advertising or other material carrying the logo. Furthermore all material carrying the Certification Mark/s must be withdrawn within 90 days.

| Revision Date      | Certificate Number | Notes                                           |
|--------------------|--------------------|-------------------------------------------------|
| 4th November 2022  | JDC:RF02:2022:GR   | Products certified under GreenTag standard v4.0 |
| 9th July 2024      | JDC:RF02:2022:GR   | 2nd year renewal                                |
| 19th February 2025 | JDC:RF02:2022:GR   | 3rd year renewal                                |
|                    |                    |                                                 |
|                    |                    |                                                 |
|                    |                    |                                                 |
|                    |                    |                                                 |
|                    |                    |                                                 |
|                    |                    |                                                 |
|                    |                    |                                                 |

19 February 2025

The Directors  
Jiangsu Changlong Decorative Materials Technical Co., Ltd  
No.168 Baichuan Road,  
Hai'An City,  
Jiangsu Province, China

Attention: Dashuai Xi  
By email: [QC06@changlongflor.com](mailto:QC06@changlongflor.com)

Dear Dashuai,

The purpose of this letter is to clarify matters set out in the Global GreenTag certificate. It provides additional information on the scope, limitations and exclusions of the certificate issued based on information provided by the applicant and its suppliers. It is not a certificate and is not a substitute for the certificate, nor does it include any product variants apart from those specifically shown below.

|                             |                                                                                                                                                        |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope definition            | The certificate applies to SPC, LVT, Loose Lay and Click produced by Jiangsu Changlong Decorative Material Technical Co., Ltd at their Hai'An factory. |
| GreenTag Standard           | Global GreenTag International Standard v4.0                                                                                                            |
| Certification level         | GreenRate Level A                                                                                                                                      |
| Licence number              | JDC:RF01:2022:GR<br>JDC:RF02:2022:GR                                                                                                                   |
| Limitations and exclusions: | NA                                                                                                                                                     |

This section of the letter clarifies the credit verification types for the awarded certificates.

| Rating Tool                                            | Credit                                                                                 | Verification Type <sup>1</sup> | Certificates      |
|--------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------|-------------------|
| Green Star® Australia<br>"Buildings" v1.0 Rating Tool  | Credit 9: Responsible Finishes                                                         | Recognised Compliant           | GreenRate Level A |
| Green Star® Australia<br>"Interiors v1.3" Rating Tool  | Credit 12: Indoor Pollutants                                                           | Recognised Compliant           | GreenRate Level A |
| Green Star® Australia<br>"Interiors v1.3" Rating Tool  | Credit 21: Sustainable Products                                                        | Recognised Compliant           | GreenRate Level A |
| Green Star® Australia<br>"Performance 1.2" Rating Tool | Credit 21: Procurement and Purchasing (Consumable Materials   Refurbishment Materials) | Recognised Evidence            | GreenRate Level A |
| IWBI® WELL v1.0 Features                               | Feature 04 VOC Reduction (Part 3)                                                      | Audit Verified Compliant       | GreenRate Level A |
| IS® v1.2                                               | Mat-2 Environmentally Labelled Products and Supply Chain                               | Recognised Compliant           | GreenRate Level A |
| IS® Design & As-Built v2.1                             | Rso-7 Sustainability Labelled Products and Supply Chains                               | Recognised Compliant           | GreenRate Level A |
| EarthCheck Rating Tool                                 | Sustainable Materials                                                                  | Recognised Compliant           | GreenRate Level A |

This information is provided to the applicant in complement to the certificate in order to ensure that the Global GreenTag Certification Mark is used only for products covered by the scope defined therein and in accordance to the current GreenTag Rules for the Use of Mark, and Terms and Conditions documents.

Global GreenTag Pty Ltd *per*



**David Baggs**

CEO

LFAIA, Green Star AP, LEED AP, MACS, Exemplar Global Lead Auditor

**Recognized Compliant:** The verification given to a Global GreenTag certificate credit statement when the Green Building Rating Tool has formally recognised the Global GreenTag certification services' individual product compliance with the credit, or the Global GreenTag standard, and the certificate meets the minimum individual product credit achievement requirements.

**Recognized Evidence:** The verification given to a Global GreenTag certificate credit statement when the Green Building Rating Tool has formally recognized the Global GreenTag certification services' individual product compliance with the credit, or the Global GreenTag standard, but the certificate doesn't meet the minimum individual product credit achievement requirements because it requires additional steps of evidence collection and evaluation to determine individual product credit achievement requirements.

**Audit Verified Compliant:** The verification given to a Global GreenTag certificate credit statement when the certification service conformity requirements outlined in the Global GreenTag International Standard meet the minimum individual product credit achievement requirements of the Green Building Rating Tool, but the Green Building Rating Tool has not yet recognised the certification services' individual product compliance with the credit.





## Jiangsu Changlong Decorative Materials Technical Co., Ltd

No.168 Baichuan Road, Hai'An City, Jiangsu Province, China

This is to Certify that the following Product/s have been found in conformance with the Global GreenTagCERT™ Scheme Standard for the Tier and Level noted herein:

**SPC**

**GreenRate Level A**

**Licensed Sites:**

Jiangsu, China

**Licence No.:**

JDC:RF01:2022:GR

**Licence Date:**

4th November 2022

**Latest Revision Date:**

19th February 2025

**Valid to:**

4th November 2025

**GreenTag<sup>CERT™</sup> Standard:**

Standard Version 4.0



[www.globalgreentag.com](http://www.globalgreentag.com)

David Baggs  
Global GreenTag<sup>Cert™</sup> Chief Executive Officer



green product certification  
trust brands

## Conditions of Licence

The conditions of licence are contained in full in the Global GreenTag<sup>CERT</sup>™ Standard, Terms and Conditions, and Rules for the Use of the Mark Documents as sighted and/or executed by the Licensee.

In summary it is the responsibility of the licensee in particular to:

1. always comply with the relevant provisions of the GreenTag certification program;
2. make all necessary arrangements for the conduct of the future evaluation, including provision for examining documentation and access to all areas, records (including internal audit reports) and personnel for the purposes of evaluation (e.g. testing, inspection, assessment surveillance, reassessment) and resolution of complaints;
3. make claims regarding certification only in respect of the scope for which certification has been granted;
4. not use its product certification in such a manner as to bring the GreenTag into disrepute and not make any statement regarding its product certification which the certification body may consider misleading or unauthorized;
5. upon suspension or cancellation of certification, discontinue its use of all advertising matter that contains any reference thereto and returns any certification documents as required by GreenTag;
6. use certification only to indicate that products are certified as being in conformity with specified GreenTag standards;
7. endeavour to ensure that no certificate or report nor any part thereof is used in a misleading manner;
8. make comment or inclusions solely in accordance with license requirements in making reference to its product certification in communication media such as online, emails, documents, brochures or advertising;
9. Inform GreenTag of any change in the Certified product or manufacturing process that is likely to significantly affect the product's design or specification, or changes in the ownership, structure or management of the Licensee, if relevant, or any other information that indicates the product may no longer comply with the requirements of this Standard;
10. In the event of GreenTag determining changes have been made to product or supplier details and not notified to GreenTag, the Licensee will, on receipt of a GreenTag 'Notice to Rectify', immediately provide GreenTag with the required details and any fees necessary to allow recertification. Failure to do so may result in the withdrawal of the Licence. If the product Licence is withdrawn, the Licensee must, within 7 days, cease to further promulgate all product marketing, packaging, advertising or other material carrying the logo. Furthermore all material carrying the Certification Mark/s must be withdrawn within 90 days.

| Revision Date      | Certificate Number | Notes                                           |
|--------------------|--------------------|-------------------------------------------------|
| 4th November 2022  | JDC:RF01:2022:GR   | Products certified under GreenTag standard v4.0 |
| 9th July 2024      | JDC:RF01:2022:GR   | 2nd year renewal                                |
| 19th February 2025 | JDC:RF01:2022:GR   | 3rd year renewal                                |
|                    |                    |                                                 |
|                    |                    |                                                 |
|                    |                    |                                                 |
|                    |                    |                                                 |
|                    |                    |                                                 |
|                    |                    |                                                 |
|                    |                    |                                                 |



## Jiangsu Changlong Decorative Materials Technical Co., Ltd

No.168 Baichuan Road, Hai'An City, Jiangsu Province, China

This is to Certify that the following Product/s have been found in conformance with the Global GreenTagCERT™ Scheme Standard for the Tier and Level noted herein:

**SPC**

**GreenRate Level A**

**Licenced Sites:**

Jiangsu, China

**Licence No.:**

JDC:RF01:2022:GR

**Licence Date:**

4th November 2022

**Latest Revision Date:**

19th February 2025

**Valid to:**

4th November 2025

**GreenTag<sup>CERT™</sup> Standard:**

Standard Version 4.0



David Baggs  
Global GreenTag<sup>Cert™</sup> Chief Executive Officer



## Jiangsu Changlong Decorative Materials Technical Co., Ltd

No.168 Baichuan Road, Hai'an City, Jiangsu Province, China

This is to Certify that the following Product/s have been found in conformance with the Global GreenTagCERT™ Scheme Standard for the Tier and Level noted herein:

### LVT, Loose Lay, Click

#### GreenRate Level A

Green Star® 'Buildings v1.0' Rating Tools Credits:

Recognized for Credit/s: Credit 9: Responsible Finishes

Green Star® 'Design & As Built' AU v1.3 and 'Interiors' AU v1.3 Rating Tools Credits:

Recognized for Credit/s: Indoor Pollutants; Sustainable Products

Green Star® 'Performance' AU v1.2 Rating Tool Credit:

Recognized for Credit/s: Credit 21: Procurement and Purchasing (Refurbishment Materials)

IS® v1.2 Credit:

Recognized for Credit/s: Mat-2 Environmentally Labelled Products and Supply Chain

IS® Design & As-Built v2.1 Credit:

Recognized for Credit/s: Rso-7 Sustainability Labelled Products and Supply Chains

Earthcheck® Rating Tool Credit:

Recognized for Credit/s: Sustainable Materials

IWBI® WELL v1.0 and v2.0 Features:

Compliant Technical Document (Audited) for Feature/s: Feature 04 (Part 3)

#### Licensed Sites:

Jiangsu, China

Licence No.:

JDC:RF02:2022:GR

Licence Date:

4th November 2022

Latest Revision Date:

9th July 2024

Valid to:

4th November 2024

GreenTag CERT™ Standard:

Standard Version 4.0



www.globalgreentag.com

David Baggs  
Chief Executive Officer  
Global GreenTag CERT™ Program Director



**GLOBAL  
GREENTAG®**

green product certification  
trust brands

## *Conditions of Licence*

The conditions of licence are contained in full in the Global GreenTag<sup>CERT™</sup> Standard, Terms and Conditions, and Rules for the Use of the Mark Documents as sighted and/or executed by the Licensee.

In summary it is the responsibility of the licensee in particular to:

1. always comply with the relevant provisions of the GreenTag certification program;
2. make all necessary arrangements for the conduct of the future evaluation, including provision for examining documentation and access to all areas, records (including internal audit reports) and personnel for the purposes of evaluation (e.g. testing, inspection, assessment surveillance, reassessment) and resolution of complaints;
3. make claims regarding certification only in respect of the scope for which certification has been granted;
4. not use its product certification in such a manner as to bring the GreenTag into disrepute and not make any statement regarding its product certification which the certification body may consider misleading or unauthorized;
5. upon suspension or cancellation of certification, discontinue its use of all advertising matter that contains any reference thereto and returns any certification documents as required by GreenTag;
6. use certification only to indicate that products are certified as being in conformity with specified GreenTag standards;
7. endeavour to ensure that no certificate or report nor any part thereof is used in a misleading manner;
8. make comment or inclusions solely in accordance with license requirements in making reference to its product certification in communication media such as online, emails, documents, brochures or advertising;
9. Inform GreenTag of any change in the Certified product or manufacturing process that is likely to significantly affect the product's design or specification, or changes in the ownership, structure or management of the Licensee, if relevant, or any other information that indicates the product may no longer comply with the requirements of this Standard;
10. In the event of GreenTag determining changes have been made to product or supplier details and not notified to GreenTag, the Licensee will, on receipt of a GreenTag 'Notice to Rectify', immediately provide GreenTag with the required details and any fees necessary to allow recertification. Failure to do so may result in the withdrawal of the Licence. If the product Licence is withdrawn, the Licensee must, within 7 days, cease to further promulgate all product marketing, packaging, advertising or other material carrying the logo. Furthermore all material carrying the Certification Mark/s must be withdrawn within 90 days.

| Revision Date     | Certificate Number | Notes                                           |
|-------------------|--------------------|-------------------------------------------------|
| 4th November 2022 | JDC:RF02:2022:GR   | Products certified under GreenTag standard v4.0 |
| 9th July 2024     | JDC:RF02:2022:GR   | 2nd year renewal                                |
|                   |                    |                                                 |
|                   |                    |                                                 |
|                   |                    |                                                 |
|                   |                    |                                                 |
|                   |                    |                                                 |
|                   |                    |                                                 |
|                   |                    |                                                 |
|                   |                    |                                                 |



**Global  
GreenTag**  
International  
green product certification  
trust brands

GlobalGreenTag.com  
Global GreenTag International  
ABN 44 600 051 554

Level 38, 71 Eagle Street, Brisbane  
Queensland 4000 Australia

1300 263 586 +61 7 3399 9686  
admin@globalgreentag.com



12 July 2024

The Directors  
Jiangsu Changlong Decorative Materials Technical Co., Ltd  
No.168 Baichuan Road,  
Hai'An City,  
Jiangsu Province, China

Attention: Dashuai Xi  
By email: [QC06@changlongflor.com](mailto:QC06@changlongflor.com)

Dear Dashuai,

The purpose of this letter is to clarify matters set out in the Global GreenTag certificate. It provides additional information on the scope, limitations and exclusions of the certificate issued based on information provided by the applicant and its suppliers. It is not a certificate and is not a substitute for the certificate, nor does it include any product variants apart from those specifically shown below.

|                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope definition                                                 | The certificate applies to SPC, LVT, Loose Lay and Click produced by Jiangsu Changlong Decorative Material Technical Co., Ltd at their Hai'An factory.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| GreenTag Standard                                                | Global GreenTag International Standard v4.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Certification level                                              | GreenRate Level A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Licence number                                                   | JDC:RF01:2022:GR<br>JDC:RF02:2022:GR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Limitations and exclusions:                                      | NA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Greenstar Building v1.0<br>Responsible Product Value<br>Credit/s | <p><b>Your GreenTag certificate/s are recognised to meet the minimum RPV credit criteria for:</b></p> <p>Responsible Finishes - Good Practice products (RPV <math>\geq</math> 7)</p> <ul style="list-style-type: none"><li>GreenRate Level A</li></ul> <p><b>Your GreenTag certificate/s can be used in the following initiative combinations to meet the minimum RPV credit criteria for:</b></p> <p>Responsible Finishes - Best Practice products (RPV <math>\geq</math> 12)</p> <ul style="list-style-type: none"><li>GreenRate Level A + (Climate Active Carbon Neutral for Products &amp; Services   Industry Wide EPD [EN 15804]   Product Specific EPD [EN 15804] )</li></ul> |

This information is provided to the applicant in complement to the certificate in order to ensure that the Global GreenTag Certification Mark is used only for products covered by the scope defined therein and in accordance to the current GreenTag Rules for the Use of Mark, and Terms and Conditions documents.

Global GreenTag Pty Ltd per

David Baggs  
CEO  
LFAIA, Green Star AP, LEED AP, MACS, Exemplar Global Lead Auditor



**TRUSTED  
BRANDS**  
2015 2016 2019  
ARCHITECTURE//DESIGN  
TOP 100 BRAND



GLOBAL  
GREEN TAG  
INTERNATIONAL  
green product certification  
trust brands

Global GreenTag International Pty Ltd

ABN 44 600 051 554

Level 54/111 Eagle Street, Brisbane City,  
Queensland 4000, Australia

1300 263 586 ☎ +617 3399 9686

✉ admin@globalgreentag.com

@globalgreentag

www.globalgreentag.com

2 November 2022

The Directors

Gina Wu

No 168, Baichuan Rd, Hai'An City, Jiangsu, China

Attention: Gina Wu

By email: [ginawu@changlongflor.com](mailto:ginawu@changlongflor.com)

Dear Gina Wu

The purpose of this letter is to clarify matters set out in the Global GreenTag certificate. It provides additional information on the scope, limitations and exclusions of the certificate issued based on information provided by the applicant and its suppliers. It is not a certificate and is not a substitute for the certificate, nor does it include any product variants apart from those specifically shown below.

|                                   |                                                                                                                                                      |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope definition                  | The certificate applies to SPC, LVT, Loose Lay & Click produced by Jiangsu Changlong Decorative Material Technical Co., Ltd at their Hai'An factory. |
| GreenTag Standard                 | Global GreenTag International Standard v4.0                                                                                                          |
| Certification level               | GreenRate Level A                                                                                                                                    |
| Licence number                    | JDC:RF01:2022:GR<br>JDC:RF02:2022:GR                                                                                                                 |
| Limitations and exclusions:       | NA                                                                                                                                                   |
| Greenstar Building v1.0<br>Credit | <b>Responsible Finishes</b><br>Good Practice: RPV = > 7 <ul style="list-style-type: none"><li>Greenrate level A</li></ul>                            |

This information is provided to the applicant in complement to the certificate in order to ensure that the Global GreenTag Certification Mark is used only for products covered by the scope defined therein and in accordance to the current GreenTag Rules for the Use of Mark, and Terms and Conditions documents.

Global GreenTag Pty Ltd per

David Baggs

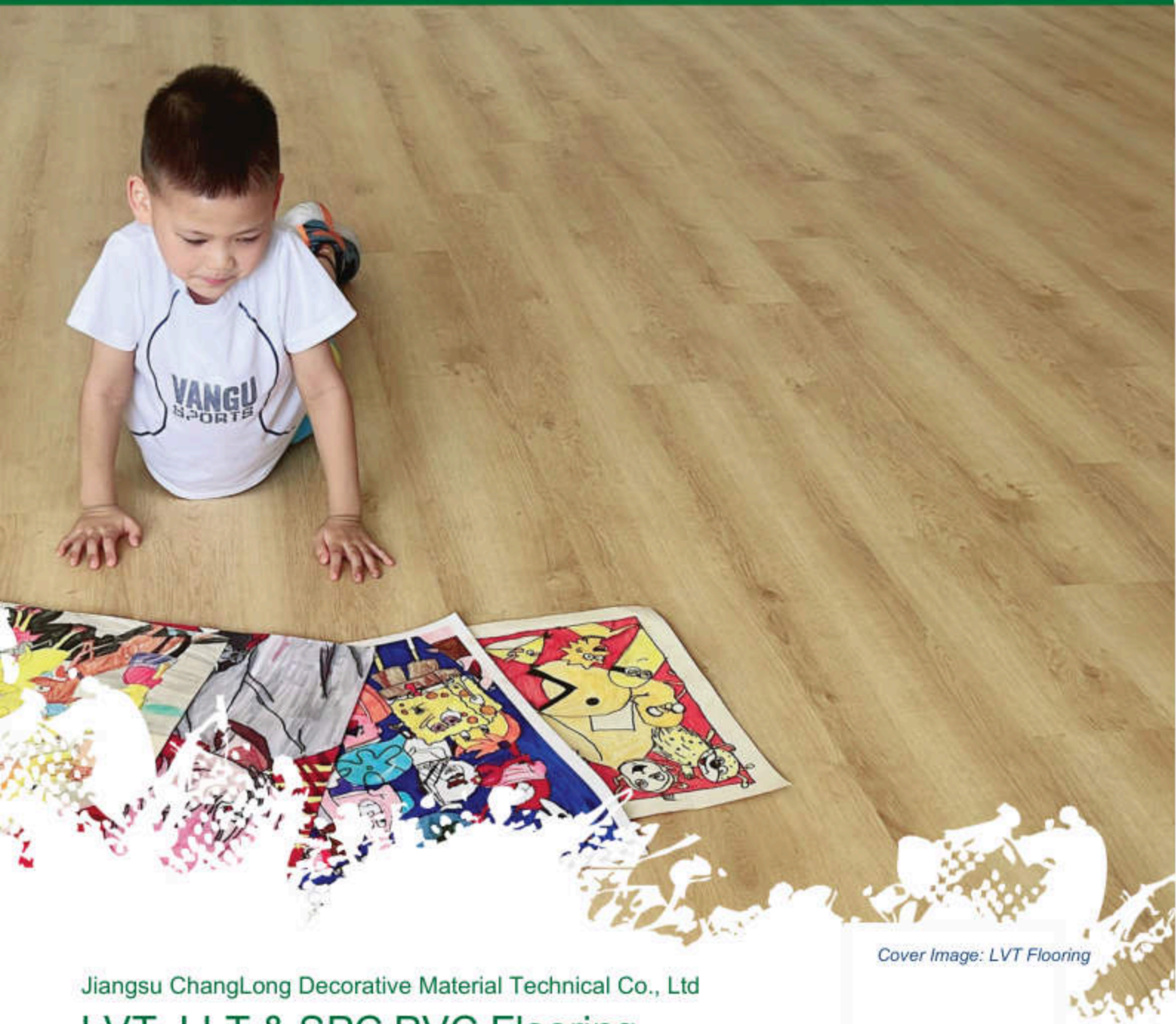
CEO & GreenTag Program Director

LFAIA, Green Star AP, LEED AP, MACS, Exemplar Global Lead Auditor



## Environmental Product Declaration

in accordance with ISO 14025 and EN 15804



Cover Image: LVT Flooring

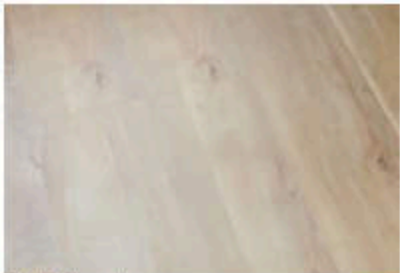
### Jiangsu ChangLong Decorative Material Technical Co., Ltd LVT, LLT & SPC PVC Flooring

|                   |                                                             |
|-------------------|-------------------------------------------------------------|
| Company Address:  | No.168, BaiChuan Road, Hai'An City, Jiangsu Province, China |
| Issue Date:       | 02 August 2024                                              |
| Valid to:         | 02 August 2029                                              |
| Document Version: | 1.0                                                         |
| Revision Date:    | 02 August 2024                                              |





### Environmental Product Declaration Details

|                                              |                                                                                                                                                                                                                              |                                                                                                                                                                    |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EPD Type                                     | Cradle to Gate with modules C1–C4 and module D (A1–A3, C and D)                                                                                                                                                              | <div>Product Image</div> <div></div> <div>LVT Flooring</div>                     |
| EPD Number                                   | JDC:FL01:2024:EP                                                                                                                                                                                                             |                                                                                                                                                                    |
| Issue Date                                   | 02 August 2024                                                                                                                                                                                                               |                                                                                                                                                                    |
| Valid Until                                  | 02 August 2029                                                                                                                                                                                                               |                                                                                                                                                                    |
| GPI Version                                  | Version 2.1                                                                                                                                                                                                                  |                                                                                                                                                                    |
| Demonstration of Verification                |                                                                                                                                                                                                                              |                                                                                                                                                                    |
| PCR                                          | CEN Standard EN 15804+A2 2019 serves as core Product Category Rules (PCR). Sub-PCR FC:2022 V1 Interior Floor Coverings V1                                                                                                    |                                                                                                                                                                    |
| <input checked="" type="checkbox"/> Internal | <div><br/>2024.02.08<br/><br/>2024.08.02</div>             | <div>LCA Developed by AnKang, Chengdu IKE Environmental Technology Co., Ltd.</div> <div>EPD Reviewed by Dr Nana Bortsie-Aryee, Global GreenTag International</div> |
| <input checked="" type="checkbox"/> External | <div><br/>2024.08.01</div>                                                                                                                | Third Party Verifier. Direshni Naiker, Gaia Conscious Consulting                                                                                                   |
| Verification                                 | Independent external verification of the declaration and data, mandatory for business-to-consumer communication according to ISO 14025:2010.                                                                                 |                                                                                                                                                                    |
| Communication                                | This EPD discloses potential environmental outcomes compliant with EN 15804 for business-to-business communication.                                                                                                          |                                                                                                                                                                    |
| Comparability                                | EPD of construction products may not be comparable if they do not comply with EN 15804. Different program EPDs may not be comparable. Comparability is further dependent on the product category rules and data source used. |                                                                                                                                                                    |
| Reliability                                  | LCIA results are relative expressions that do not predict impacts on category endpoints, exceeding of thresholds, safety margins or risks.                                                                                   |                                                                                                                                                                    |
| Owner                                        | This EPD is the property of the declared manufacturer.                                                                                                                                                                       |                                                                                                                                                                    |
| Explanations                                 | Further explanatory information is available at <a href="mailto:info@globalgreentag.com">info@globalgreentag.com</a> or by contacting <a href="mailto:epd@globalgreentag.com">epd@globalgreentag.com</a> .                   |                                                                                                                                                                    |

| EPD Program Operator                                                                                                                                                                                                              | LCA and EPD Producer                                                                                                                                                                               | Declaration Owner                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Global GreenTag International Pty Ltd</b><br><br>Level 38, 71 Eagle Street<br>Brisbane City, QLD 4000<br>Australia<br><br>Phone: +61 1300 263 586<br><a href="http://www.globalgreentag.com">http://www.globalgreentag.com</a> | <b>IKE Environmental Technology Co. Ltd.</b><br><br>No.139 Kehua Middle Road,<br>Wuhou District<br><br>Phone: +86 18280064252<br><a href="http://www.ike-global.com">http://www.ike-global.com</a> | <b>Jiangsu ChangLong Decorative Material Technical Co., Ltd</b><br><br>No.168, Baichuan Road, Haian City, Jiangsu<br>Province, P.R.China<br><br>Email: <a href="mailto:sales@changlongflor.com">sales@changlongflor.com</a><br><a href="http://www.changlongflor.com/">http://www.changlongflor.com/</a> |



GLOBAL  
**GREENTAG**  
INTERNATIONALS  
green product certification  
trust brands

**IKE** Integrated Knowledge  
for our Environment  
亿科环境科技





## Program Description

|                        |                                                                                                                                                                                                                                                 |           |             |           |           |        |          |        |         |           |            |           |             |           |               |                              |       |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|-----------|-----------|--------|----------|--------|---------|-----------|------------|-----------|-------------|-----------|---------------|------------------------------|-------|
| EPD Scope              | Cradle to gate with options (A1 to A3, C1-C4 and D)                                                                                                                                                                                             |           |             |           |           |        |          |        |         |           |            |           |             |           |               |                              |       |
| EPD Type               | EPD based on specific site and product data                                                                                                                                                                                                     |           |             |           |           |        |          |        |         |           |            |           |             |           |               |                              |       |
| System boundary        | The system boundary with nature includes processing material and energy system inputs, transport to factory gate, manufacturing plus packing, waste disposal, as well as waste removal and waste disposal after the expiration of product life. |           |             |           |           |        |          |        |         |           |            |           |             |           |               |                              |       |
| Stages included        | A1-A3, C1-C4, D                                                                                                                                                                                                                                 |           |             |           |           |        |          |        |         |           |            |           |             |           |               |                              |       |
| Stages excluded        | A4-A5, B1-B7                                                                                                                                                                                                                                    |           |             |           |           |        |          |        |         |           |            |           |             |           |               |                              |       |
| Information Modules    | Figure 1 depicts all modules being declared including some with zero results. Any module not declared (MND) does not indicate a zero result.                                                                                                    |           |             |           |           |        |          |        |         |           |            |           |             |           |               |                              |       |
| Model                  | ActualScenariosPotential                                                                                                                                                                                                                        |           |             |           |           |        |          |        |         |           |            |           |             |           |               |                              |       |
| Information            | Life Cycle AssessmentSupplementary                                                                                                                                                                                                              |           |             |           |           |        |          |        |         |           |            |           |             |           |               |                              |       |
| Stages                 | Product                                                                                                                                                                                                                                         |           |             | Construct |           | Use    |          |        |         |           |            |           | End-of-Life |           |               | Benefit & load beyond system |       |
| Modules                |                                                                                                                                                                                                                                                 |           |             |           |           | Fabric |          |        |         |           | Operation  |           |             |           |               |                              |       |
| Unit Operations        | A1                                                                                                                                                                                                                                              | A2        | A3          | A4        | A5        | B1     | B2       | B3     | B4      | B5        | B6         | B7        | C1          | C2        | C3            | C4                           | D     |
| Cradle to grave phases | Resources                                                                                                                                                                                                                                       | Transport | Manufacture | Transport | Construct | Use    | Maintain | Repair | Replace | Refurbish | Energy Use | Water use | Demolish    | Transport | Process Waste | Disposal                     | Reuse |
| Modules Declared       | ✓                                                                                                                                                                                                                                               | ✓         | ✓           | ND        | ND        | ND     | ND       | ND     | ND      | ND        | ND         | ND        | ✓           | ✓         | ✓             | ✓                            | ✓     |

Note: ND = Module not declared ✓ = included

Figure 1 EPD Life Cycle Modules Cradle to Grave



Product Information

General Information

|                                 |                                                                                                                                                                                                                                                                                                                                           |                                                                      |                     |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|---------------------|
| Brand Name & Code               | CHANGLONG PVC flooring                                                                                                                                                                                                                                                                                                                    |                                                                      |                     |
| Range Names                     | LVT, LLT, SPC flooring                                                                                                                                                                                                                                                                                                                    |                                                                      |                     |
| Factory warranty                | 10-20 Years                                                                                                                                                                                                                                                                                                                               |                                                                      |                     |
| Geographical Area               | China                                                                                                                                                                                                                                                                                                                                     |                                                                      |                     |
| Application                     | Indoor flooring                                                                                                                                                                                                                                                                                                                           |                                                                      |                     |
| Function in Building            | Flooring covering                                                                                                                                                                                                                                                                                                                         |                                                                      |                     |
| Reference Service Life          | 10 Years                                                                                                                                                                                                                                                                                                                                  |                                                                      |                     |
| Declared Unit                   | <p>3.8 kg of LVT interior floor covering per square metre covered in any building sectors cradle to gate.</p> <p>8.2 kg of SPC interior floor covering per square metre covered in any building sectors cradle to gate.</p> <p>9.4 kg of LLT interior floor covering per square metre covered in any building sectors cradle to gate.</p> |                                                                      |                     |
| Manufacturer Warranty           | 10-20 Years                                                                                                                                                                                                                                                                                                                               |                                                                      |                     |
| Manufacturing Site              | No.168, Baichuan Road, Haian City, Jiangsu Province, P.R.China                                                                                                                                                                                                                                                                            |                                                                      |                     |
| Site Representation & Geography | Jiangsu, China                                                                                                                                                                                                                                                                                                                            |                                                                      |                     |
| Substances Of Very High Concern | None                                                                                                                                                                                                                                                                                                                                      |                                                                      |                     |
| Test Reports                    | Standard/Certification                                                                                                                                                                                                                                                                                                                    | Status                                                               | Last Date Completed |
|                                 | EN 13501-1:2018 Fire                                                                                                                                                                                                                                                                                                                      | Br-s1, passed                                                        | Apr 12, 2023        |
|                                 | ISO 8301:1991 Thermal Conductivity and Thermal Resistance                                                                                                                                                                                                                                                                                 | <0.05 m <sup>2</sup> K/W, passed                                     | Apr 11, 2023        |
|                                 | EN 717-1:2004 Formaldehyde Emission (In air)                                                                                                                                                                                                                                                                                              | 0.080 mg/m <sup>3</sup>                                              | Apr 14, 2023        |
|                                 | BS EN 14041:2018 Annex B & EN12673:1999 Pentachlorophenol (PCP)                                                                                                                                                                                                                                                                           | 0.1 mg/kg                                                            | Apr 11, 2023        |
|                                 | ASTM E 648-19ae1 Critical Radiant Flux                                                                                                                                                                                                                                                                                                    | Min. 1.1W/cm <sup>2</sup> (Class I > 0.45W/cm <sup>2</sup> ), passed | Apr 11, 2023        |
|                                 | ASTM E662-21ae1 Smoke Density                                                                                                                                                                                                                                                                                                             | Smoke density of <450                                                | Apr 11, 2023        |
|                                 | EN ISO 26987:2012 Determination of Staining and Resistance to Chemical                                                                                                                                                                                                                                                                    | Index 0 (Not affected, passed)                                       | May 31, 2024        |
|                                 | EN ISO 105-B02:2014 Colour Fastness to Light                                                                                                                                                                                                                                                                                              | ≥6, passed                                                           | Jun 05, 2024        |



|  |                                                     |                                      |              |
|--|-----------------------------------------------------|--------------------------------------|--------------|
|  | ISO 4918:2016+Amd.1:2018 Castor Chair Test          | No visible damage after 25000 cycles | May 31, 2024 |
|  | EN 1815:2016 Method A Static electricity propensity | <=2.0KV, passed                      | May 31, 2024 |
|  | BS 7976-2:2002+A1:2013 Slip Test                    | KT 2.0 #19 emboss Dry 60, Wet 32     | May 31, 2024 |
|  | DIN EN 16165:2023-02 Annex B Slip Test              | KT 2.0 #19 emboss R9 (9.7 drgree)    | May 31, 2024 |
|  | EN 14372:2004 Diisononyl Phthalate (DINP)           | 0.010%                               | Jun 11, 2024 |
|  | EN 71-3:2019+A1:2021 Migration of certain elements  | Passed                               | Jun 11, 2024 |
|  | SGS In-House method SVHC Test                       | SVHC ≤ 0.1%(w/w), passed             | Jun 11, 2024 |
|  | ISO 16000-9:2006 / Cor1:2007                        | TVOC, SVOC not detectable            | Jul 05, 2024 |
|  | EN 660-2:1999+A1:2003 Wear Resistance               | Passed, Group T                      | May 31, 2024 |

Table 1 LVT resilient flooring specifications

| Attribute         | Comment                | Date    |
|-------------------|------------------------|---------|
| Density           | 1800 kg/m <sup>3</sup> |         |
| Product thickness | 2mm                    | 2024.02 |
| Product weight    | 3.8 kg/m <sup>2</sup>  |         |

Table 2 SPC resilient flooring specifications

| Attribute         | Comment                | Date    |
|-------------------|------------------------|---------|
| Density           | 2000 kg/m <sup>3</sup> |         |
| Product thickness | 4mm                    | 2024.02 |
| Product weight    | 8.2 kg/m <sup>2</sup>  |         |

Table 3 LLT resilient flooring specifications

| Attribute         | Comment                | Date    |
|-------------------|------------------------|---------|
| Density           | 1900 kg/m <sup>3</sup> |         |
| Product thickness | 5mm                    | 2024.02 |
| Product weight    | 9.4 kg/m <sup>2</sup>  |         |

Note: Specifications are current at time of publishing. Please check currency of specifications with manufacturer.

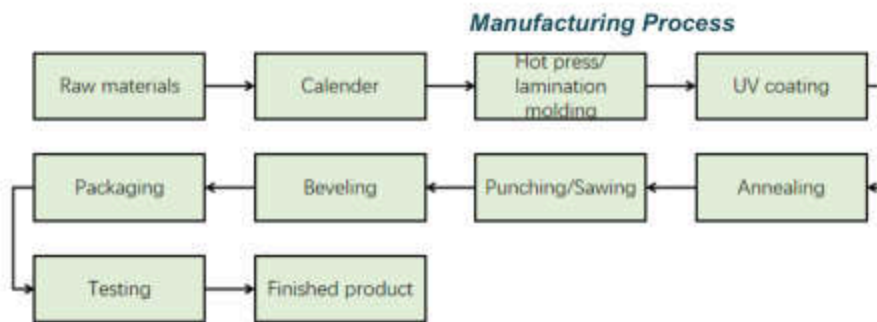


Figure 2. LVT&LLT resilient flooring Cradle to Gate System Boundary

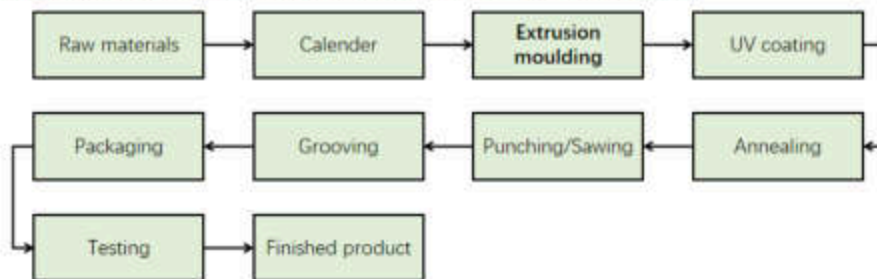


Figure 3. SPC resilient flooring Cradle to Gate System Boundary

### Product Components

In product content listed below the % mass has a  $\pm 5\%$  range and a confidence interval that is 90% certain to contain true population means at any time. Listing such  $90\pm 5\%$  certainty considers normal resource acquisition, supply chain, sedimentation, seasonal, manufacturing and product colour variation over this EPD's 5-year validity period. This also allows for intellectual property protection whilst ensuring fullest possible transparency.

Table 4 List of key components and additives by function, type, key operation, source and amount

| Function       | Component                           | Source        | Amount |
|----------------|-------------------------------------|---------------|--------|
| Fillers        | CaCO <sub>3</sub>                   | China         | 50-75% |
| Binder         | PVC                                 | Taiwan, China | 35-45% |
| Plasticizer    | DOTP                                | China         | 10-15% |
| Backing        | PRINTING FILMS                      | China         | 0-5%   |
| Backing        | Glass fiber (only for LLT flooring) | China         | 0-5%   |
| Stabilizer     | Calcium stearate                    | China         | 0-5%   |
| Pigment        | Carbon black                        | China         | 0-5%   |
| Pigment        | UV Lacquer.                         | China         | 0-5%   |
| <b>Packing</b> |                                     |               |        |
| Pallet         | Wood pallet                         | China         | 55-60% |
| Cardboard caps | Cardboard box                       | China         | 35-40% |
| PET            | PET Wrapping Film and Strapping     | China         | 0-5%   |



## Program Description

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Product Stages Included</b> | <p>A1 Raw material supply<br/>Raw material acquisition, extraction, refining and processing<br/>Electricity generated from all sources with extraction, refining &amp; transport</p> <p>A2 Transport internal and to the factory gate</p> <p>A3 Manufacture of product, co-products and plus packaging<br/>Production of ancillary material<br/>System flows leaving at end-of-waste boundary allocated as co-products</p> <p>C1 Deconstruction demolition<br/>C2 Transport to waste processing<br/>C3 Waste processing for reuse, recovery and/or recycling<br/>C4 Disposal<br/>D Reuse, recovery and/or recycling potentials, expressed as net impacts and benefits.</p>                                                                                                                                                                                                                                                                                                                                      |
| <b>Cut off criteria</b>        | <p>In this study, waste transport during the production phase is not considered, as the volume of waste is small and the transport distance is less than 100km.</p> <p>Since the system boundary of this study does not include the CONSTRUCTION PROCESS STAGE of A4-A5, the final disposal of the packaging was not considered in the study.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Data collection Year</b>    | 2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Background Data</b>         | Table 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Allocations Method</b>      | <p>For production stage allocation be distinguished between multi-input and multi-output processes</p> <ul style="list-style-type: none"> <li>- Multi-input processes: Allocation is based on physical properties and is based on weight. For example, a variety of products flooring products are produced in one factory. The consumption of the target products is obtained by dividing the total annual production weight of each product by the total weight of all the products produced in the factory, obtaining the weight ratio of target product, and then multiplying by the total data.</li> <li>- Multi-output processes: The production process generates economically valuable co-products, and its price is very different from the price of the target product (the difference about 13 times), so economic allocation is used for the co-products. Other outputs, such as solid waste, etc. follow mass allocation.</li> </ul> <p>No allocation is required for products at end-of-life.</p> |



**Scenario  
Modelling  
Assumption**

Stage A - production stage:

1. The raw material Stabilizer (calcium stearate) was used in the production of 3 PVC flooring in an amount of less than 1% which was not in the background database, so they were substituted with stearic acid from Econinvent database.

2. According to Changlong's statement, the subsequent disposal of the Off cut generated in the production stage is usually to landfill, but in Changlong's plant the Off cut is directly reused by the production line instead of being disposed of in a landfill, which is a typical closed loop, so there is no need to make any allocation. On the other hand, theoretically, the environmental benefits of not disposing to landfill should be considered, but here according to EN 15804 section 6.3.5.2, 'Therefore, as a general rule, potential loads or benefits from A1-A3 do not appear in module D.' and to be conservative, the benefits of Off cut reuse and subsequent landfill avoidance are not considered in this study.

Stage C - end of life: it is assumed that the product be deconstructed manually and transported from building site to waste processing is 161 km (100 miles) by diesel-powered truck (unspecified).

Stage D - benefits and loads beyond the system boundary: PVC flooring is typically not reused or recycled following removal from a building. Thus, reuse, recycling, and energy recovery are not applicable for this product.

**Table 5 Data sources for LVT, LLT, SPC flooring**

| Component                     | Material Description            | Material Dataset                                               | Data Source | Publication Date |
|-------------------------------|---------------------------------|----------------------------------------------------------------|-------------|------------------|
| <b>LVT, LLT, SPC flooring</b> |                                 |                                                                |             |                  |
| <b>PVC</b>                    | polyvinyl chloride              | polyvinylchloride production, suspension polymerisation        | EI 3.9.1    | 2022             |
| <b>Filler</b>                 | calcium carbonate stone powder  | calcium carbonate, precipitated                                | EI 3.9.1    | 2022             |
| <b>Stabilizer</b>             | calcium stearate                | stearic acid                                                   | EI 3.9.1    | 2022             |
| <b>UV LACQUER</b>             | polyurethane acrylic resin      | acrylic varnish production, with water, in 53% solution state  | EI 3.9.1    | 2022             |
| <b>DOTP</b>                   | dioctyl terephthala             | market for dioctyl terephthalate                               | EI 3.9.1    | 2022             |
| <b>Pigment</b>                | carbon black                    | market for carbon black                                        | EI 3.9.1    | 2022             |
| <b>Pigment Paste</b>          | pigment mixture                 | market for toner, colour, powder                               | EI 3.9.1    | 2022             |
| <b>Printing films</b>         | polypropylene film              | polypropylene production, granulate                            | EI 3.9.1    | 2022             |
| <b>Glass fibre</b>            | glass fibre                     | market for glass fibre                                         | EI 3.9.1    | 2022             |
| <b>Packing</b>                |                                 |                                                                |             |                  |
| <b>Cardboard caps</b>         | cardboard                       | market for folding boxboard carton                             | EI 3.9.1    | 2022             |
| <b>Wooden pallet</b>          | wood                            | market for furniture, wooden                                   | EI 3.9.1    | 2022             |
| <b>PET</b>                    | PET wrapping film and strapping | polyethylene terephthalate production, granulate, bottle grade | EI 3.9.1    | 2022             |



### Transportation

|                                                      |                      |                                        |          |      |
|------------------------------------------------------|----------------------|----------------------------------------|----------|------|
| <b>Transportation of raw and auxiliary materials</b> | truck transportation | Transport, freight, lorry, unspecified | EI 3.9.1 | 2022 |
| <b>Transportation of waste flooring</b>              | truck transportation | transport, freight, lorry, unspecified | EI 3.9.1 | 2022 |

### Energy

|                                           |                                                                   |                                                                                 |          |      |
|-------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------|----------|------|
| <b>Grid Electricity for producing</b>     | Electricity production, transmission                              | East China power grid, transmission to the user                                 | CLCD 0.9 | 2020 |
| <b>Solar Electricity for producing</b>    | Factory self-generated solar electricity production, transmission | electricity production, photovoltaic, 570kWp open ground installation, multi-Si | EI 3.9.1 | 2022 |
| <b>Natural gas for producing</b>          | Natural gas                                                       | Natural gas acquisition                                                         | CLCD 0.9 | 2020 |
| <b>Steam for producing</b>                | 0.8MPa steam                                                      | Steam (1Mpa, 183°C) production                                                  | CLCD 0.9 | 2020 |
| <b>Diesel oil for warehouse forklifts</b> | Diesel oil                                                        | Diesel oil acquisition (Market average)                                         | CLCD 0.9 | 2020 |

### Waste treatment

|                         |                                                                   |                                                        |          |      |
|-------------------------|-------------------------------------------------------------------|--------------------------------------------------------|----------|------|
| <b>Landfilled floor</b> | flooring                                                          | treatment of waste plastic, mixture, sanitary landfill | EI 3.9.1 | 2022 |
| <b>Hazardous waste</b>  | spray tower sludge, cloth and gloves containing color paste, etc. | market for hazardous waste, for underground deposit    | EI 3.9.1 | 2022 |

### Data Quality Assessment

The data quality assessment addressed the following parameters: time-related coverage, geographical coverage, technological coverage, precision, completeness, representativeness, consistency, reproducibility, sources of data, and uncertainty.

**Table 6 Data quality assessment for the Product Name product system**

| Data Quality Parameter                                                                                                       | Data Quality Discussion                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Time-Related Coverage:<br>Age of data and the minimum length of time over which data is collected                            | The most recent available data are used, based on other considerations such as data quality and similarity to the actual operations. Typically, these data are less than 2 years old (typically 2022). Manufacturer-supplied data (primary data) are based on half annualized production for 2023.                                                                                                    |
| Geographical Coverage:<br>Geographical area from which data for unit processes is collected to satisfy the goal of the study | The data used in the analysis provides the best possible representation available with current data. Electricity use for product manufacture is modeled using representative data for China. Surrogate data used in the assessment are representative of global or rest of world operations. Data representative of rest of world operations are considered sufficiently similar to actual processes. |
| Technology Coverage:<br>Specific technology or                                                                               | For the most part, data is representative of the actual technologies used for processing, transportation, and manufacturing operations.                                                                                                                                                                                                                                                               |



|                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| technology mix                                                                                                                                                                                             | Representative fabrication datasets, specific to the type of material, are used to represent the actual processes, as appropriate.                                                                                                                                                                                                                                                                                                                                                   |
| Precision:<br>Measure of the variability of the data values for each data expressed                                                                                                                        | All relevant foreground data is primary data, which is collected from on-site reviewing and supported by professional data input document. The activity data of the enterprise are all from enterprise statistics or on-site measured data, with high precision.                                                                                                                                                                                                                     |
| Completeness:<br>Percentage of flow that is measured or estimated                                                                                                                                          | The LCA model included all known mass and energy flows for production of 3 PVC flooring. No known processes or activities contributing to more than 1% of the total environmental impact for each indicator are excluded.                                                                                                                                                                                                                                                            |
| Representativeness:<br>Qualitative assessment of the degree to which the data set reflects the true population of interest                                                                                 | In this study, for all background processes representative primary data input based on specific industry averages which derived from reliable databases and the data input for foreground processes all obtained from on-site product related precise investigation. Considerable deviation may exist among actual processes on a site-specific basis; however, such a determination would require detailed data collection throughout the supply chain back to resource extraction. |
| Consistency:<br>Qualitative assessment of whether the study methodology is applied uniformly to the various components of the analysis                                                                     | In order to figure out that the LCA methodology can be uniformly applied or not, various component's qualitative assessment is conducted. The primary data input provided by manufacturers is re-checked and recalculated.                                                                                                                                                                                                                                                           |
| Reproducibility:<br>Qualitative assessment of the extent to which information about the methodology and data values would allow an independent practitioner to reproduce the results reported in the study | Based on the description of data and assumptions used, this assessment would be reproducible by other practitioners. All assumptions, models, and data sources are documented.                                                                                                                                                                                                                                                                                                       |
| Sources of the Data:<br>Description of all primary and secondary data sources                                                                                                                              | Data representing energy use, raw and auxiliary material consumption, and emissions, etc. at CHANGLONG's facility in China represent an annual average and are considered of high quality due to the length of time over which these data are collected. For secondary LCI datasets, Ecoinvent v3.9.1 and CLCD 0.9 LCI data are used.                                                                                                                                                |
| Uncertainty of the Information:<br>Uncertainty related to data, models, and assumptions                                                                                                                    | Uncertainty related to materials in the 3 PVC flooring product is low. Actual supplier data for upstream operations was not available for all suppliers and the study relied upon the use of existing representative datasets. These datasets contained relatively recent data (<4 years).                                                                                                                                                                                           |

### LCA Scenarios

#### End of Life stages (C1 - C4, D)

At the end of life, for 3 PVC products, no emissions are generated during demolition (C1) while no waste processing (C3) is required for landfill disposal. Transportation of waste materials at end-of-life (C2) assumes a 161 km (100 miles) average distance to disposal. No recycling of the product



materials is assumed at end-of-life.

**Table 7 C1-C4 and D Scenario Information**

| Processes                         | Unit                                         | LVT Scenario Value | SPC Scenario Value | LLT Scenario Value |
|-----------------------------------|----------------------------------------------|--------------------|--------------------|--------------------|
| <b>Collection process by type</b> | kg collected separately                      | 3.8                | 8.2                | 9.4                |
|                                   | kg collected with mixed construction waste   | 0                  | 0                  | 0                  |
| <b>Recovery system by type</b>    | Kg for re-use                                | 0                  | 0                  | 0                  |
|                                   | Kg for recycling                             | 0                  | 0                  | 0                  |
|                                   | Kg for energy recovery                       | 0                  | 0                  | 0                  |
| <b>Safe disposal</b>              | Kg or product or material for final disposal | 3.8                | 8.2                | 9.4                |
| <b>transportation</b>             | km                                           | 161                | 161                | 161                |

#### **Additional Technical Information**

The environmental impact category indicators are also reported based on the CML-IA characterization factors according to EN15804.

No substances required to be reported as hazardous (as determined under the Resource Conservation and Recovery Act (RCRA (EPA, n.d.)) are associated with the production of flooring.

Please visit <https://www.changlongflor.com> for additional information regarding product.

#### **Additional Environmental Information**

The flooring products in this EPD comply with the Indoor Air Comfort GOLD requirements. Low VOC cleaning materials are available for use in maintaining flooring.

#### **Product Results**

**Table 8 LCA impact indicators, resource use, waste and other measured flows**

Acronyms, methods and units of impact potentials plus inventory inputs and outputs, are defined below:

| Impact Potentials       | Acronym              | Description of Methods       | Units                |
|-------------------------|----------------------|------------------------------|----------------------|
| Climate Change biogenic | GWP <sub>bio</sub>   | GWP biogenic [7]             | kg CO <sub>2eq</sub> |
| Climate Change luluc    | GWP <sub>luluc</sub> | GWP land use & change [7]    | kg CO <sub>2eq</sub> |
| Climate Change fossil   | GWP <sub>ff</sub>    | GWP fossil fuels [7]         | kg CO <sub>2eq</sub> |
| Climate Change total    | GWP <sub>t</sub>     | Global Warming Potential [7] | kg CO <sub>2eq</sub> |



| Stratospheric Ozone Depletion                                                                                       | ODP                  | Stratospheric Ozone Loss [8]       | kg CFC <sub>11eq</sub>           |
|---------------------------------------------------------------------------------------------------------------------|----------------------|------------------------------------|----------------------------------|
| Photochemical Ozone Creation                                                                                        | POCP                 | Summer Smog [9]                    | kg NMOC <sub>eq</sub>            |
| Acidification Potential                                                                                             | AP                   | Accumulated Exceedance [10]        | mol H <sup>+</sup> <sub>eq</sub> |
| Eutrophication Freshwater                                                                                           | EP <sub>fresh</sub>  | Excess nutrients freshwater [11]   | kg Po <sub>4</sub> <sub>eq</sub> |
| Eutrophication Marine                                                                                               | EP <sub>marine</sub> | Excess marine nutrients [11]       | kg N <sub>eq</sub>               |
| Eutrophication Terrestrial                                                                                          | EP <sub>land</sub>   | Excess Terrestrial nutrients [11]  | mol N <sub>eq</sub>              |
| Mineral & Metal Depletion <sup>1</sup>                                                                              | ADP <sub>min</sub>   | Abiotic Depletion minerals [12]    | kg Sb <sub>eq</sub>              |
| Fossil Fuel Depletion <sup>1</sup>                                                                                  | ADP <sub>ff</sub>    | Abiotic Depletion fossil fuel [13] | MJ <sub>ncv</sub>                |
| Water Depletion <sup>1</sup>                                                                                        | WDP                  | Water Deprivation Scarcity [14,15] | m <sup>3</sup> <sub>WDP eq</sub> |
| Particulate Matter Emissions                                                                                        | PM                   | SETAC-UNE [26]                     | Disease incidence                |
| Ionizing Radiation, Human Health <sup>2</sup>                                                                       | IRP                  | Human health effect model [27]     | kBq U235 eq                      |
| Eco-toxicity (freshwater) <sup>1</sup>                                                                              | ETP-fw               | USEtox [28]                        | CTU <sub>e</sub>                 |
| Human toxicity, cancer effects <sup>1</sup>                                                                         | HTP-c                | USEtox [28]                        | CTU <sub>h</sub>                 |
| Human toxicity, non-cancer effects <sup>1</sup>                                                                     | HTP-nc               | USEtox [28]                        | CTU <sub>h</sub>                 |
| Land use related impacts/ Soil quality <sup>1</sup>                                                                 | SQP                  | Soil quality index                 | dimensionless                    |
| Resource Use                                                                                                        | Acronym              | Units                              |                                  |
| Use of renewable primary energy excluding renewable primary energy resources used as raw materials                  | PERE                 | MJ <sub>NCV</sub>                  |                                  |
| Use of renewable primary energy resources used as raw materials                                                     | PERM                 | MJ <sub>NCV</sub>                  |                                  |
| Total use of renewable primary energy resources (primary energy and primary energy resources used as raw materials) | PERT                 | MJ <sub>NCV</sub>                  |                                  |
| Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials          | PENRE                | MJ <sub>NCV</sub>                  |                                  |
| Use of non-renewable primary energy resources used as raw materials                                                 | PENRM                | MJ <sub>NCV</sub>                  |                                  |

<sup>1</sup> The results of this environmental impact indicator shall be used with care as uncertainties on these results are high or as there is limited experience with the indicator.

<sup>2</sup> This impact category deals mainly with the eventual impact of low dose ionizing radiation on human health of the nuclear fuel cycle. It does not consider effects due to possible nuclear accidents, occupational exposure nor due to radioactive waste disposal in underground facilities. Potential ionizing radiation from the soil, from radon and from some construction materials is also not measured by this indicator.



|                                                                                                                         |                |                   |
|-------------------------------------------------------------------------------------------------------------------------|----------------|-------------------|
| Total use of non-renewable primary energy resources (primary energy and primary energy resources used as raw materials) | PENRT          | MJ <sub>NCV</sub> |
| Use of secondary material                                                                                               | SM             | kg                |
| Use of renewable secondary fuels                                                                                        | RSF            | MJ <sub>NCV</sub> |
| Use of non-renewable secondary fuels                                                                                    | NRSF           | MJ <sub>NCV</sub> |
| Use of net fresh water                                                                                                  | FW             | m <sup>3</sup>    |
| <b>Waste Type</b>                                                                                                       | <b>Acronym</b> | <b>Units</b>      |
| Hazardous waste disposed                                                                                                | HWD            | kg                |
| Non-hazardous waste disposed                                                                                            | NHWD           | kg                |
| Radioactive waste disposed                                                                                              | RWD            | kg                |
| <b>Other Outputs</b>                                                                                                    | <b>Acronym</b> | <b>Units</b>      |
| Components for re-use                                                                                                   | CRU            | kg                |
| Materials for recycling                                                                                                 | MFR            | kg                |
| Materials for energy recovery                                                                                           | MER            | kg                |
| Exported energy                                                                                                         | EE             | Mj <sub>pec</sub> |

Note: MJ<sub>NCV</sub> is MJ, net calorific value, Mj<sub>pec</sub> is Mj, per energy carrier



## Cradle to Gate + modules C1–C4 and module D Inventory

Table 9 Inventory Resource Use Results/1 m<sup>2</sup>-LVT flooring

|                       |                         | Product stage | End of life stage          |           |                  |          | Resource recovery stage    |
|-----------------------|-------------------------|---------------|----------------------------|-----------|------------------|----------|----------------------------|
|                       |                         | A1-A3         | C1                         | C2        | C3               | C4       | D                          |
| Module Codes          | Unit                    | Production    | De-construction demolition | Transport | Waste processing | Disposal | Reuse, Recovery, Recycling |
| GWP-total             | kg CO <sub>2</sub> eq   | 1.12E+01      | 0.00E+00                   | 5.83E-02  | 0.00E+00         | 3.85E-01 | 0.00E+00                   |
| GWP-luluc             | kg CO <sub>2</sub> eq   | 7.61E-03      | 0.00E+00                   | 3.04E-05  | 0.00E+00         | 3.75E-05 | 0.00E+00                   |
| GWP-biogenic          | kg CO <sub>2</sub> eq   | 2.13E-02      | 0.00E+00                   | 1.59E-05  | 0.00E+00         | 3.05E-05 | 0.00E+00                   |
| GWP-fossil            | kg CO <sub>2</sub> eq   | 1.12E+01      | 0.00E+00                   | 5.82E-02  | 0.00E+00         | 3.85E-01 | 0.00E+00                   |
| ADP-fossil            | MJ, net calorific value | 1.94E+02      | 0.00E+00                   | 8.37E-01  | 0.00E+00         | 1.00E+00 | 0.00E+00                   |
| ADP-minerals & metals | kg Sb eq.               | 7.96E-05      | 0.00E+00                   | 1.81E-07  | 0.00E+00         | 1.03E-07 | 0.00E+00                   |
| EP-freshwater         | kg P eq.                | 1.93E-03      | 0.00E+00                   | 4.83E-06  | 0.00E+00         | 6.77E-06 | 0.00E+00                   |
| POCP                  | kg NMVOC eq.            | 3.50E-02      | 0.00E+00                   | 3.82E-04  | 0.00E+00         | 5.36E-04 | 0.00E+00                   |
| AP                    | mol H+eq.               | 4.82E-02      | 0.00E+00                   | 2.77E-04  | 0.00E+00         | 3.45E-04 | 0.00E+00                   |
| EP-terrestrial        | mol N eq                | 9.77E-02      | 0.00E+00                   | 1.12E-03  | 0.00E+00         | 1.32E-03 | 0.00E+00                   |
| EP-marine             | kg N eq.                | 9.22E-03      | 0.00E+00                   | 1.05E-04  | 0.00E+00         | 8.13E-03 | 0.00E+00                   |
| ODP                   | kg CFC 11 eq.           | 4.83E-06      | 0.00E+00                   | 9.01E-10  | 0.00E+00         | 1.07E-09 | 0.00E+00                   |
| WDP                   | m <sup>3</sup> world eq | 3.16E+00      | 0.00E+00                   | 4.09E-03  | 0.00E+00         | 6.00E-03 | 0.00E+00                   |

See table 8 for additional information



Table 2 Inventory Resource Use Results/1 m<sup>2</sup>-LLT flooring

|                       |                         | Product stage | End of life stage          |           |                  |          | Resource recovery stage    |
|-----------------------|-------------------------|---------------|----------------------------|-----------|------------------|----------|----------------------------|
|                       |                         | A1-A3         | C1                         | C2        | C3               | C4       | D                          |
| Module Codes          | Unit                    | Production    | De-construction demolition | Transport | Waste processing | Disposal | Reuse, Recovery, Recycling |
| GWP-total             | kg CO2 eq               | 2.69E+01      | 0.00E+00                   | 1.44E-01  | 0.00E+00         | 9.53E-01 | 0.00E+00                   |
| GWP-luluc             | kg CO2 eq               | 1.83E-02      | 0.00E+00                   | 7.51E-05  | 0.00E+00         | 9.29E-05 | 0.00E+00                   |
| GWP-biogenic          | kg CO2 eq               | 5.05E-02      | 0.00E+00                   | 3.92E-05  | 0.00E+00         | 7.54E-05 | 0.00E+00                   |
| GWP-fossil            | kg CO2 eq               | 2.68E+01      | 0.00E+00                   | 1.44E-01  | 0.00E+00         | 9.53E-01 | 0.00E+00                   |
| ADP-fossil            | MJ, net calorific value | 4.63E+02      | 0.00E+00                   | 2.07E+00  | 0.00E+00         | 2.48E+00 | 0.00E+00                   |
| ADP-minerals & metals | kg Sb eq.               | 2.22E-04      | 0.00E+00                   | 4.48E-07  | 0.00E+00         | 2.55E-07 | 0.00E+00                   |
| EP-freshwater         | kg P eq.                | 4.64E-03      | 0.00E+00                   | 1.20E-05  | 0.00E+00         | 1.67E-05 | 0.00E+00                   |
| POCP                  | kg NMVOC eq.            | 8.42E-02      | 0.00E+00                   | 9.46E-04  | 0.00E+00         | 1.33E-03 | 0.00E+00                   |
| AP                    | mol H+eq.               | 1.16E-01      | 0.00E+00                   | 6.86E-04  | 0.00E+00         | 8.52E-04 | 0.00E+00                   |
| EP-terrestrial        | mol N eq                | 2.36E-01      | 0.00E+00                   | 2.78E-03  | 0.00E+00         | 3.27E-03 | 0.00E+00                   |
| EP-marine             | kg N eq.                | 2.22E-02      | 0.00E+00                   | 2.59E-04  | 0.00E+00         | 2.01E-02 | 0.00E+00                   |
| ODP                   | kg CFC 11 eq.           | 1.14E-05      | 0.00E+00                   | 2.23E-09  | 0.00E+00         | 2.64E-09 | 0.00E+00                   |
| WDP                   | m3 world eq             | 7.54E+00      | 0.00E+00                   | 1.01E-02  | 0.00E+00         | 1.48E-02 | 0.00E+00                   |

See table 8 for additional information



Table 3 Inventory Resource Use Results/1 m<sup>2</sup>-SPC flooring

|                       |                         | Product stage | End of life stage          |           |                  |          | Resource recovery stage    |
|-----------------------|-------------------------|---------------|----------------------------|-----------|------------------|----------|----------------------------|
|                       |                         | A1-A3         | C1                         | C2        | C3               | C4       | D                          |
| Module Codes          | Unit                    | Production    | De-construction demolition | Transport | Waste processing | Disposal | Reuse, Recovery, Recycling |
| GWP-total             | kg CO <sub>2</sub> eq   | 2.26E+01      | 0.00E+00                   | 1.26E-01  | 0.00E+00         | 8.32E-01 | 0.00E+00                   |
| GWP-luluc             | kg CO <sub>2</sub> eq   | 1.60E-02      | 0.00E+00                   | 6.55E-05  | 0.00E+00         | 8.10E-05 | 0.00E+00                   |
| GWP-biogenic          | kg CO <sub>2</sub> eq   | 4.47E-02      | 0.00E+00                   | 3.42E-05  | 0.00E+00         | 6.58E-05 | 0.00E+00                   |
| GWP-fossil            | kg CO <sub>2</sub> eq   | 2.25E+01      | 0.00E+00                   | 1.26E-01  | 0.00E+00         | 8.31E-01 | 0.00E+00                   |
| ADP-fossil            | MJ, net calorific value | 3.97E+02      | 0.00E+00                   | 1.81E+00  | 0.00E+00         | 2.17E+00 | 0.00E+00                   |
| ADP-minerals & metals | kg Sb eq.               | 1.68E-04      | 0.00E+00                   | 3.91E-07  | 0.00E+00         | 2.23E-07 | 0.00E+00                   |
| EP-freshwater         | kg P eq.                | 4.08E-03      | 0.00E+00                   | 1.04E-05  | 0.00E+00         | 1.46E-05 | 0.00E+00                   |
| POCP                  | kg NMVOC eq.            | 7.16E-02      | 0.00E+00                   | 8.25E-04  | 0.00E+00         | 1.16E-03 | 0.00E+00                   |
| AP                    | mol H <sup>+</sup> eq.  | 9.77E-02      | 0.00E+00                   | 5.99E-04  | 0.00E+00         | 7.43E-04 | 0.00E+00                   |
| EP-terrestrial        | mol N eq                | 1.97E-01      | 0.00E+00                   | 2.42E-03  | 0.00E+00         | 2.85E-03 | 0.00E+00                   |
| EP-marine             | kg N eq.                | 1.86E-02      | 0.00E+00                   | 2.26E-04  | 0.00E+00         | 1.75E-02 | 0.00E+00                   |
| ODP                   | kg CFC 11 eq.           | 1.02E-05      | 0.00E+00                   | 1.94E-09  | 0.00E+00         | 2.30E-09 | 0.00E+00                   |
| WDP                   | m <sup>3</sup> world eq | 6.66E+00      | 0.00E+00                   | 8.82E-03  | 0.00E+00         | 1.29E-02 | 0.00E+00                   |

See table 8 for additional information



**Table 4 Optional Indicators -LVT flooring**

|              |                   | Product stage | End of life stage          |           |                  |          | Resource recovery stage    |
|--------------|-------------------|---------------|----------------------------|-----------|------------------|----------|----------------------------|
|              |                   | A1-A3         | C1                         | C2        | C3               | C4       | D                          |
| Module Codes | Unit              | Production    | De-construction demolition | Transport | Waste processing | Disposal | Reuse, Recovery, Recycling |
| ETP-fw       | CTUe              | 5.09E+01      | 0.00E+00                   | 4.56E-01  | 0.00E+00         | 1.90E+00 | 0.00E+00                   |
| HTP-c        | CTUh              | 9.70E-09      | 0.00E+00                   | 3.07E-11  | 0.00E+00         | 2.88E-11 | 0.00E+00                   |
| HTP-nc       | CTUh              | 7.44E-07      | 0.00E+00                   | 6.52E-10  | 0.00E+00         | 1.05E-09 | 0.00E+00                   |
| SQP          | dimensionless     | 4.77E+01      | 0.00E+00                   | 6.23E-01  | 0.00E+00         | 2.21E+00 | 0.00E+00                   |
| PM           | Disease incidence | 2.41E-06      | 0.00E+00                   | 5.61E-09  | 0.00E+00         | 6.96E-09 | 0.00E+00                   |
| IRP          | kBq U235 eq       | 4.90E-01      | 0.00E+00                   | 7.46E-04  | 0.00E+00         | 1.57E-03 | 0.00E+00                   |

See table 8 for additional information

**Table 5 Optional Indicators -LLT flooring**

|              |                   | Product stage | End of life stage          |           |                  |          | Resource recovery stage    |
|--------------|-------------------|---------------|----------------------------|-----------|------------------|----------|----------------------------|
|              |                   | A1-A3         | C1                         | C2        | C3               | C4       | D                          |
| Module Codes | Unit              | Production    | De-construction demolition | Transport | Waste processing | Disposal | Reuse, Recovery, Recycling |
| ETP-fw       | CTUe              | 1.21E+02      | 0.00E+00                   | 1.13E+00  | 0.00E+00         | 4.71E+00 | 0.00E+00                   |
| HTP-c        | CTUh              | 2.32E-08      | 0.00E+00                   | 7.58E-11  | 0.00E+00         | 7.12E-11 | 0.00E+00                   |
| HTP-nc       | CTUh              | 1.77E-06      | 0.00E+00                   | 1.61E-09  | 0.00E+00         | 2.59E-09 | 0.00E+00                   |
| SQP          | dimensionless     | 1.13E+02      | 0.00E+00                   | 1.54E+00  | 0.00E+00         | 5.46E+00 | 0.00E+00                   |
| PM           | Disease incidence | 5.71E-06      | 0.00E+00                   | 1.39E-08  | 0.00E+00         | 1.72E-08 | 0.00E+00                   |
| IRP          | kBq U235 eq       | 1.18E+00      | 0.00E+00                   | 1.85E-03  | 0.00E+00         | 3.89E-03 | 0.00E+00                   |

See table 8 for additional information



Table 6 Optional Indicators -SPC flooring

| Module Codes | Unit              | Product stage | End of life stage          |           |                  |          | Resource recovery stage    |
|--------------|-------------------|---------------|----------------------------|-----------|------------------|----------|----------------------------|
|              |                   | A1-A3         | C1                         | C2        | C3               | C4       | D                          |
|              |                   | Production    | De-construction demolition | Transport | Waste processing | Disposal | Reuse, Recovery, Recycling |
| ETP-fw       | CTUe              | 1.02E+02      | 0.00E+00                   | 9.84E-01  | 0.00E+00         | 4.10E+00 | 0.00E+00                   |
| HTP-c        | CTUh              | 1.90E-08      | 0.00E+00                   | 6.62E-11  | 0.00E+00         | 6.21E-11 | 0.00E+00                   |
| HTP-nc       | CTUh              | 1.40E-06      | 0.00E+00                   | 1.41E-09  | 0.00E+00         | 2.26E-09 | 0.00E+00                   |
| SQP          | dimensionless     | 9.69E+01      | 0.00E+00                   | 1.34E+00  | 0.00E+00         | 4.77E+00 | 0.00E+00                   |
| PM           | Disease incidence | 4.54E-06      | 0.00E+00                   | 1.21E-08  | 0.00E+00         | 1.50E-08 | 0.00E+00                   |
| IRP          | kBq U235 eq       | 1.01E+00      | 0.00E+00                   | 1.61E-03  | 0.00E+00         | 3.39E-03 | 0.00E+00                   |

See table 8 for additional information

Table 7 Resource Use and other environmental flows -LVT

| Resource Use | Unit              | Product stage | End of life stage          |           |                  |          | Resource recovery stage    |
|--------------|-------------------|---------------|----------------------------|-----------|------------------|----------|----------------------------|
|              |                   | A1-A3         | C1                         | C2        | C3               | C4       | D                          |
|              |                   | Production    | De-construction demolition | Transport | Waste processing | Disposal | Reuse, Recovery, Recycling |
| PERE         | MJ <sub>NCV</sub> | 8.98E+00      | 0.00E+00                   | 2.08E-02  | 0.00E+00         | 1.10E-02 | MNR                        |
| PERM         | MJ <sub>NCV</sub> | 5.02E+00      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| PERT         | MJ <sub>NCV</sub> | 1.40E+01      | 0.00E+00                   | 2.08E-02  | 0.00E+00         | 1.10E-02 | MNR                        |
| PENRE        | MJ <sub>NCV</sub> | 1.36E+02      | 0.00E+00                   | 1.00E+00  | 0.00E+00         | 8.37E-01 | MNR                        |
| PENRM        | MJ <sub>NCV</sub> | 5.80E+01      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| PENRT        | MJ <sub>NCV</sub> | 1.94E+02      | 0.00E+00                   | 1.00E+00  | 0.00E+00         | 8.37E-01 | MNR                        |
| SM           | kg                | 0.00E+00      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| RSF          | MJ <sub>NCV</sub> | 3.82E+00      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| NRSF         | MJ <sub>NCV</sub> | 2.68E+01      | 0.00E+00                   | 4.86E-01  | 0.00E+00         | 0.00E+00 | MNR                        |
| FW           | m <sup>3</sup>    | 5.06E-02      | 0.00E+00                   | 5.35E-05  | 0.00E+00         | 1.56E-04 | MNR                        |
| Waste        | Unit              |               |                            |           |                  |          |                            |
| HWD          | kg                | 6.70E-03      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |



|                |                   |          |          |          |          |          |     |
|----------------|-------------------|----------|----------|----------|----------|----------|-----|
| NHWD           | kg                | 4.72E-01 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | MNR |
| RWD            | kg                | 5.17E-05 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | MNR |
| <b>Outputs</b> | <b>Unit</b>       |          |          |          |          |          |     |
| CRU            | kg                | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | MNR |
| MFR            | kg                | 4.60E-01 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | MNR |
| MER            | kg                | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | MNR |
| EE             | M <sub>Jpec</sub> | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | MNR |

See table 8 for additional information

**Table 8 Resource Use and other environmental flows -LLT**

|                     |                   | Product stage | End of life stage          |           |                  |          | Resource recovery stage    |
|---------------------|-------------------|---------------|----------------------------|-----------|------------------|----------|----------------------------|
|                     |                   | A1-A3         | C1                         | C2        | C3               | C4       | D                          |
|                     |                   | Production    | De-construction demolition | Transport | Waste processing | Disposal | Reuse, Recovery, Recycling |
| <b>Resource Use</b> | <b>Unit</b>       |               |                            |           |                  |          |                            |
| PERE                | MJ <sub>NCV</sub> | 2.14E+01      | 0.00E+00                   | 2.71E-02  | 0.00E+00         | 5.14E-02 | MNR                        |
| PERM                | MJ <sub>NCV</sub> | 1.19E+01      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| PERT                | MJ <sub>NCV</sub> | 3.33E+01      | 0.00E+00                   | 2.71E-02  | 0.00E+00         | 5.14E-02 | MNR                        |
| PENRE               | MJ <sub>NCV</sub> | 3.24E+02      | 0.00E+00                   | 2.07E+00  | 0.00E+00         | 2.48E+00 | MNR                        |
| PENRM               | MJ <sub>NCV</sub> | 1.39E+02      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| PENRT               | MJ <sub>NCV</sub> | 4.63E+02      | 0.00E+00                   | 2.07E+00  | 0.00E+00         | 2.48E+00 | MNR                        |
| SM                  | kg                | 0.00E+00      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| RSF                 | MJ <sub>NCV</sub> | 9.43E+00      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| NRSF                | MJ <sub>NCV</sub> | 6.64E+01      | 0.00E+00                   | 1.20E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| FW                  | m <sup>3</sup>    | 1.22E-01      | 0.00E+00                   | 1.32E-04  | 0.00E+00         | 3.86E-04 | MNR                        |
| <b>Waste</b>        | <b>Unit</b>       |               |                            |           |                  |          |                            |
| HWD                 | kg                | 1.87E-02      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| NHWD                | kg                | 1.17E+00      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| RWD                 | kg                | 5.17E-05      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| <b>Outputs</b>      | <b>Unit</b>       |               |                            |           |                  |          |                            |
| CRU                 | kg                | 0.00E+00      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| MFR                 | kg                | 1.14E+00      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| MER                 | kg                | 0.00E+00      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| EE                  | M <sub>Jpec</sub> | 0.00E+00      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |

See table 8 for additional information



Table 9 Resource Use and other environmental flows -SPC

|              |                   | Product stage | End of life stage          |           |                  |          | Resource recovery stage    |
|--------------|-------------------|---------------|----------------------------|-----------|------------------|----------|----------------------------|
|              |                   | A1-A3         | C1                         | C2        | C3               | C4       | D                          |
|              |                   | Production    | De-construction demolition | Transport | Waste processing | Disposal | Reuse, Recovery, Recycling |
| Resource Use | Unit              |               |                            |           |                  |          |                            |
| PERE         | MJ <sub>NCV</sub> | 1.78E+01      | 0.00E+00                   | 2.36E-02  | 0.00E+00         | 4.48E-02 | MNR                        |
| PERM         | MJ <sub>NCV</sub> | 1.07E+01      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| PERT         | MJ <sub>NCV</sub> | 2.85E+01      | 0.00E+00                   | 2.36E-02  | 0.00E+00         | 4.48E-02 | MNR                        |
| PENRE        | MJ <sub>NCV</sub> | 2.74E+02      | 0.00E+00                   | 1.81E+00  | 0.00E+00         | 2.17E+00 | MNR                        |
| PENRM        | MJ <sub>NCV</sub> | 1.23E+01      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| PENRT        | MJ <sub>NCV</sub> | 3.97E+02      | 0.00E+00                   | 1.81E+00  | 0.00E+00         | 2.17E+00 | MNR                        |
| SM           | kg                | 0.00E+00      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| RSF          | MJ <sub>NCV</sub> | 7.34E+00      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| NRSF         | MJ <sub>NCV</sub> | 5.17E+01      | 0.00E+00                   | 1.05E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| FW           | m <sup>3</sup>    | 1.01E-01      | 0.00E+00                   | 1.15E-04  | 0.00E+00         | 3.37E-04 | MNR                        |
| Waste        | Unit              |               |                            |           |                  |          |                            |
| HWD          | kg                | 1.78E-02      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| NHWD         | kg                | 1.02E+00      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| RWD          | kg                | 3.07E-04      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| Outputs      | Unit              |               |                            |           |                  |          |                            |
| CRU          | kg                | 0.00E+00      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| MFR          | kg                | 1.14E+00      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| MER          | kg                | 0.00E+00      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |
| EE           | MJ <sub>pec</sub> | 0.00E+00      | 0.00E+00                   | 0.00E+00  | 0.00E+00         | 0.00E+00 | MNR                        |

See table 8 for additional information

Table 18 Biogenic Carbon at Factory Gate (A1-A3)

| Biogenic Carbon                         | Unit              | LVT      | SPC    | LLT    |
|-----------------------------------------|-------------------|----------|--------|--------|
| Biogenic carbon content in product      | Kg C <sup>3</sup> | 0        | 0      | 0      |
| Biogenic carbon content in accompanying | Kg C              | 0.309375 | 0.6788 | 0.7587 |

<sup>3</sup> 1 kg biogenic carbon is equivalent to 44/12 kg of CO<sub>2</sub>.



## Interpretation

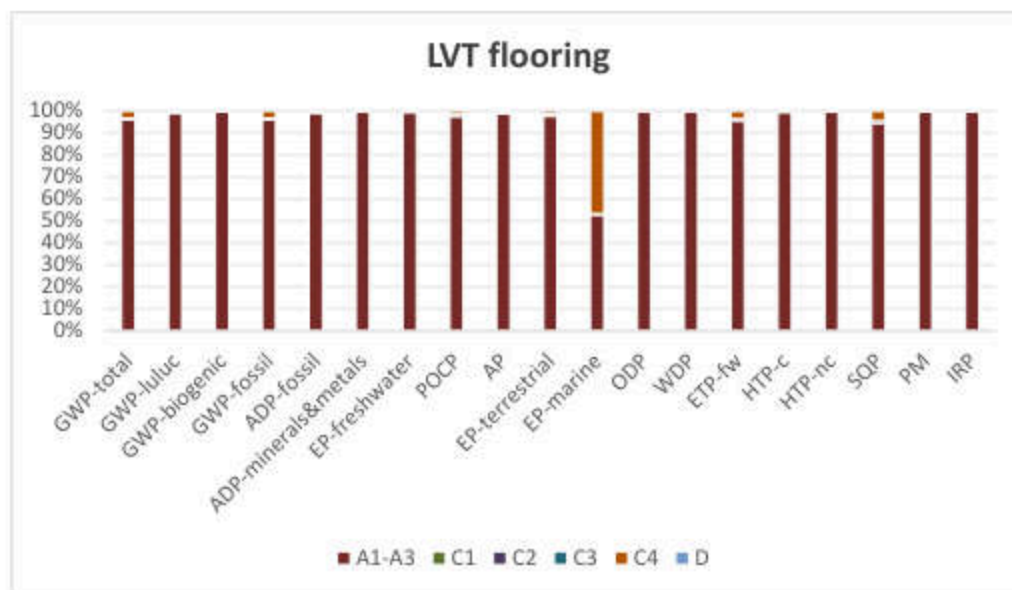


Figure 4. LVT flooring each stage contribution to LCA results

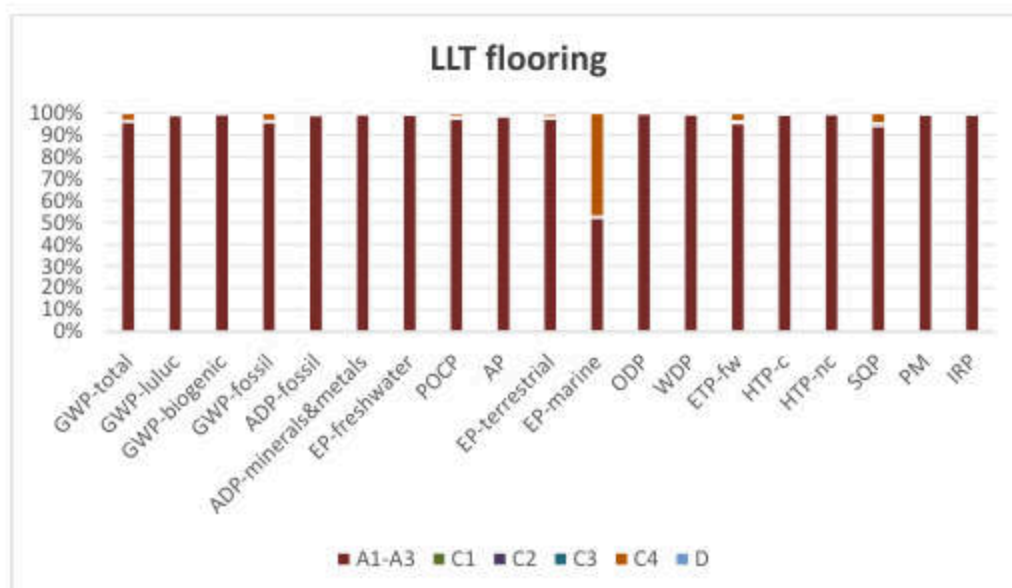


Figure 5. LLT flooring each stage contribution to LCA results

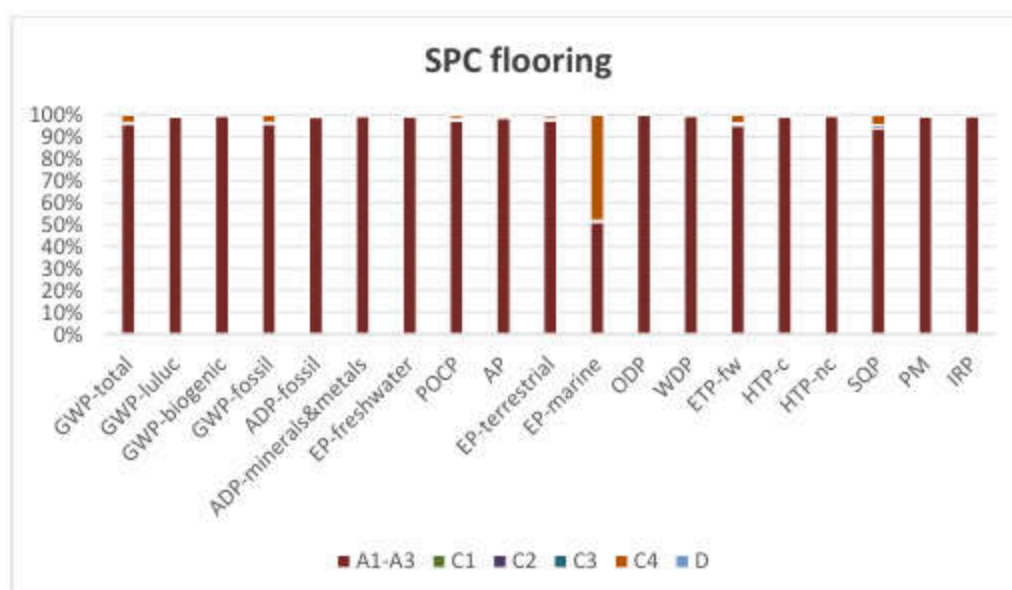


Figure 6. SPC each stage contribution to LCA results

For the indicator EP-marine, the high contribution of the C4 phase compared to other indicators is due to direct emissions such as total organic carbon during waste disposal (treatment of waste plastic, mixture, sanitary landfill).

The wastewater generated from the used water is divided into two parts. One part needs to be discharged after WWT wastewater treatment, and the other part is directly discharged into the municipal wastewater system. Therefore this part is not included in the inventory.

The LCA study has been carried out based on available data, information, regional and global knowledge and experience to achieve more possible accuracy, completeness and representative of the results.



### References for this EPD

1. Ecoinvent, Switzerland. Ecoinvent database. <http://www.ecoinvent.org/>
2. EN 15804:2012+A2:2019: Sustainability of construction works - Environmental product declarations - Core rules for the product category of construction products, CEN
3. ISO 14025:2006 Environmental labelling & declarations Type III EPDs Principles & procedures ISO 14031:1999 EM: Environmental performance evaluation: Guidelines
4. ISO 14040:2006: Life cycle assessment (LCA): Principles & framework
5. ISO 14044:2006: LCA: Requirement & guideline for data review: LCI; LCIA, Interpretation results
6. Global GreenTag International. Product Category Rules. Sub-PCR FC:2022 V1 Interior Floor Coverings V1. <http://www.globalgreentag.com/greentag-epd-program>
7. Global GreenTag International. General Program Instructions Version 2.1. <http://www.globalgreentag.com/greentag-epd-program>
8. Product Environmental Footprint Category Rules Guidance, Version 6.3, May 2018
9. IKEA, 2012a. Chinese Life Cycle Database-CLCD accessed in March 2015  
<<http://www.ike-global.com/products-2/chinese-lca-database-clcd>>

Certificate CN22/00002574



The management system of

## JIANGSU CHANGLONG DECORATIVE MATERIAL TECHNICAL CO., LTD.

Unified Social Credit Code: 91320621MA1P76946C

Business Registration Address: No. 168, Baichuan Road, Haian City

Business Operation Address: No. 168, Baichuan Road, Haian City, Jiangsu Province, P.R.  
China

has been assessed and certified as meeting the requirements of

**ISO 14001:2015**

For the following activities

Manufacture of PVC (polyvinyl chloride) flooring tiles

This certificate is valid from 26 August 2025 until 25 August 2028 and remains valid subject to satisfactory  
surveillance audits.

Issue 2. Certified since 26 August 2022

*L. Moran*

Authorised by

Liz Moran

Business Manager

SGS United Kingdom Ltd

Rossmore Business Park, Ellesmere Port, Cheshire, CH65 3EN, UK

t +44 (0)151 350-6666 - [www.sgs.com](http://www.sgs.com)



This document is an authentic electronic certificate for Client business purposes use only. Printed version of the electronic certificate are permitted and will be considered as a copy. This document is issued by the Company subject to SGS General Conditions of certification services available on [Terms and Conditions](#) | SGS. Attention is drawn to the limitation of liability, indemnification and jurisdictional clauses contained therein. This document is copyright protected and any unauthorized alteration, forgery or falsification of the content or appearance of this document is unlawful.



此为证书 CN22/00002574 译本

下述组织

## 江苏长隆装饰材料科技有限公司

统一社会信用代码: 91320621MA1P76946C

注册地址: 海安市百川路 168 号

经营地址: 中国江苏省海安市百川路 168 号

的管理体系已经过审核, 并被证明符合下述要求

**ISO 14001:2015**

所涉及的活动范围覆盖

PVC(聚氯乙烯)地板的制造

该证书的有效期自 2025 年 08 月 26 日 至 2028 年 08 月 25 日 并须经过符合要求的监督审核保持有效  
版本号 2. 初始注册日期 2022 年 08 月 26 日

*L. Moran*

签署

Liz Moran

Business Manager

SGS United Kingdom Ltd

Rossmore Business Park, Ellesmere Port, Cheshire, CH65 3EN, UK

t: +44 (0)151 350-6666 ~ [www.sgs.com](http://www.sgs.com)



本文件是真实的电子版证书, 仅供客户用于其商业用途。客户可自行打印, 视同副本。本文件根据 [Terms and Conditions | SGS](#) 中认证服务通用条款的要求颁发。请注意其中已包含的责任范围、赔偿和司法管辖事项。本文件受 SGS 版权保护。任何未经授权的对此文件的内容或外观的变更, 伪造或篡改皆属非法。



Certificate CN22/00002463



The management system of

## JIANGSU CHANGLONG DECORATIVE MATERIAL TECHNICAL CO., LTD.

Unified Social Credit Code: 91320621MA1P76946C

Business Registration Address: No. 168, Baichuan Road, Haian City

Business Operation Address: No. 168, Baichuan Road, Haian City, Jiangsu Province, P.R.  
China

has been assessed and certified as meeting the requirements of

**ISO 9001:2015**

For the following activities

Manufacture of PVC (polyvinyl chloride) flooring tiles

This certificate is valid from 22 August 2025 until 21 August 2028 and remains valid subject to satisfactory surveillance audits.

Issue 2. Certified since 22 August 2022

*L. Moran*

Authorised by

Liz Moran

Business Manager

SGS United Kingdom Ltd

Rossmore Business Park, Ellesmere Port, Cheshire, CH65 3EN, UK

t +44 (0)151 350-6666 - [www.sgs.com](http://www.sgs.com)

The certification information can be verified on the web site of Certification and Accreditation Administration of the People's Republic of China [www.cnca.gov.cn](http://www.cnca.gov.cn)



This document is an authentic electronic certificate for Client business purposes use only. Printed version of the electronic certificate are permitted and will be considered as a copy. This document is issued by the Company subject to SGS General Conditions of certification services available on [Terms and Conditions](#) | SGS. Attention is drawn to the limitation of liability, indemnification and jurisdictional clauses contained therein. This document is copyright protected and any unauthorized alteration, forgery or falsification of the content or appearance of this document is unlawful.



此为证书 CN22/00002463 译本

下述组织

## 江苏长隆装饰材料科技有限公司

统一社会信用代码: 91320621MA1P76946C

注册地址: 海安市百川路 168 号

经营地址: 中国江苏省海安市百川路 168 号

的管理体系已经过审核, 并被证明符合下述要求

**ISO 9001:2015**

所涉及的活动范围覆盖

PVC(聚氯乙烯)地板的制造

该证书的有效期自 2025 年 08 月 22 日 至 2028 年 08 月 21 日 并须经过符合要求的监督审核保持有效  
版本号 2, 初始注册日期 2022 年 08 月 22 日

*L. Moran*

签署

Liz Moran

Business Manager

SGS United Kingdom Ltd

Rossmore Business Park, Ellesmere Port, Cheshire, CH65 3EN, UK

t +44 (0)151 350-6666 - [www.sgs.com](http://www.sgs.com)

本证书信息可在国家认证认可监督管理委员会官方网站 [www.cnca.gov.cn](http://www.cnca.gov.cn) 上查询



本文件是真实的电子版证书, 仅供客户用于其商业用途。客户可自行打印, 视同副本。本文件根据 Terms and Conditions | SGS 中认证服务通用条款的要求颁发。提请注意其中已包含的责任范围、赔偿和司法管辖事项。本文件受 SGS 版权保护。任何未经授权的对此文件的内容或外观的变更, 伪造或篡改皆属非法。



此为证书 CN23/00001985 译本

SGS

下述范围

## 江苏长隆装饰材料有限公司

统一社会信用代码: 91320611MA2P19849C

注册地址: 海安镇长江路 558 号

经营地址: 中国江苏省海安市白甸镇 108 号

的管理体系已经过审核, 并被证明符合下述要求

ISO 45001:2018

所涉及的活动范围范围是

PVC(聚氯乙烯)地板的制造

证书的有效期至 2023 年 06 月 21 日 至 2026 年 06 月 20 日 并附经过符合要求的监督审核保持有效  
附录号 1, 初始发证日期 2023 年 06 月 21 日

SGS

Jonathan Hall

Global Head

Certification Services

8000, United Kingdom Ltd

Business Building Park, Ellmers Park, Dordrecht, 2600 DR, NL

T +44 (0) 131 230-6000 - [www.sgs.com](http://www.sgs.com)



本文件是真实的电子证书, 经所发用户于其政府网站, 客户网站打印, 网络副本, 本文件根据 [Terms and Conditions](#) [ 828 条款和条件 ] 提供。在网站上发布时, 请检查其品牌, 标识和认证号。本文件受法律保护, 未经本组织许可, 对本文件进行任何修改或变更, 均造成原文件无效。



Certificate CN23/00001985

SGS

The management system of

## Jiangsu Changlong Decorative Material Technology Co., Ltd.

Unified Social Credit Code: 91330621MA7PT0088C

Business Registration Address: No. 108, Baichuan Road, Hailan City

Business Operation Address: No. 108, Baichuan Road, Hailan City, Jiangsu Province, P.R. China

has been assessed and certified as meeting the requirements of  
**ISO 45001:2018**

For the following activities

Manufacture of PVC (polyvinyl chloride) flooring tiles

This certificate is valid from 21 April 2023 until 20 April 2028 and remains valid subject to satisfactory surveillance audits.  
Issue 1, Certified since 21 April 2023

Authorized by  
Jonathan Hull  
Global Head - Certification Services

SGS United Kingdom Ltd  
Rosemore Business Park, Ebbw Vale, Chepstow, CN23 3EN, UK  
t +44 (0)157 250-6000 - [www.sgs.com](http://www.sgs.com)



This document is an authentic electronic certificate for Client security purposes only. Printed version of the electronic certificate are controlled and will be considered as invalid.  
This document is issued by the Company subject to its General Conditions of certification services available on [www.sgs.com/terms](http://www.sgs.com/terms) (2023). Attention is drawn to the disclaimer of liability, indemnification and optional clauses contained therein. This document is copyright protected and any unauthorised alterations, copies or falsification of the content or appearance of the document is prohibited.



# Certificate of Compliance

**Certificate**

436173-4270

**Issue Date**

10 Jul 2026

**Expiration Date**

10 Jul 2029



Ranee Valles  
Director and General Manager

UL Verification Services Inc.  
2211 Newmarket Parkway, ste 106  
Marietta, GA 30067 USA

UL Verification Services does hereby certify that an independent assessment has been conducted on behalf of:

**Jiangsu Changlong Decorative Material Technical Co Ltd**

for the following product:

**PVC Flooring**

meets the requirements for:

**ECOLOGO®**

NSF 332 - 2022 Level 1 - Sustainability Assessment for Resilient Floor Coverings



UL Solutions evaluated representative samples of the identified product, process or facility to the identified Standard or other requirements in accordance with the agreements and any applicable program service terms in place between UL Solutions and the Client (collectively "Agreement"). The Client is authorized to use the UL Mark for the identified Product, process or facility covered by this certificate, in accordance with the terms of the Agreement. This Certificate is valid for the identified dates unless there is non-compliance with the Agreement. This certificate is subject to modification, suspension and withdrawal by UL Solutions see [SPOT.ul.com](https://SPOT.ul.com), to authenticate the certificate.

N° 370-002.01/23



**Le CSTB accorde le droit d'usage de la marque QB à la société :**

*CSTB grants the right to use the QB Mark to the company :*

Nom du titulaire  
Holder name

**JIANGSU CHANGLONG DECORATIVE MATERIAL  
TECHNICAL Co., Ltd**

168 BAICHUAN Rd Hai An City  
226600 JIANGSU CHINA

Chine  
**370**

Site  
Site

168 BAICHUAN Rd Hai An City  
226600 JIANGSU CHINA  
Chine

**Pour les produits listés ci-après, certifiés conformes aux exigences du référentiel de certification,  
par le CSTB.**

*For the products listed below, certified conform to the certification reference system requirements by CSTB*

Désignation  
Name

**GRAND MASTER COLLECTION**



**La validité de ce certificat et la liste des produits certifiés sont vérifiables sur le site Internet  
ou en flashant le QR-code ci-contre :**

*The validity of this certificate and the certified product list can both be checked on the website or by flashing the QR-Code:*

**<https://evaluation.cstb.fr>**

Décision d'admission initiale / Admission decision N° 7010/370- du 20/02/2024

Président du CSTB  
Etienne CREPON

Fait à : **Marne-la-Vallée, France.**  
Done at

Date : **26/02/2024**  
Date



En vertu de la présente décision, le CSTB accorde le droit d'usage de la Marque QB à la société qui en est titulaire pour les produits visés par ce certificat, dans les conditions définies par les Exigences générales de la Marque QB et par les référentiels de certification QB, pour autant que les contrôles réguliers de la fabrication et les vérifications par tierce partie soient satisfaisants.

On the strength of the present decision, CSTB grants the right to use the QB Mark to the licence holder for the products mentioned in this certificate, within the frame of the general requirements of the QB Mark and of the QB certification reference system, as far as the regular checking and third party verifications of the production are satisfactory.



<https://evaluation.cstb.fr>

N° **370-002.01/23**

Date de publication : 26/02/2024

## Produits certifiés par le CSTB

CSTB Certified products

### Revêtements de sol hétérogènes en polychlorure de vinyle / Resilient floor coverings - heterogeneous polyvinyl chloride floor coverings

Normes / Standards: NF EN ISO 10582

#### Dalle / Tiles

Désignation Commerciale / Commercial Designation : GRAND MASTER COLLECTION

Épaisseur totale (mm) / Total thickness (mm) : 2.5 ( + 0,13 ; - 0,10)

Épaisseur couche usure (mm) / Wear layer thickness (mm) : 0.70 ( + 13% - 10% limité à ± 0,1)

Masse surfacique totale (g/m²) / Total mass per unit area (g/m²) : 4077 ( + 13% ; - 10%)

| Largeur (mm) | Longueur (mm) | Pose lés chape ciment ou béton | Classement chape ciment ou béton | Pose lés bois ou à base de bois | Classement bois ou à base de bois |
|--------------|---------------|--------------------------------|----------------------------------|---------------------------------|-----------------------------------|
| Width (mm)   | Length (mm)   | Lay out on cement or concrete  | Cement or concrete class         | Lay out on wood or wooden       | Wood or wooden class              |
| 304.8        | 304.8         | a                              | U4 P3 E2 C2                      | a                               | U4 P3 E1 C2                       |
| 457.2        | 457.2         | a                              | U4 P3 E2 C2                      | a                               | U4 P3 E1 C2                       |
| 457.2        | 914.4         | a                              | U4 P3 E2 C2                      | a                               | U4 P3 E1 C2                       |
| 457.2        | 1219.2        | a                              | U4 P3 E2 C2                      | a                               | U4 P3 E1 C2                       |
| 600.0        | 600.0         | a                              | U4 P3 E2 C2                      | a                               | U4 P3 E1 C2                       |

### Revêtements de sol hétérogènes en polychlorure de vinyle / Resilient floor coverings - heterogeneous polyvinyl chloride floor coverings

Normes / Standards: NF EN ISO 10582

#### Lame / Planks

Désignation Commerciale / Commercial Designation : GRAND MASTER COLLECTION

Épaisseur totale (mm) / Total thickness (mm) : 2.5 ( + 0,13 ; - 0,10)

Épaisseur couche usure (mm) / Wear layer thickness (mm) : 0.70 ( + 13% - 10% limité à ± 0,1)

Masse surfacique totale (g/m²) / Total mass per unit area (g/m²) : 4077 ( + 13% ; - 10%)

| Largeur (mm) | Longueur (mm) | Pose lés chape ciment ou béton | Classement chape ciment ou béton | Pose lés bois ou à base de bois | Classement bois ou à base de bois |
|--------------|---------------|--------------------------------|----------------------------------|---------------------------------|-----------------------------------|
| Width (mm)   | Length (mm)   | Lay out on cement or concrete  | Cement or concrete class         | Lay out on wood or wooden       | Wood or wooden class              |
| 76.2         | 304.8         | a                              | U4 P3 E2 C2                      | a                               | U4 P3 E1 C2                       |
| 144.0        | 720.0         | a                              | U4 P3 E2 C2                      | a                               | U4 P3 E1 C2                       |
| 184.1        | 1219.2        | a                              | U4 P3 E2 C2                      | a                               | U4 P3 E1 C2                       |
| 228.6        | 1219.2        | a                              | U4 P3 E2 C2                      | a                               | U4 P3 E1 C2                       |
| 228.6        | 1422.4        | a                              | U4 P3 E2 C2                      | a                               | U4 P3 E1 C2                       |

N° 370-003.01/23



**Le CSTB accorde le droit d'usage de la marque QB à la société :**

*CSTB grants the right to use the QB Mark to the company :*

Nom du titulaire  
Holder name

**JIANGSU CHANGLONG DECORATIVE MATERIAL  
TECHNICAL Co., Ltd**

168 BAICHUAN Rd Hai An City  
226600 JIANGSU CHINA

Chine  
**370**

Site  
Site

168 BAICHUAN Rd Hai An City  
226600 JIANGSU CHINA  
Chine

**Pour les produits listés ci-après, certifiés conformes aux exigences du référentiel de certification,  
par le CSTB.**

*For the products listed below, certified conform to the certification reference system requirements by CSTB*

Désignation  
Name

**DESIGN COLLECTION**



**La validité de ce certificat et la liste des produits certifiés sont vérifiables sur le site Internet  
ou en flashant le QR-code ci-contre :**

*The validity of this certificate and the certified product list can both be checked on the website or by flashing the QR-Code:*

**<https://evaluation.cstb.fr>**

Décision d'admission initiale / Admission decision N° 370-003.01 du 20/02/2024

  
Président du CSTB  
Etienne CREPON

Fait à : **Marne-la-Vallée, France.**  
Done at

Date : **26/02/2024**  
Date



En vertu de la présente décision, le CSTB accorde le droit d'usage de la Marque QB à la société qui en est titulaire pour les produits visés par ce certificat, dans les conditions définies par les Exigences générales de la Marque QB et par les référentiels de certification QB, pour autant que les contrôles réguliers de la fabrication et les vérifications par tierce partie soient satisfaisants.

On the strength of the present decision, CSTB grants the right to use the QB Mark to the licence holder for the products mentioned in this certificate, within the frame of the general requirements of the QB Mark and of the QB certification reference system, as far as the regular checking and third party verifications of the production are satisfactory.



<https://evaluation.cstb.fr>



N° **370-003.01/23**

Date de publication : 26/02/2024

## Produits certifiés par le CSTB

CSTB Certified products

### Revêtements de sol hétérogènes en polychlorure de vinyle / Resilient floor coverings - heterogeneous polyvinyl chloride floor coverings

Normes / Standards: NF EN ISO 10582

#### Dalle / Tiles

Désignation Commerciale / Commercial Designation : DESIGN COLLECTION

Épaisseur totale (mm) / Total thickness (mm) : 2.5 ( + 0,13 ; - 0,10)

Épaisseur couche usure (mm) / Wear layer thickness (mm) : 0.55 ( + 13% - 10% limité à ± 0,1)

Masse surfacique totale (g/m²) / Total mass per unit area (g/m²) : 4398 ( + 13% ; - 10%)

| Largeur (mm) | Longueur (mm) | Pose lés chape ciment ou béton | Classement chape ciment ou béton | Pose lés bois ou à base de bois | Classement bois ou à base de bois |
|--------------|---------------|--------------------------------|----------------------------------|---------------------------------|-----------------------------------|
| Width (mm)   | Length (mm)   | Lay out on cement or concrete  | Cement or concrete class         | Lay out on wood or wooden       | Wood or wooden class              |
| 304.8        | 304.8         | a                              | U3 P3 E2 C2                      | a                               | U3 P3 E1 C2                       |
| 457.2        | 457.2         | a                              | U3 P3 E2 C2                      | a                               | U3 P3 E1 C2                       |
| 457.2        | 914.4         | a                              | U3 P3 E2 C2                      | a                               | U3 P3 E1 C2                       |
| 457.2        | 1219.2        | a                              | U3 P3 E2 C2                      | a                               | U3 P3 E1 C2                       |
| 600.0        | 600.0         | a                              | U3 P3 E2 C2                      | a                               | U3 P3 E1 C2                       |

### Revêtements de sol hétérogènes en polychlorure de vinyle / Resilient floor coverings - heterogeneous polyvinyl chloride floor coverings

Normes / Standards: NF EN ISO 10582

#### Lame / Planks

Désignation Commerciale / Commercial Designation : DESIGN COLLECTION

Épaisseur totale (mm) / Total thickness (mm) : 2.5 ( + 0,13 ; - 0,10)

Épaisseur couche usure (mm) / Wear layer thickness (mm) : 0.55 ( + 13% - 10% limité à ± 0,1)

Masse surfacique totale (g/m²) / Total mass per unit area (g/m²) : 4398 ( + 13% ; - 10%)

| Largeur (mm) | Longueur (mm) | Pose lés chape ciment ou béton | Classement chape ciment ou béton | Pose lés bois ou à base de bois | Classement bois ou à base de bois |
|--------------|---------------|--------------------------------|----------------------------------|---------------------------------|-----------------------------------|
| Width (mm)   | Length (mm)   | Lay out on cement or concrete  | Cement or concrete class         | Lay out on wood or wooden       | Wood or wooden class              |
| 76.2         | 304.8         | a                              | U3 P3 E2 C2                      | a                               | U3 P3 E1 C2                       |
| 144.0        | 720.0         | a                              | U3 P3 E2 C2                      | a                               | U4 P3 E1 C2                       |
| 184.1        | 1219.2        | a                              | U3 P3 E2 C2                      | a                               | U3 P3 E1 C2                       |
| 228.6        | 1219.2        |                                |                                  |                                 |                                   |
| 228.6        | 1422.4        | a                              | U3 P3 E2 C2                      | a                               | U3 P3 E1 C2                       |

## Test Report

No.: CANIN24010654101

Date: Jun 11, 2024

Page 1 of 3

Client Name: JIANGSU CHANGLONG DECORATIVE MATERIAL TECHNICAL CO., LTD.

Client Address: NO.168 BAICHUAN ROAD, HAIAN CITY, JIANGSU PROVINCE, CHINA

Sample Name: PVC FLOOR COVERING

Product Specification: 3.0/0.5mm

The above sample(s) and information were provided by the client.

SGS Job No.: GZCCM2405000808

Sample Receiving Date: May 22, 2024

Testing Period: May 22, 2024 ~ May 29, 2024

Test Requested: Select test(s) as requested by the client.

Test Method(s): Please refer to next page(s).

Test Result(s): Please refer to next page(s).

| Test Requirement                                                                                 | Conclusion |
|--------------------------------------------------------------------------------------------------|------------|
| EN 71-3:2019+A1:2021- Migration of certain elements (Category III:for scrapped-off toy material) | Pass       |

Signed for and on behalf of  
SGS-CSTC Standards Technical Services Co., Ltd. Guangzhou Branch

Violet Shi

Violet Shi

Approved Signatory

Scan to see the report



CANIN24010654101

Verification:  
check.sgsonline.com.cn

Unless otherwise agreed in writing, this document is issued by the Company subject to its General Conditions of Service printed overleaf, available on request or accessible at <https://www.sgs.com/en/Terms-and-Conditions>. Attention is drawn to the limitation of liability, indemnification and jurisdiction issues defined therein. Any holder of this document is advised that information contained hereon reflects the Company's findings at the time of its intervention only and within the limits of Client's instructions, if any. The Company's sole responsibility is to its Client and this document does not exonerate parties to a transaction from exercising all their rights and obligations under the transaction documents. This document cannot be reproduced except in full, without prior written approval of the Company. Any unauthorized alteration, forgery or falsification of the content or appearance of this document is unlawful and offenders may be prosecuted to the fullest extent of the law. Unless otherwise stated the results shown in this test report refer only to the sample(s) tested.

Attention: To check the authenticity of testing (inspection report & certificate, please contact us at telephone: (86-755) 8307 1443, or email: CN.Doccheck@sgs.com

No.168, Baichuan Road, Science City, Economic & Technological Development Area, Guangzhou, Guangdong, China 510663  
中国·广东·广州高新技术产业开发区科学城珠路168号 邮编: 510663

T: (86-20) 82155555 www.sgs.com.cn  
T: (86-20) 82155555 sgs.china@sgs.com

Member of the SGS Group (SGS SA)

## Test Report

No.: CANIN24010654101

Date: Jun 11, 2024

Page 2 of 3

### Test Result(s):

#### Test Part Description:

| SN ID | Sample No. | SGS Sample ID           | Description          |
|-------|------------|-------------------------|----------------------|
| SN1   | 001        | CAN24-0106541-0001.C001 | "PVC FLOOR COVERING" |

#### Remarks:

- (1) 1 mg/kg = 1 ppm = 0.0001%
- (2) MDL = Method Detection Limit
- (3) ND = Not Detected (< MDL)
- (4) "-" = Not Regulated

### EN 71-3:2019+A1:2021- Migration of certain elements (Category III:for scrapped-off toy material)

Test Method: With reference to EN 71-3:2019+A1:2021, analysis was performed by ICP-OES, IC-UV or LC-ICP-MS.  
Soluble Organic Tin was analyzed by GC-MS.

| Test Item(s)                 | Limit  | Unit(s) | MDL   | 001         |
|------------------------------|--------|---------|-------|-------------|
| Soluble Aluminum(Al)         | 28130  | mg/kg   | 50    | ND          |
| Soluble Antimony(Sb)         | 560    | mg/kg   | 10    | ND          |
| Soluble Arsenic(As)          | 47     | mg/kg   | 10    | ND          |
| Soluble Barium(Ba)           | 18750  | mg/kg   | 50    | ND          |
| Soluble Boron(B)             | 15000  | mg/kg   | 50    | ND          |
| Soluble Cadmium(Cd)          | 17     | mg/kg   | 5     | ND          |
| Soluble Chromium(III)(CrIII) | 460    | mg/kg   | 5     | ND          |
| Soluble Chromium(VI)(Cr VI)  | 0.053  | mg/kg   | 0.025 | ND          |
| Soluble Cobalt(Co)           | 130    | mg/kg   | 10    | ND          |
| Soluble Copper(Cu)           | 7700   | mg/kg   | 50    | ND          |
| Soluble Lead(Pb)             | 23     | mg/kg   | 5     | ND          |
| Soluble Manganese(Mn)        | 15000  | mg/kg   | 50    | ND          |
| Soluble Mercury(Hg)          | 94     | mg/kg   | 10    | ND          |
| Soluble Nickel(Ni)           | 930    | mg/kg   | 10    | ND          |
| Soluble Selenium(Se)         | 460    | mg/kg   | 10    | ND          |
| Soluble Strontium(Sr)        | 56000  | mg/kg   | 50    | ND          |
| Soluble Tin(Sn)              | 180000 | mg/kg   | 3     | ND          |
| Soluble Zinc(Zn)             | 46000  | mg/kg   | 50    | ND          |
| Soluble Organic Tin*         | 12     | mg/kg   | -     | ND          |
| <b>Conclusion</b>            |        |         |       | <b>Pass</b> |

#### Notes:

(1) According to Chapter of 10.1.2 of EN 71-3:2019+A1:2021, Chromium (III) is calculated by the following formula:

Soluble Chromium (III) = Soluble Total Chromium - Soluble Chromium (VI)

(2) \*Confirmation test of soluble organic tin is not required in case of soluble tin, after conversion, does not exceed the soluble organic tin requirement as specified in EN 71-3:2019+A1:2021.

Unless otherwise stated, the decision rule for conformity reporting is based on Binary Statement for Simple Acceptance Rule (w=0) stated in ILAC-G8:09/2019.



Unless otherwise agreed in writing, this document is issued by the Company subject to its General Conditions of Service printed overleaf, available on request or accessible at <https://www.sgs.com/en/Terms-and-Conditions>. Attention is drawn to the limitation of liability, indemnification and jurisdiction issues defined therein. Any holder of this document is advised that information contained hereon reflects the Company's findings at the time of its intervention only and within the limits of Client's instructions, if any. The Company's sole responsibility is to its Client and this document does not exonerate parties to a transaction from exercising all their rights and obligations under the transaction documents. This document cannot be reproduced except in full, without prior written approval of the Company. Any unauthorized alteration, forgery or falsification of the content or appearance of this document is unlawful and offenders may be prosecuted to the fullest extent of the law. Unless otherwise stated the results shown in this test report refer only to the sample(s) tested.  
Attention: To check the authenticity of testing / inspection report & certificate, please contact us at telephone: (86-755) 8307 1443, or email: [CN.Doccheck@sgs.com](mailto:CN.Doccheck@sgs.com)

SGS-CTI Technical Service Co., Ltd.  
Guangzhou Branch / 广州分公司

No.198, Kechu Road, Science City, Economic & Technological Development Area, Guangzhou, Guangdong, China 510663  
中国·广东·广州高新技术产业开发区科学城科珠路198号 邮编: 510663

T (86-20) 82155555 [www.sgs.com.cn](http://www.sgs.com.cn)  
T (86-20) 82155555 [sgs.china@sgs.com](mailto:sgs.china@sgs.com)

Member of the SGS Group (SGS SA)



## Test Report

No.: CANIN24010654101

Date: Jun 11, 2024

Page 3 of 3

Sample Photo:



SGS authenticate the photo on original report only  
\*\*\* End of Report \*\*\*



SGS-CHN (Guangzhou) Technical Services Co., Ltd.  
Guangzhou Branch Technical Laboratory

Unless otherwise agreed in writing, this document is issued by the Company subject to its General Conditions of Service printed overleaf, available on request or accessible at <https://www.sgs.com/en/Terms-and-Conditions>. Attention is drawn to the limitation of liability, indemnification and jurisdiction issues defined therein. Any holder of this document is advised that information contained hereon reflects the Company's findings at the time of its intervention only and within the limits of Client's instructions, if any. The Company's sole responsibility is to its Client and this document does not exonerate parties to a transaction from exercising all their rights and obligations under the transaction documents. This document cannot be reproduced except in full, without prior written approval of the Company. Any unauthorized alteration, forgery or falsification of the content or appearance of this document is unlawful and offenders may be prosecuted to the fullest extent of the law. Unless otherwise stated the results shown in this test report refer only to the sample(s) tested.  
Attention: To check the authenticity of testing/inspection report & certificate, please contact us at telephone: (86-755) 8307 1443, or email: [CN.Doccheck@sgs.com](mailto:CN.Doccheck@sgs.com)

No.198, Kechu Road, Science City, Economic & Technological Development Area, Guangzhou, Guangdong, China 510663  
中国·广东·广州高新技术产业开发区科学城科珠路198号 邮编: 510663

T (86-20) 82155555 [www.sgsgroup.com.cn](http://www.sgsgroup.com.cn)  
T (86-20) 82155555 [sgs.china@sgs.com](mailto:sgs.china@sgs.com)

Member of the SGS Group (SGS SA)